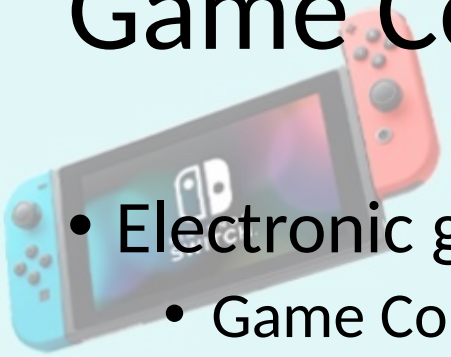


Game Design

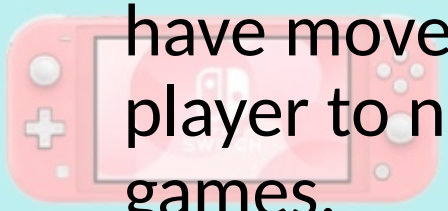
Mr Butler's

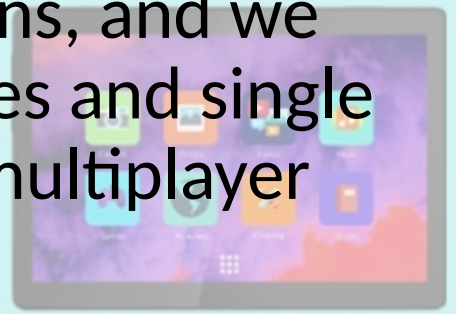
Game Consoles and Devices






- Electronic games can be played on devices including:
  - Game Consoles
  - Mobile Phones
  - Computers
- These are also known as Game **platforms**.
- These platforms have been developed for many generations, and we have moved from the installation of discs on game consoles and single player to now using cloud technology and playing online multiplayer games.



Educational Gaming

- Educational gaming applications allow users to apply their knowledge and support their learning within a virtual world.
- Features such as leader boards help motivate individuals to improve their scores
- Can take on different roles within the game and reflect on what happened in certain situations, enhancing learning
- Games are designed with rules and increasing levels of difficulty – the user must comply with the rules and cannot move to the next level before completing the previous.



Discussion: What makes a good game?





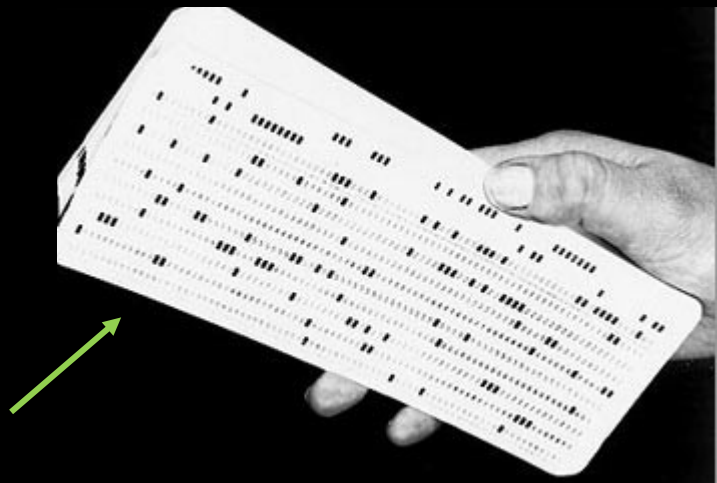
THE HISTORY OF GAMING

- The history of video games goes as far back as the early 1950s, when academic computer scientists began designing simple games and simulations as part of their research or just for fun.
- At M.I.T. in the 1960s, professors and students played games such as 3D tic-tac-toe and Moon Landing.
- These games were played on computer such as the IBM 1560, and moves were made by means of punch cards.
- Video gaming did not reach mainstream popularity until the 1970s and 1980s, when video arcade games and gaming consoles using joysticks, buttons, and other controllers, along with graphics on computer screens and home computer games were introduced to the general public.

The IBM 1560



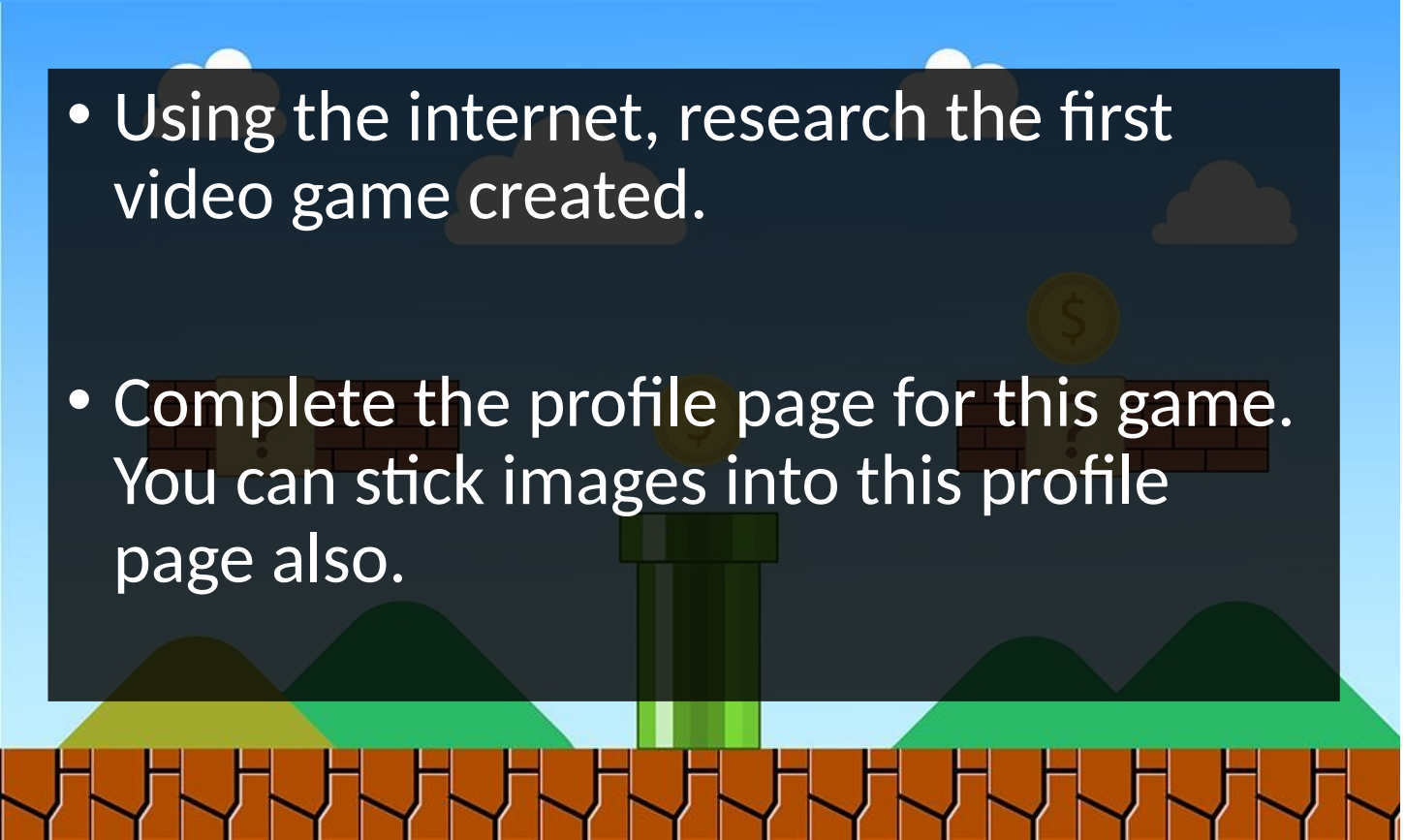
Name:	IBM 1560
Year of Manufacture:	August 12, 1981
Type:	Personal Computer
Memory:	16kb-640kb
Sound:	1-channel PC speaker
OS:	IBM BASIC / PC DOS



Punch cards held the game

So what was the first game?

- Using the internet, research the first video game created.
- Complete the profile page for this game. You can stick images into this profile page also.



TASK: Profile Page: The First Game

Game Maker 8 Features

PICTURE	COVER PHOTO
	NAME OF GAME
DOB	WHY WAS IT CREATED?
KEY TERMS / KEY NAMES / WHO CREATED IT?	HOW DID PLAYERS INTERACT WITH IT?
	HOW DID YOU WIN THE GAME?



- 2D game engine which allows us to design, build (and play!) a 2D game.
- The engine allows us to use drag and drop options, meaning that a game will not need to be programmed in the way that we can still develop a game without needing to develop the code behind these drag and drop features.
- The Game Maker software splits its game creation into several areas/categories

Theory: Sprites

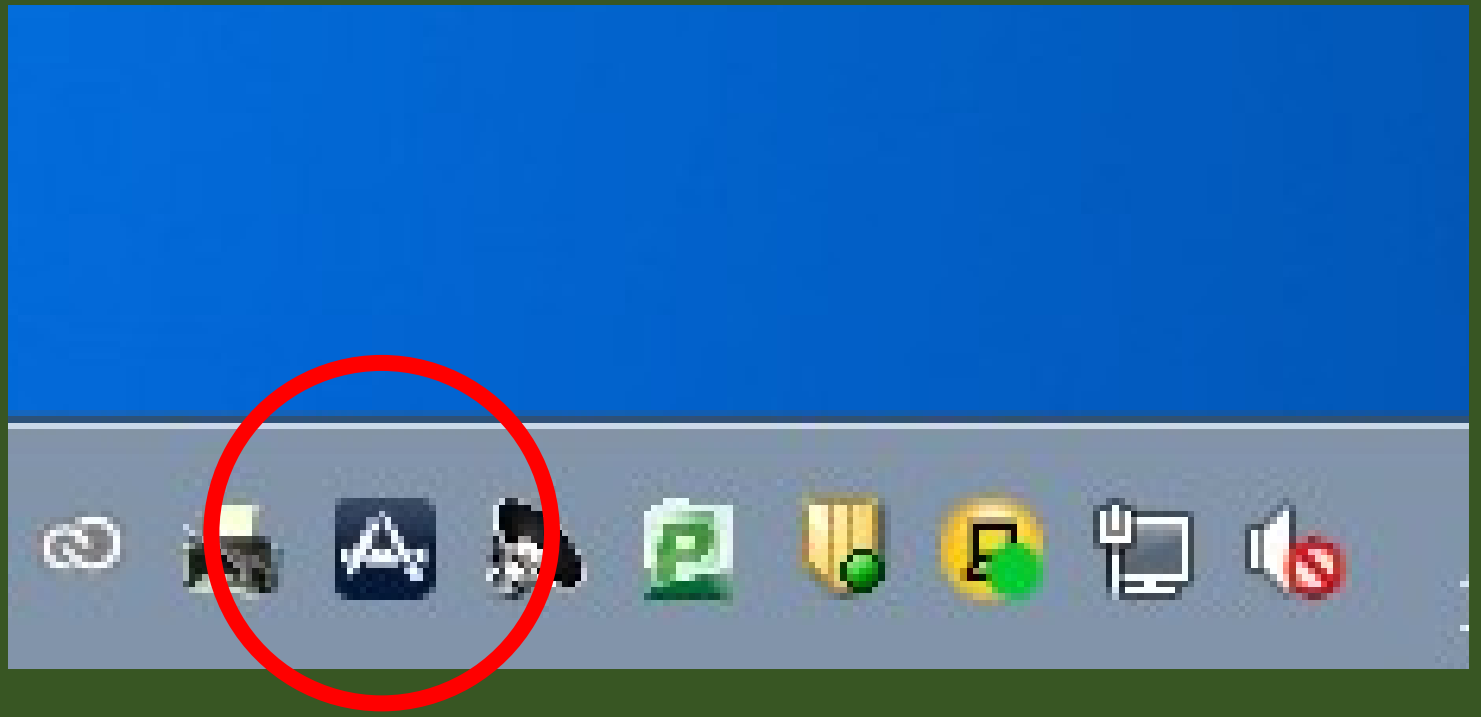
- A sprite is an image, or a number of images, that represents something in the game.
  - If the sprite is not animated, will be one picture
  - If it is animated, the sprite has a number of sub-images.
- Many games use multiple sprites with different animations each to represent one character performing different actions such as walking, jumping and falling.



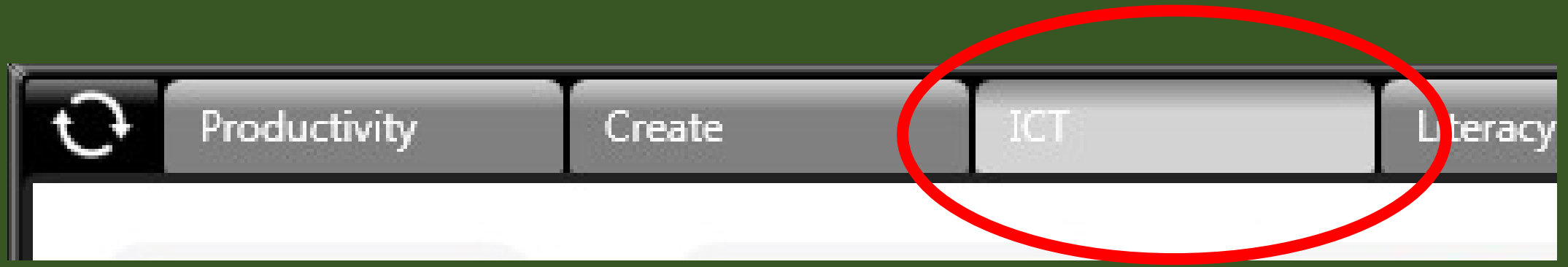
Let’s Create a Sprite

Tutorial Part 1

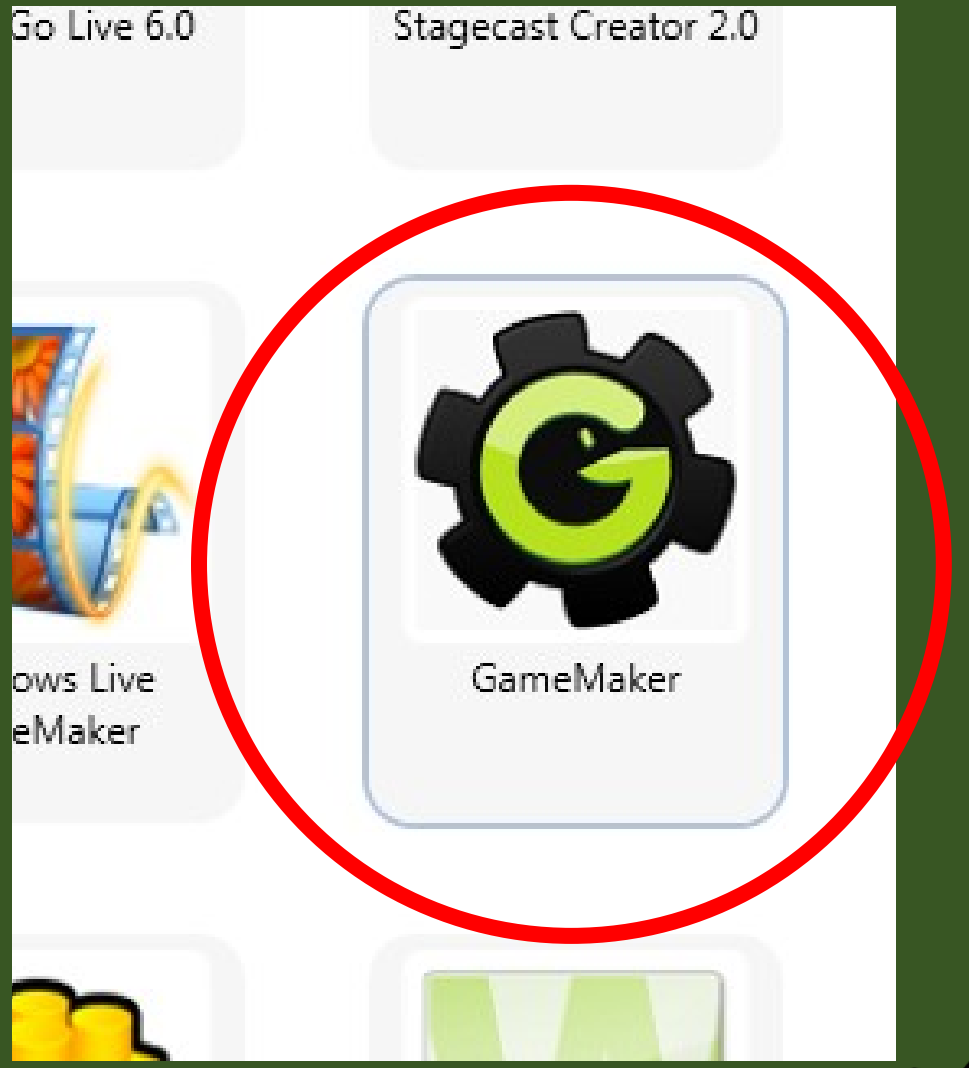
Open My Apps.



Select the ICT Tab



Double click on GameMaker



Click on
Continue Using the Lite Edition



<new game> - Game Maker 8.0 Lite (Simple Mode)

File Edit Resources Run Window Help

- Sprites
- Sounds
- Backgrounds
- Objects
- Rooms
- Game Information
- Global Game Settings

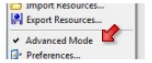
Tutorial

Page 1 of 15

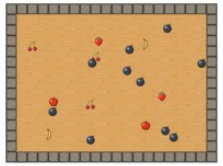
Your First Game

Welcome to *Game Maker*. This tutorial shows you how to create your first game. It will take just 30 minutes to complete and will teach you the important basics of *Game Maker*.

We assume you run *Game Maker* in **Simple** mode. (You can see this in the window caption.) If not, click on the **File** menu and press **Advanced Mode** to remove the checkmark:



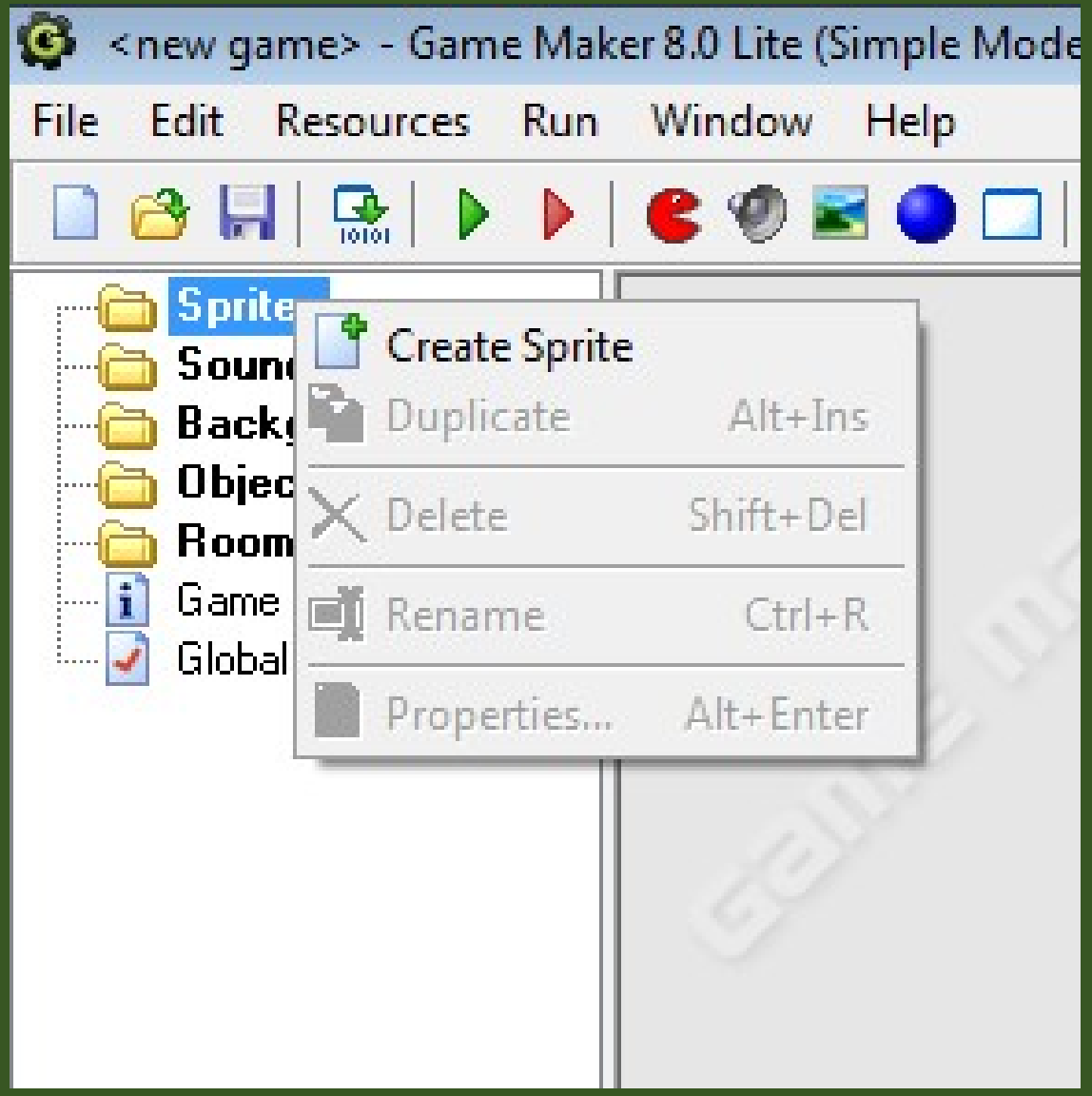
In the game we create, some pieces of fruit move around. The player must click on the fruits with the mouse to collect them. For each collected fruit, points are scored. Fast moving fruits give more points than slow fruits. And there are bombs lying around that you should avoid. The game will look as follows:



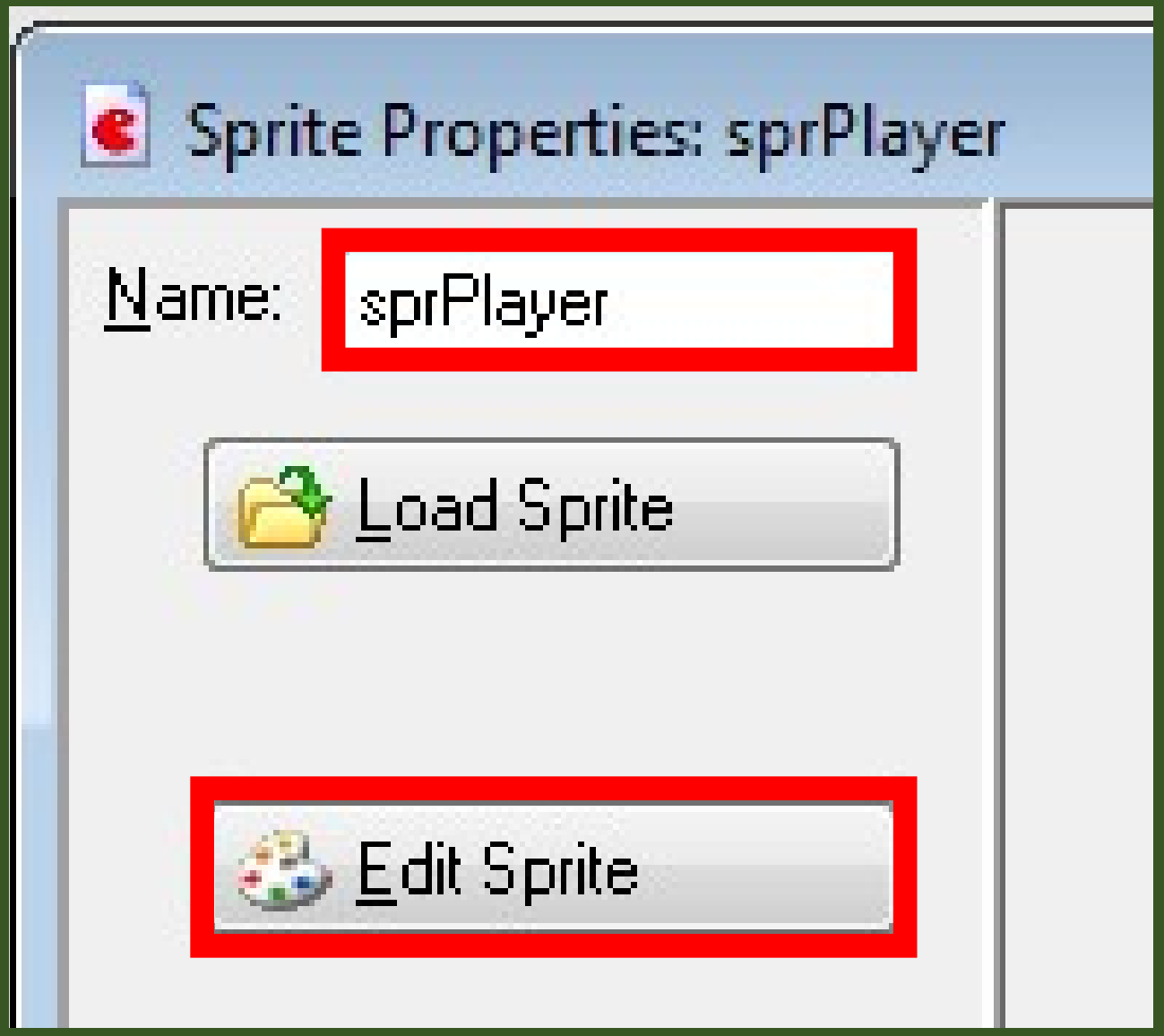
*Click on the **Next** button to go to the next page of the tutorial.*

Next Hide

Show the tutorials at startup.

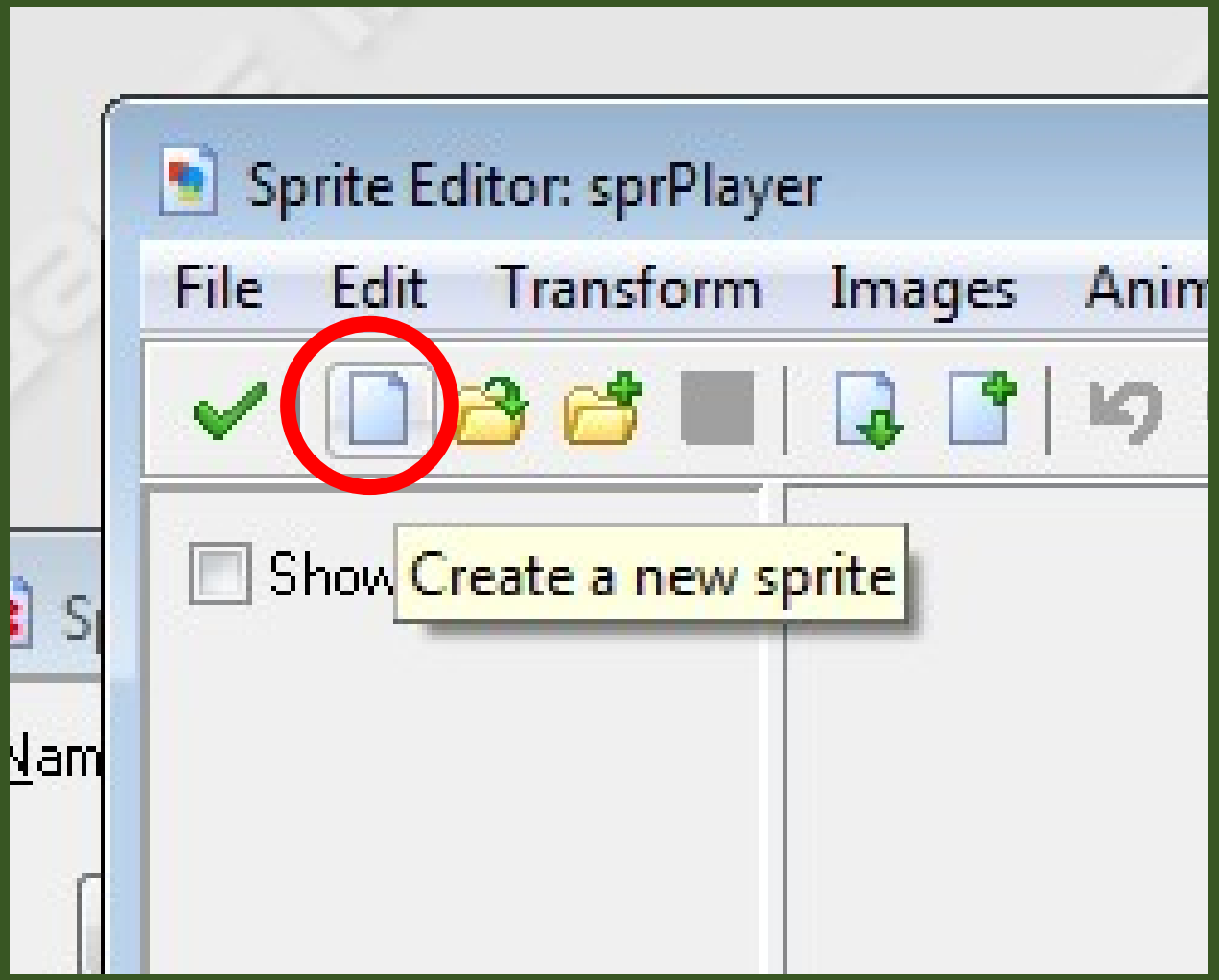


Right Click on **Sprite**.
Click **Create Sprite**



We need to name this Sprite.
Name this one sprPlayer.

Then Click **Edit Sprite**.

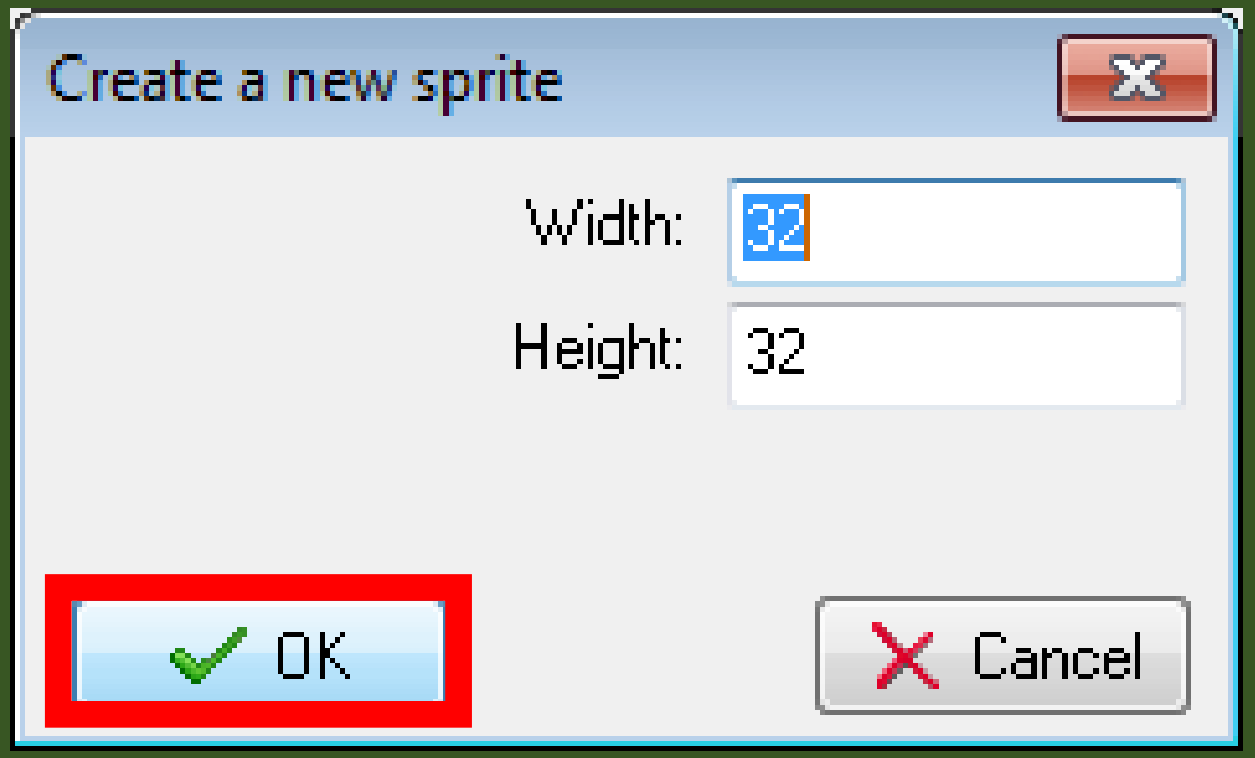


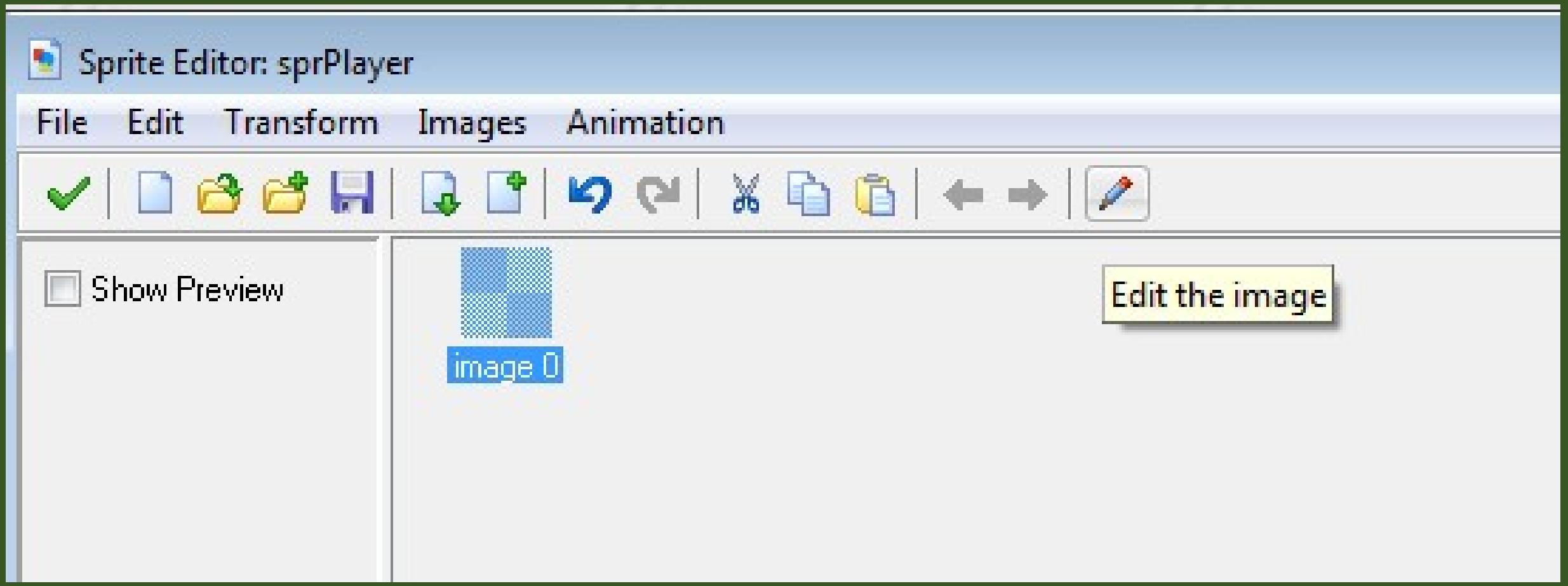
Click The Sheet of Paper.

This will create a ‘New’ sprite / image

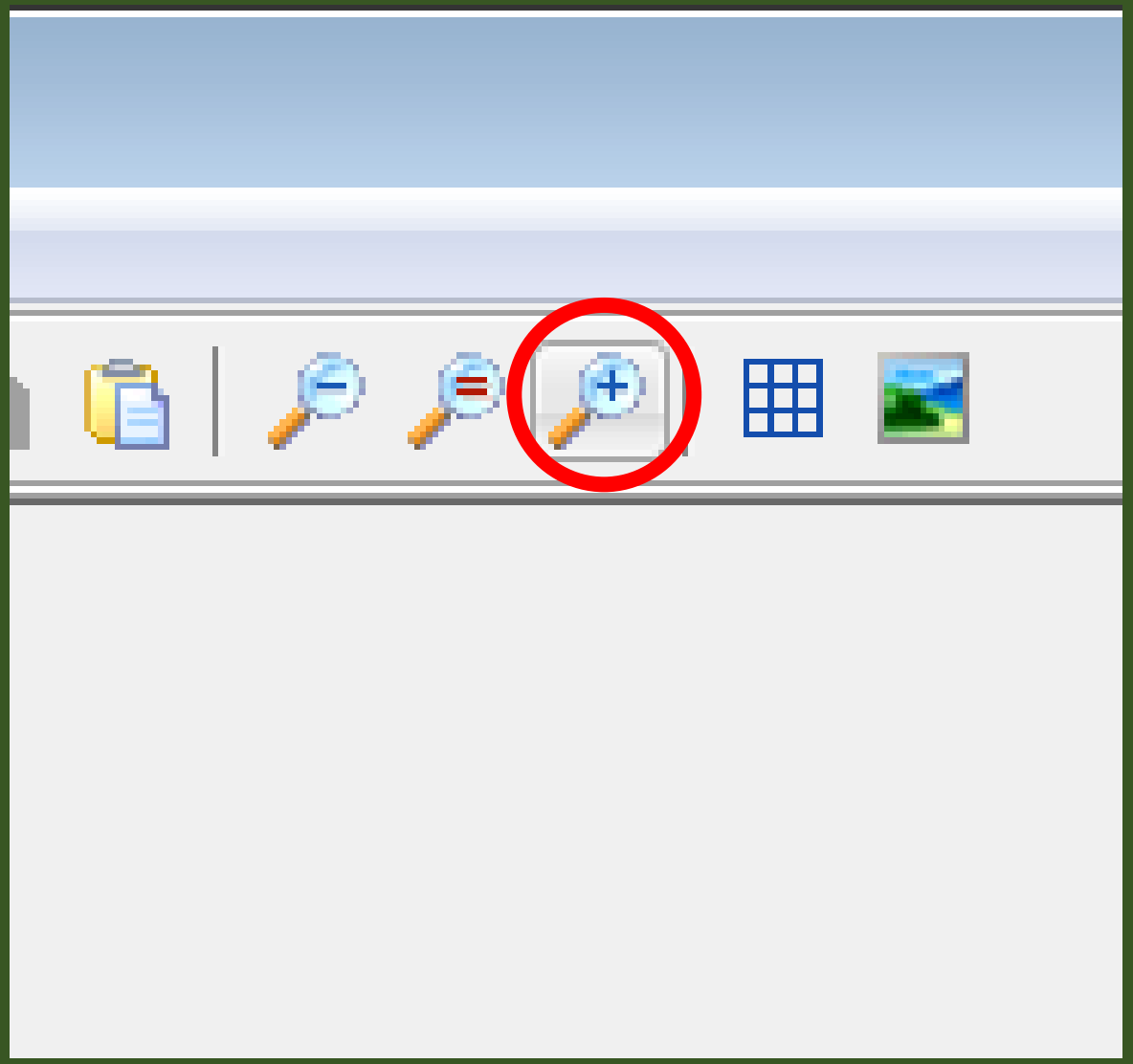
Leave the Width and Height as 32.

Click OK to start editing!

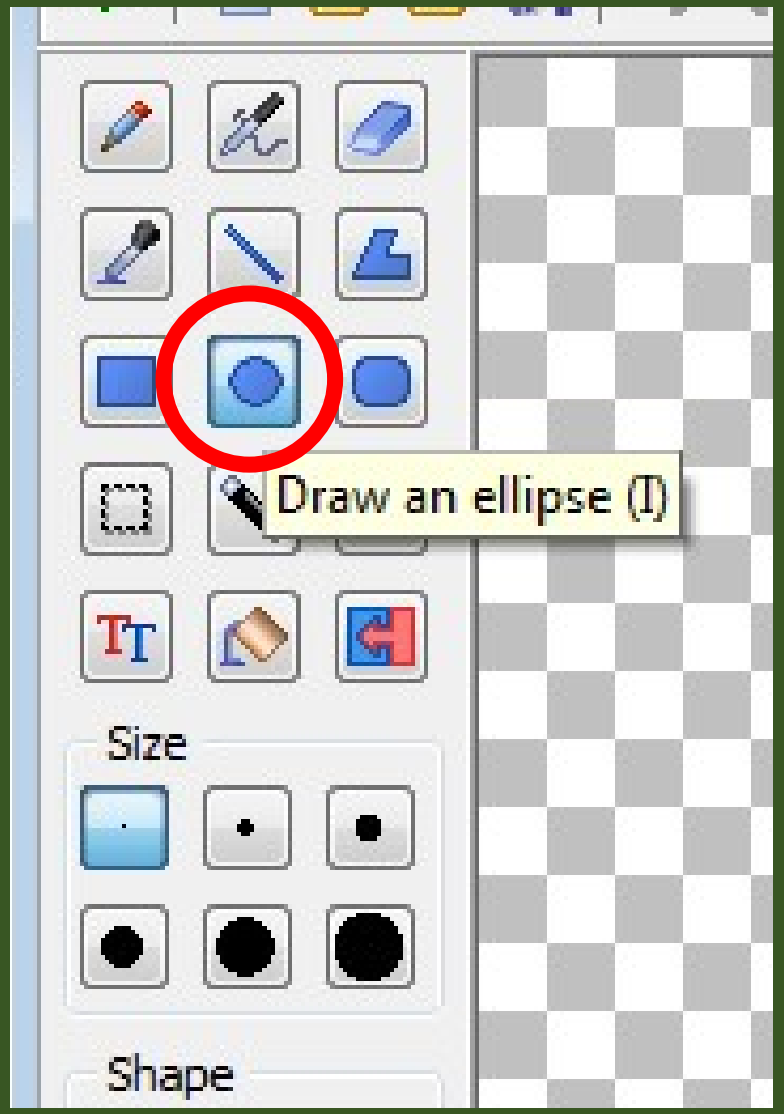




Click on the Pen to start editing this sprite.

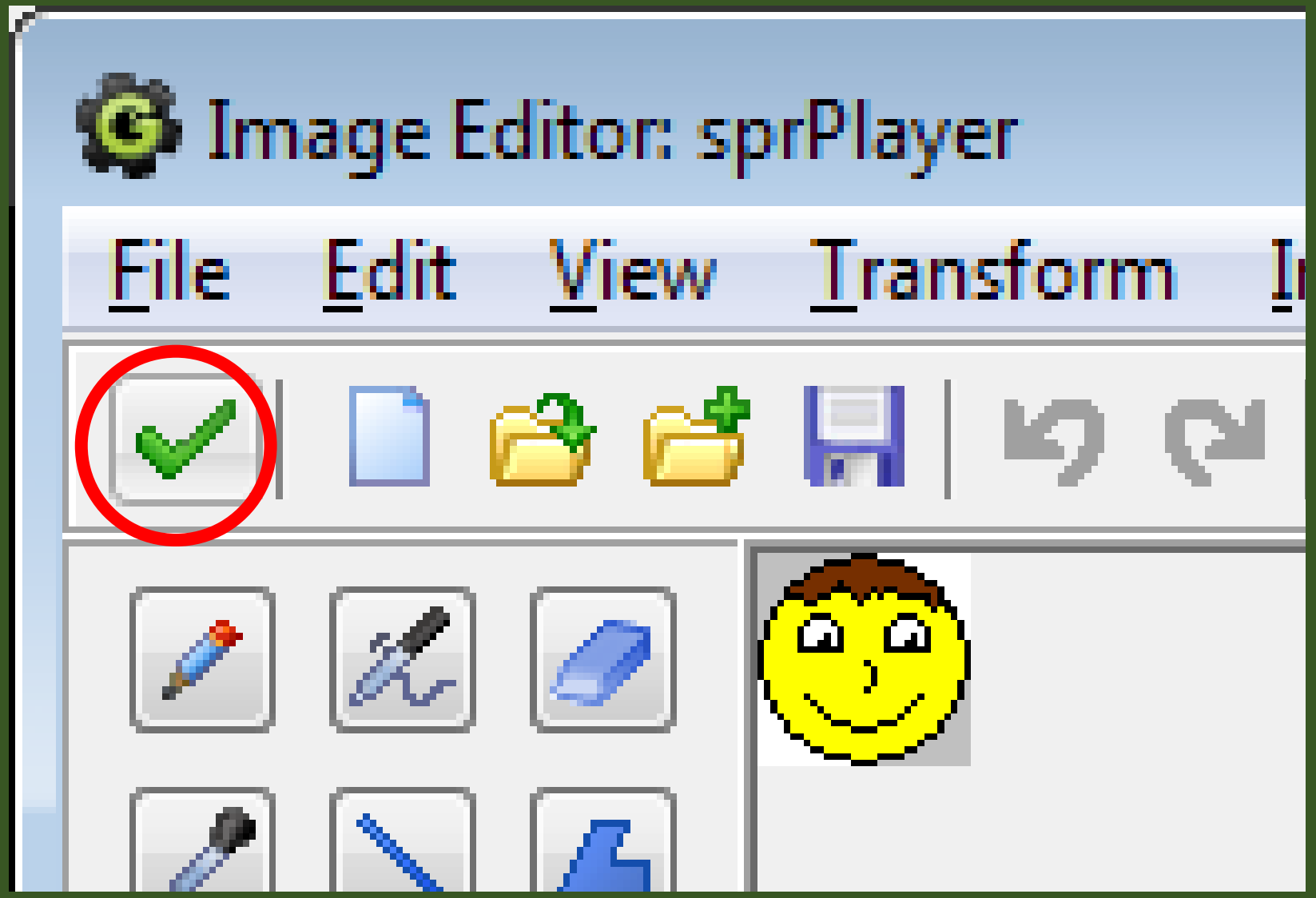


The image will be very small. Click the magnifying glass with the + sign to make the image bigger.



Use the toolbox to create an image for your player.

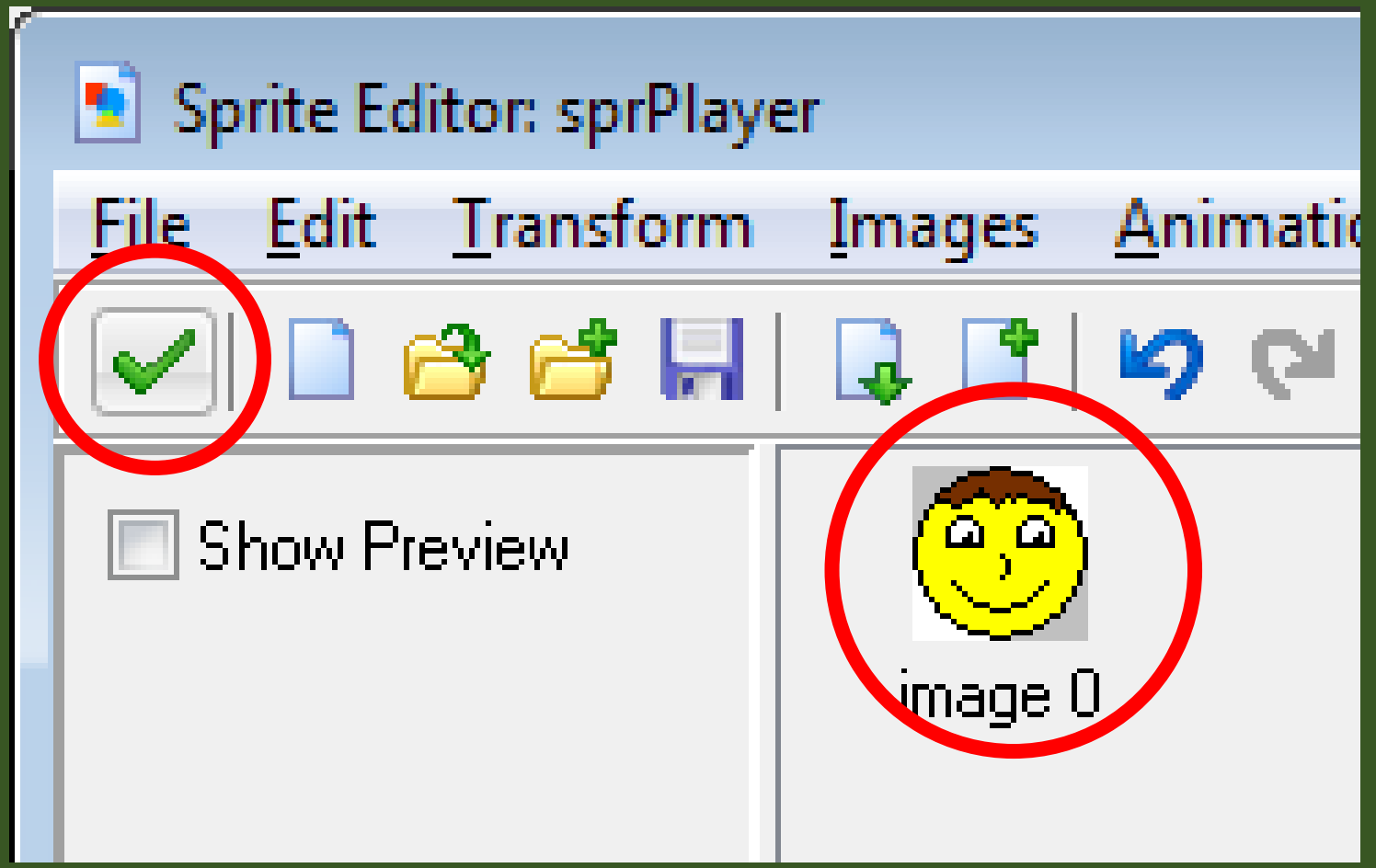
Make Sure you fill the area. Remember this will be very small on screen!



Once we have made the player, we can click the Green Tick to confirm

We can now see a preview of our sprite/character.

Click the Green Tick Again to save.



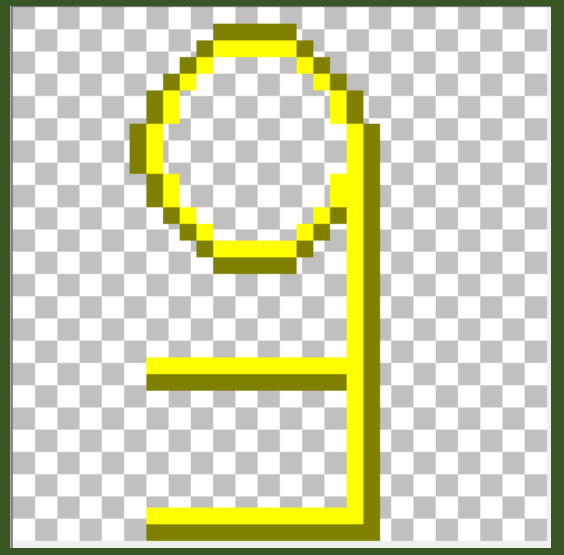
We now need to create:

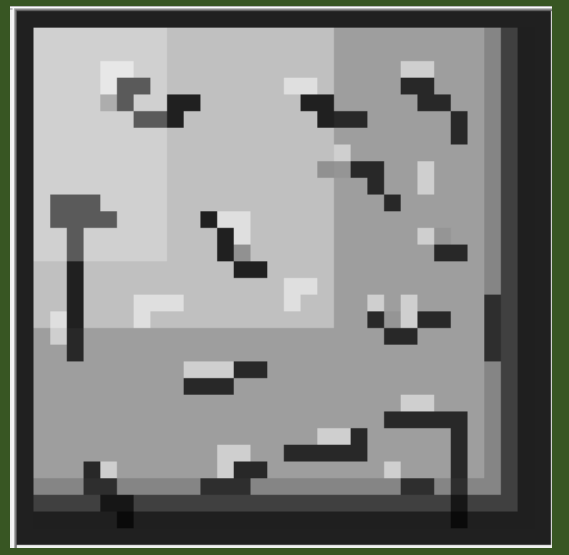
- A Square Wall (labelled as sprWall)
- A Bad Guy (labelled as sprEnemy)
- A Collectible (a key of some description, labelled as sprKey)

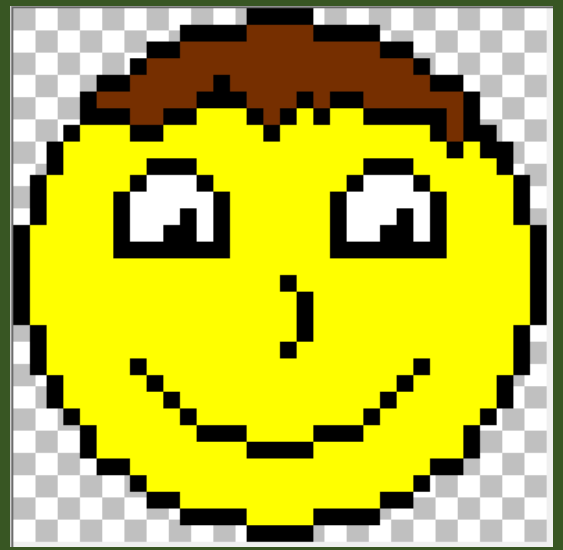
Remember! You only have this lesson to create all of these! So don't spend too much time perfecting them – we can do this at the end!




Sprites
sprPlayer
sprWall
sprEnemy
sprKey
Sounds
Backgrounds











Theory: Objects

- In order to interact or see a sprite in the game, you need to make it an **object** – however not all objects need sprites and there are cases we will create an object without an image!
- This means you can perform actions on the object, move an object or simply be able to see it on the screen.



However, this object is not placed into the game, but rather **acts as a template for instances which are placed into rooms**.



An instance is basically **a copy of the object that is placed in the room (level) of your game.**

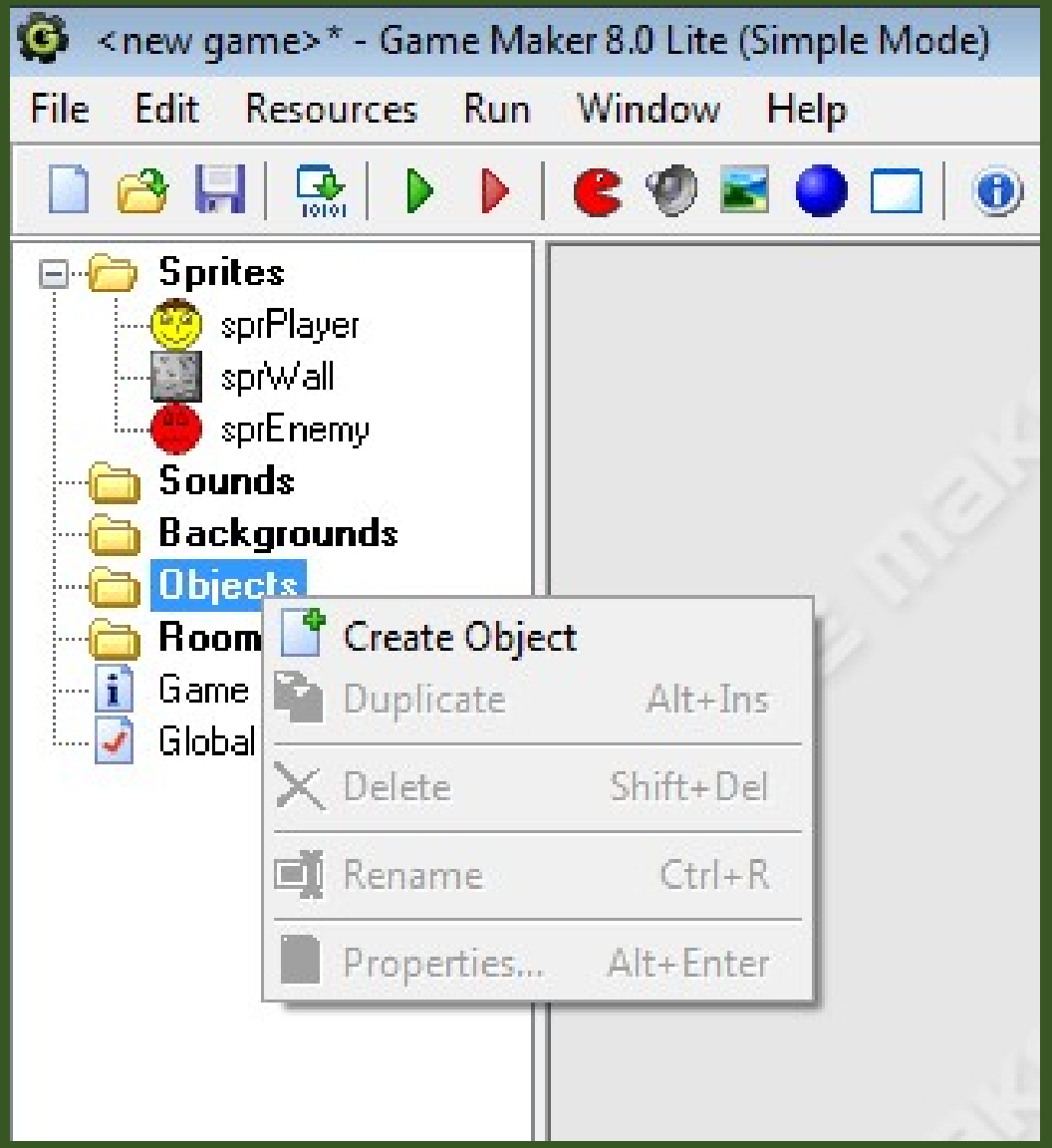
So, you could make **one** tree object and then place **one hundred** instances of this tree in the room.

Let’s Create an Object

Tutorial Part 2

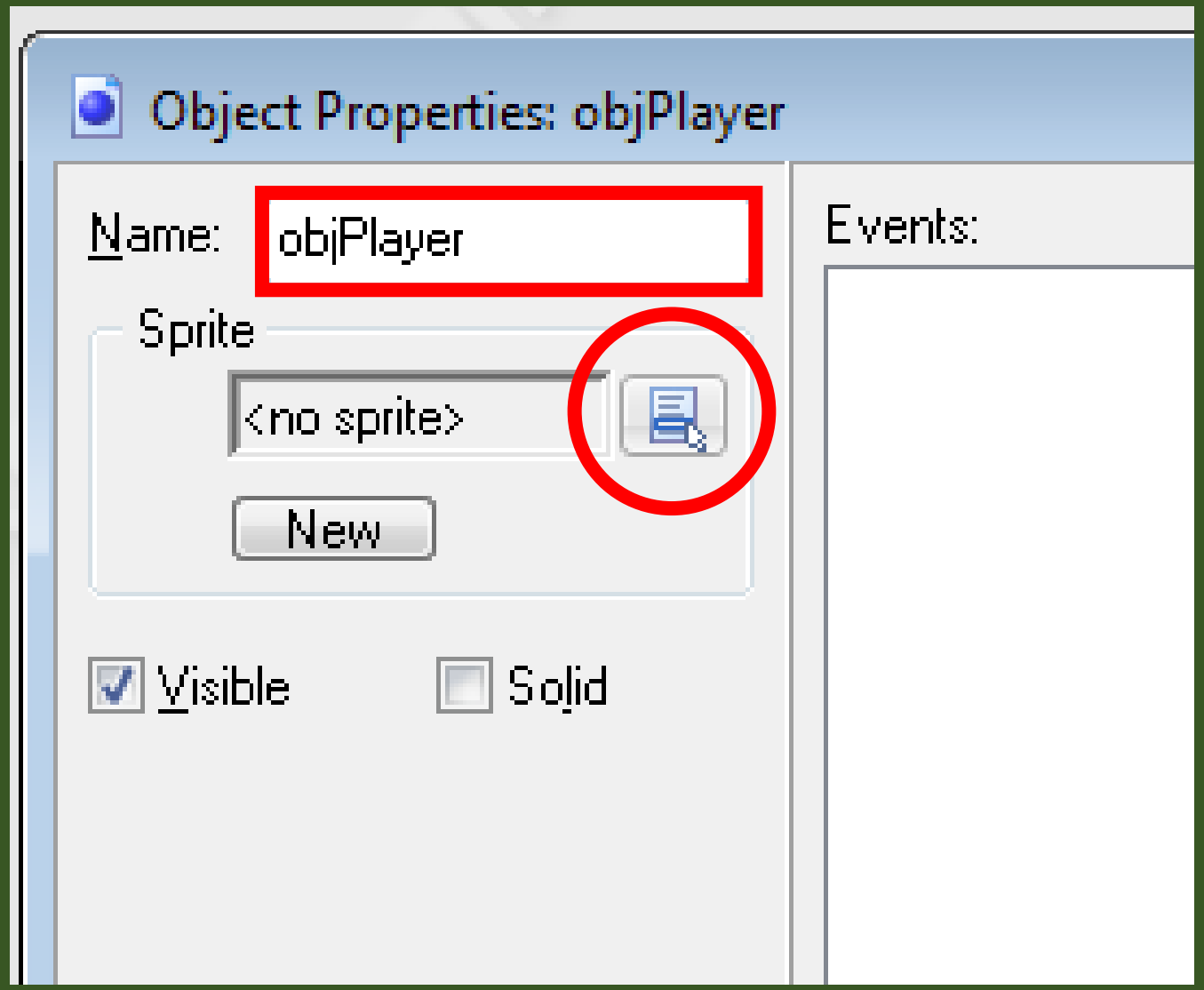
Let’s Make The Player Move





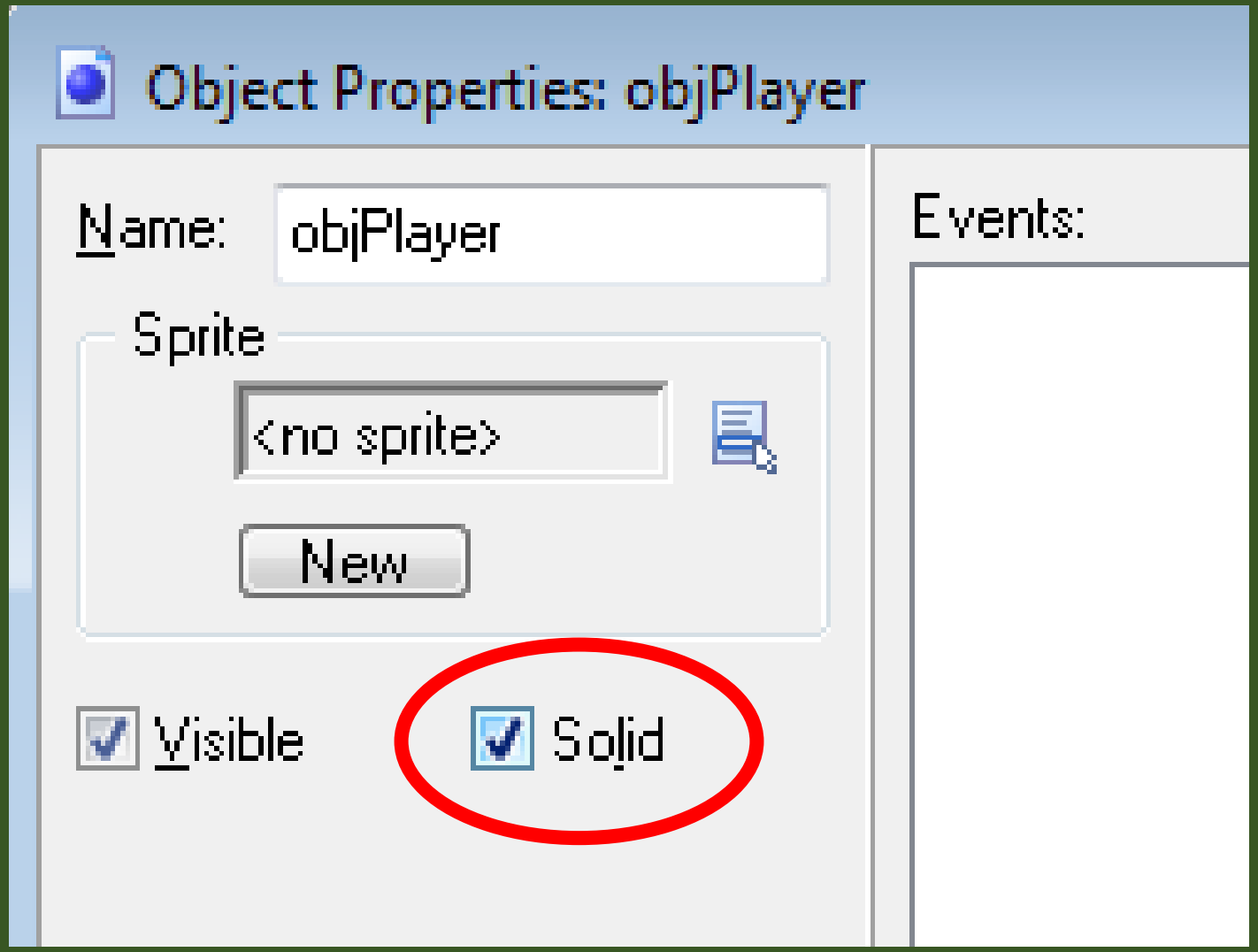
Firstly, we need to make a New Object.

An Object is something we can put into the game itself.



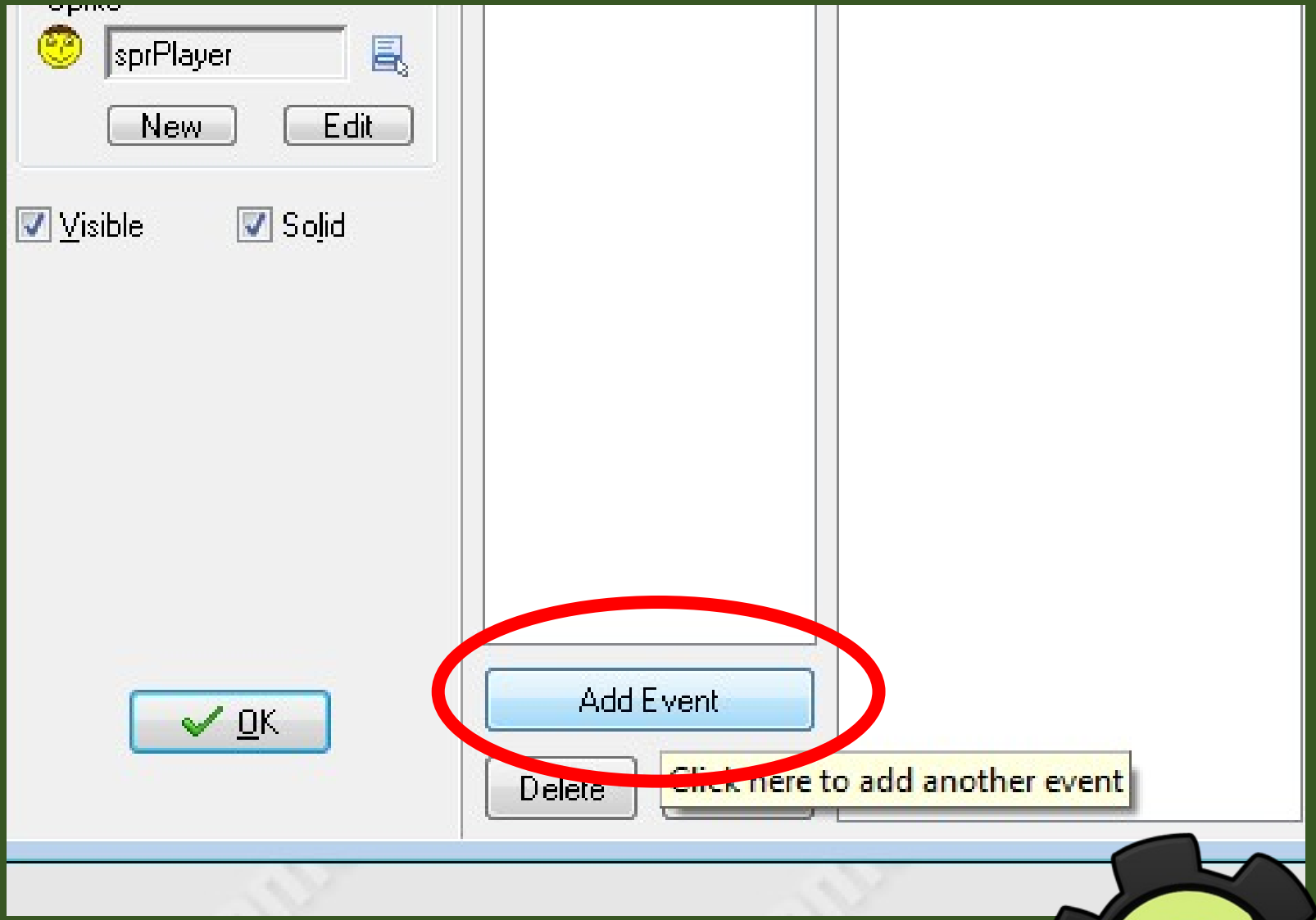
Now Click on this Icon next to the <no sprite> text.

Select your sprPlayer

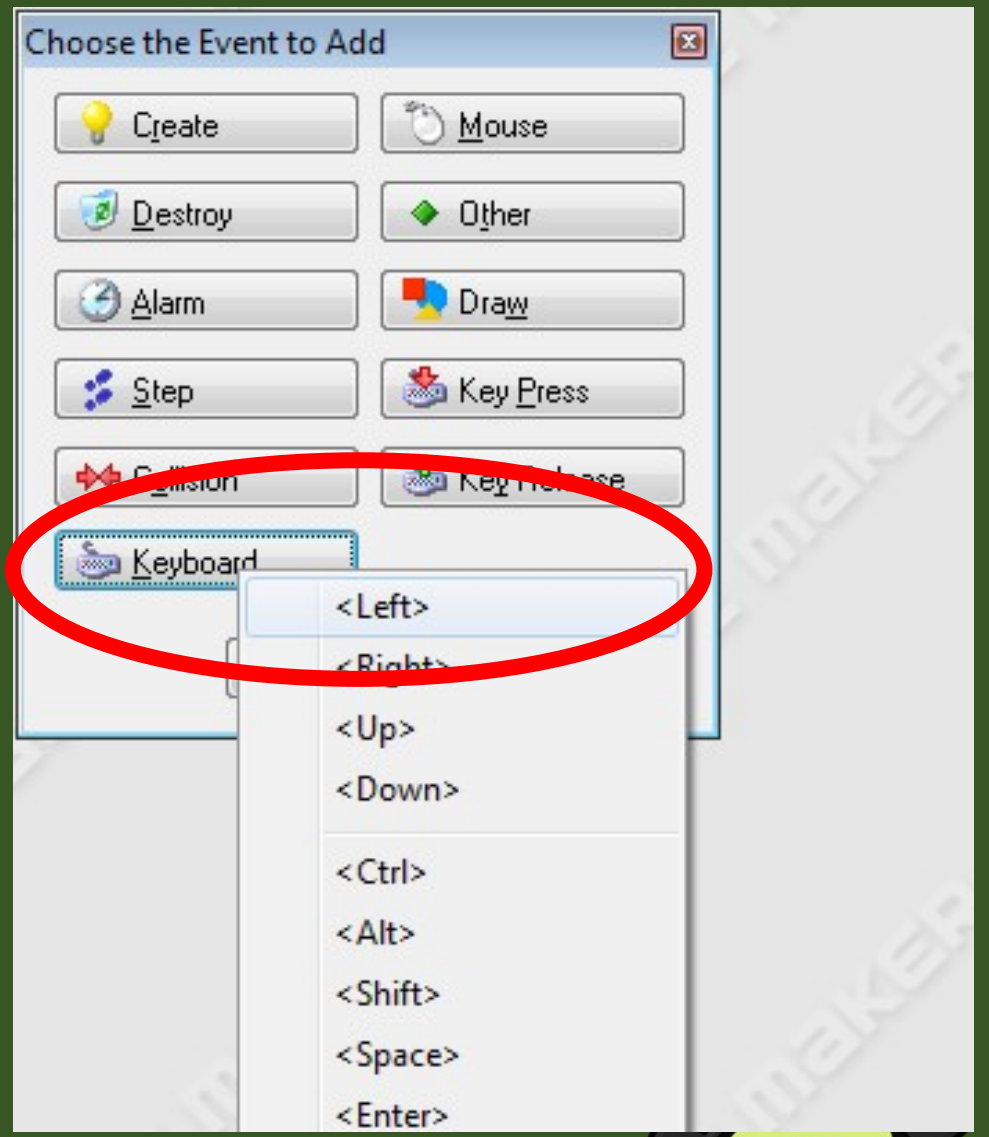


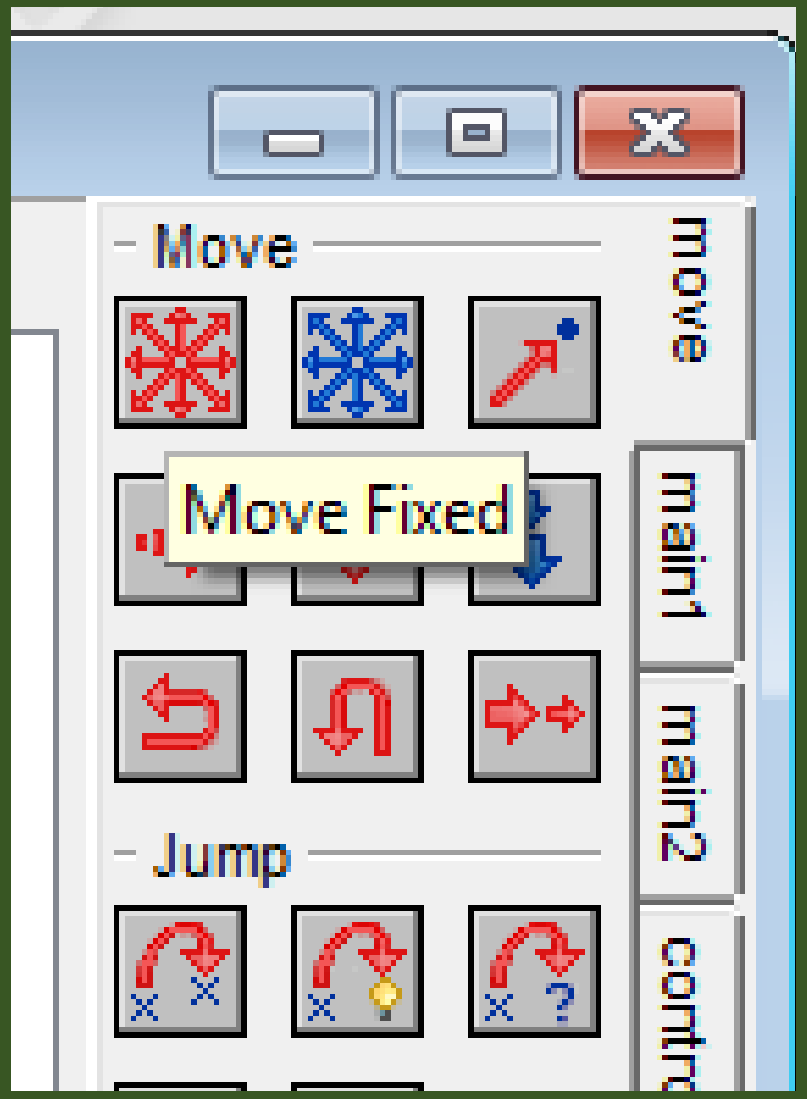
We also want to ensure **Solid** is checked.

Click on the **Add Event**.

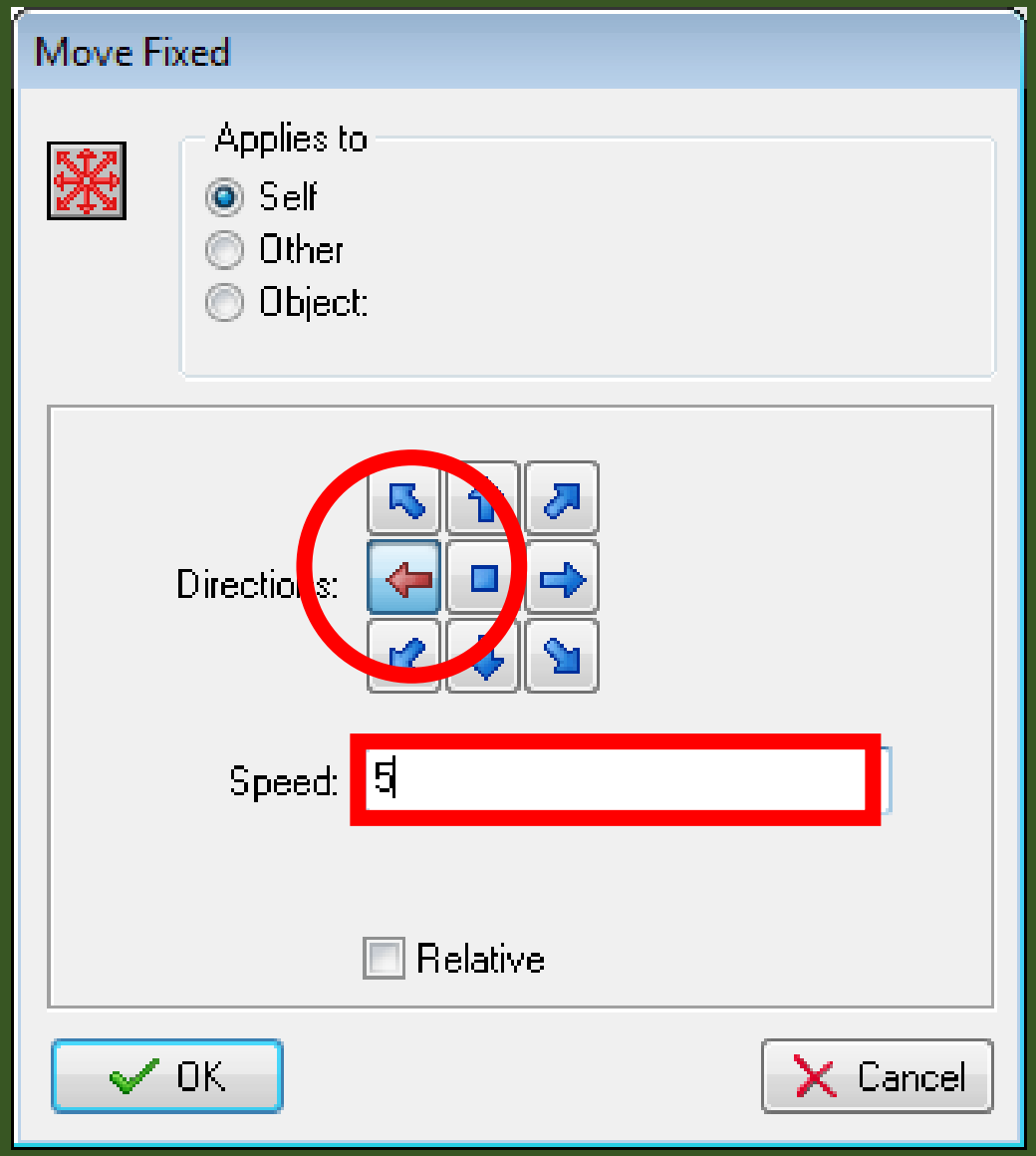


In the Popup, select
Keyboard > Left

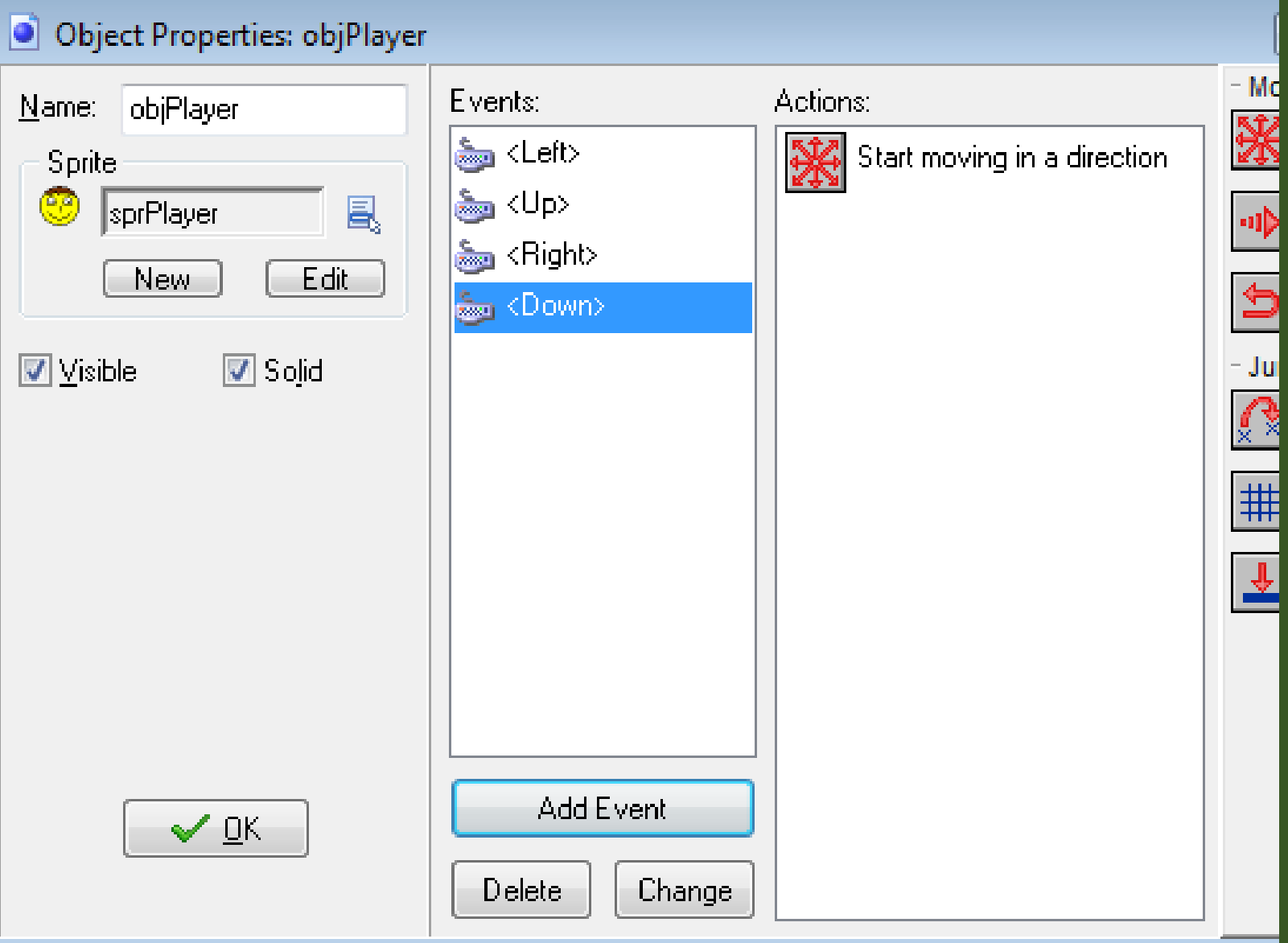




In the Move tab, we want to use the red arrows icon. The tooltip reads ‘Move Fixed’



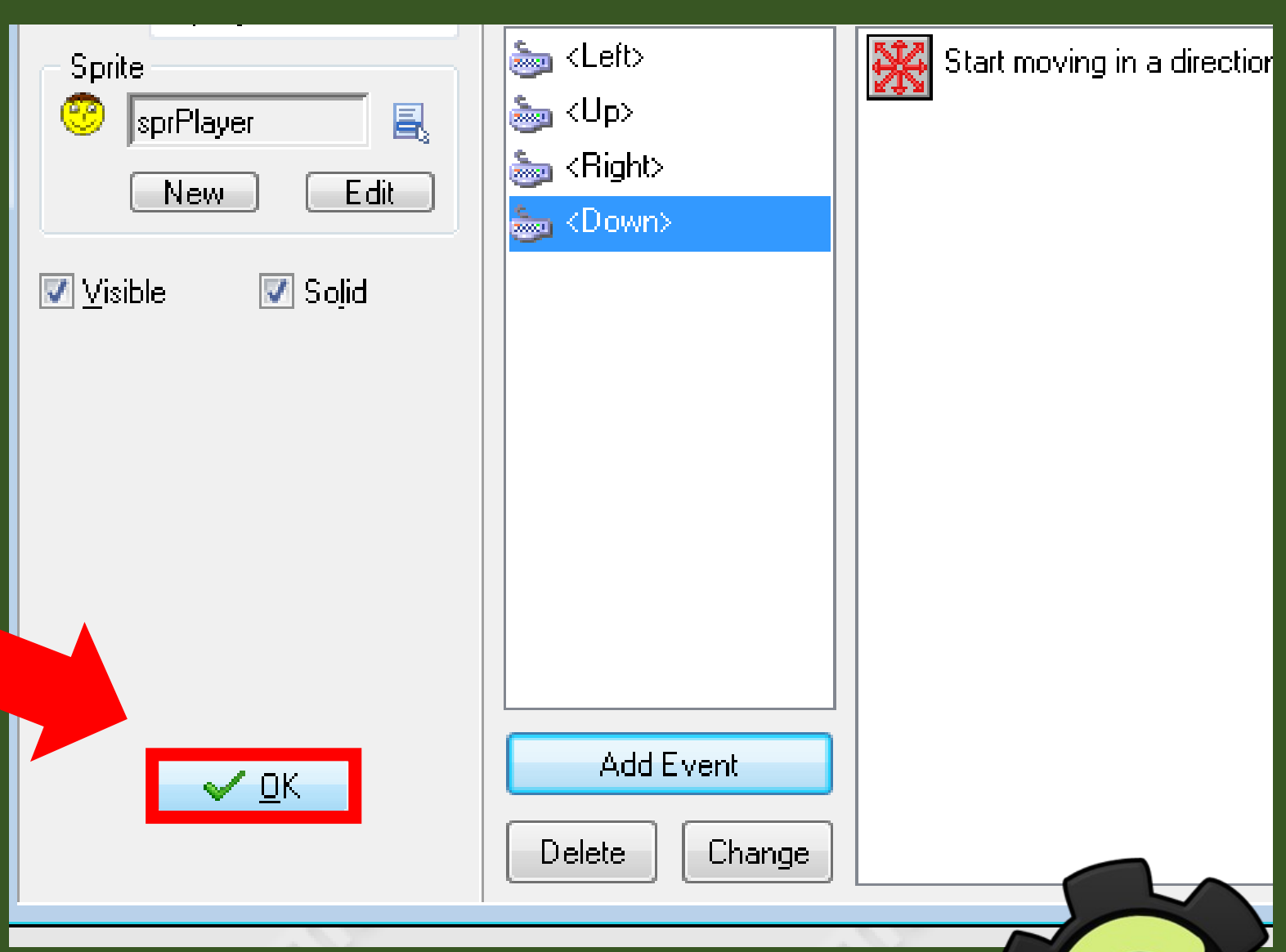
Once you have dragged the Move Fixed Icon over, you need to click the **Left Arrow**. **Set it with a Speed of 5. Then Click OK.**



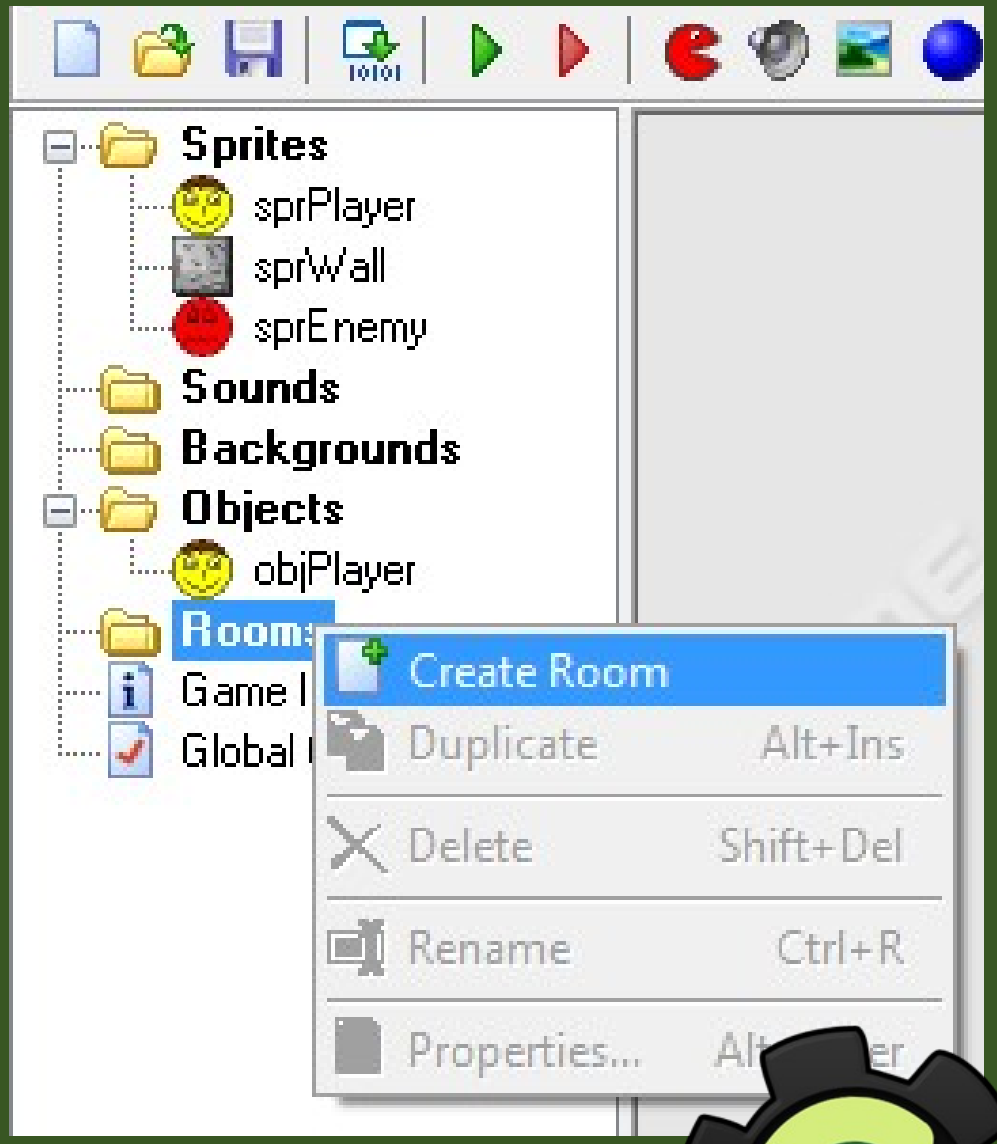
Repeat this for Up, Down and Right.

Q. Can you think of what will happen if no Key is pressed?

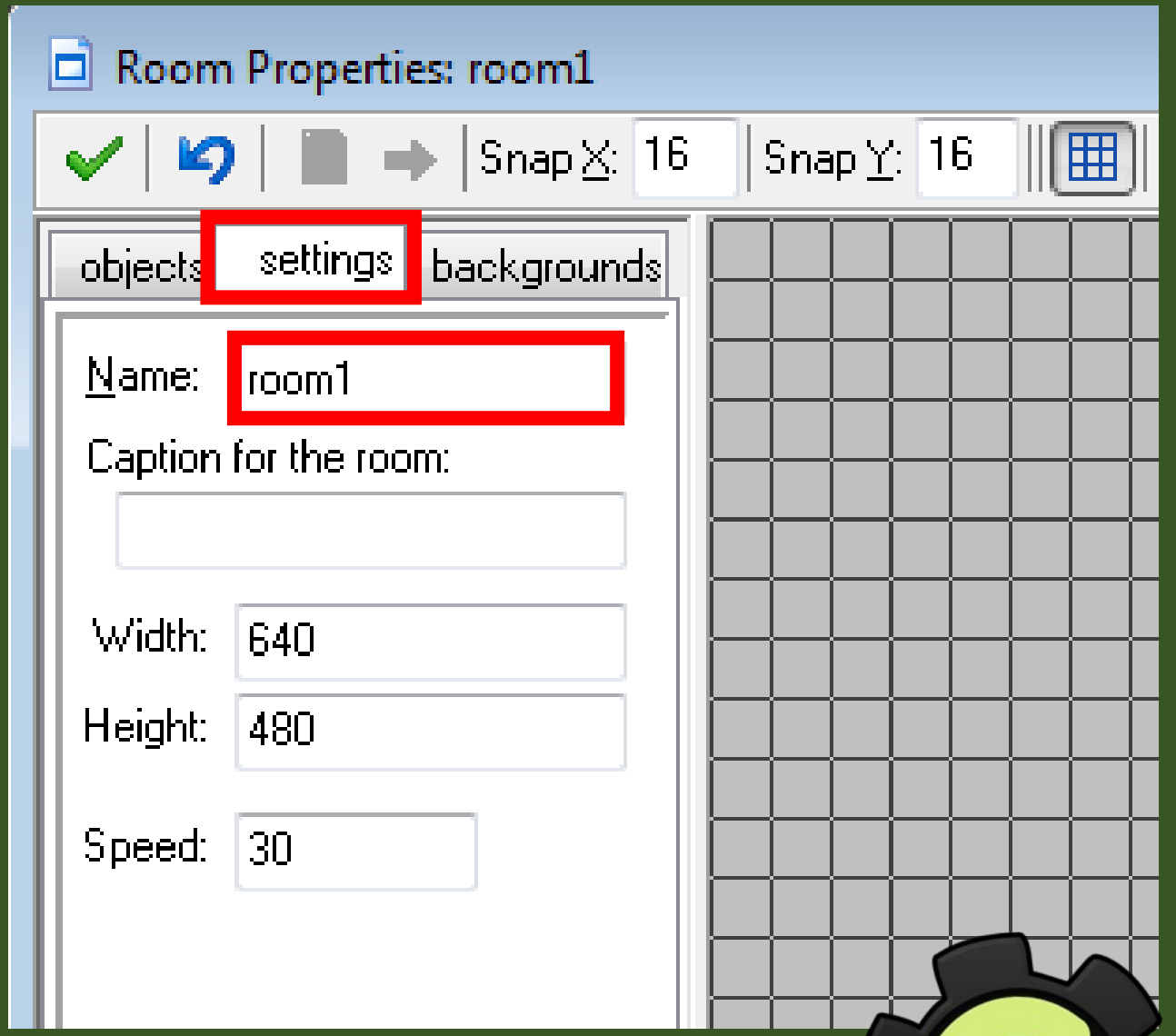
Click OK to save.



We now need to create a Room for our Character.



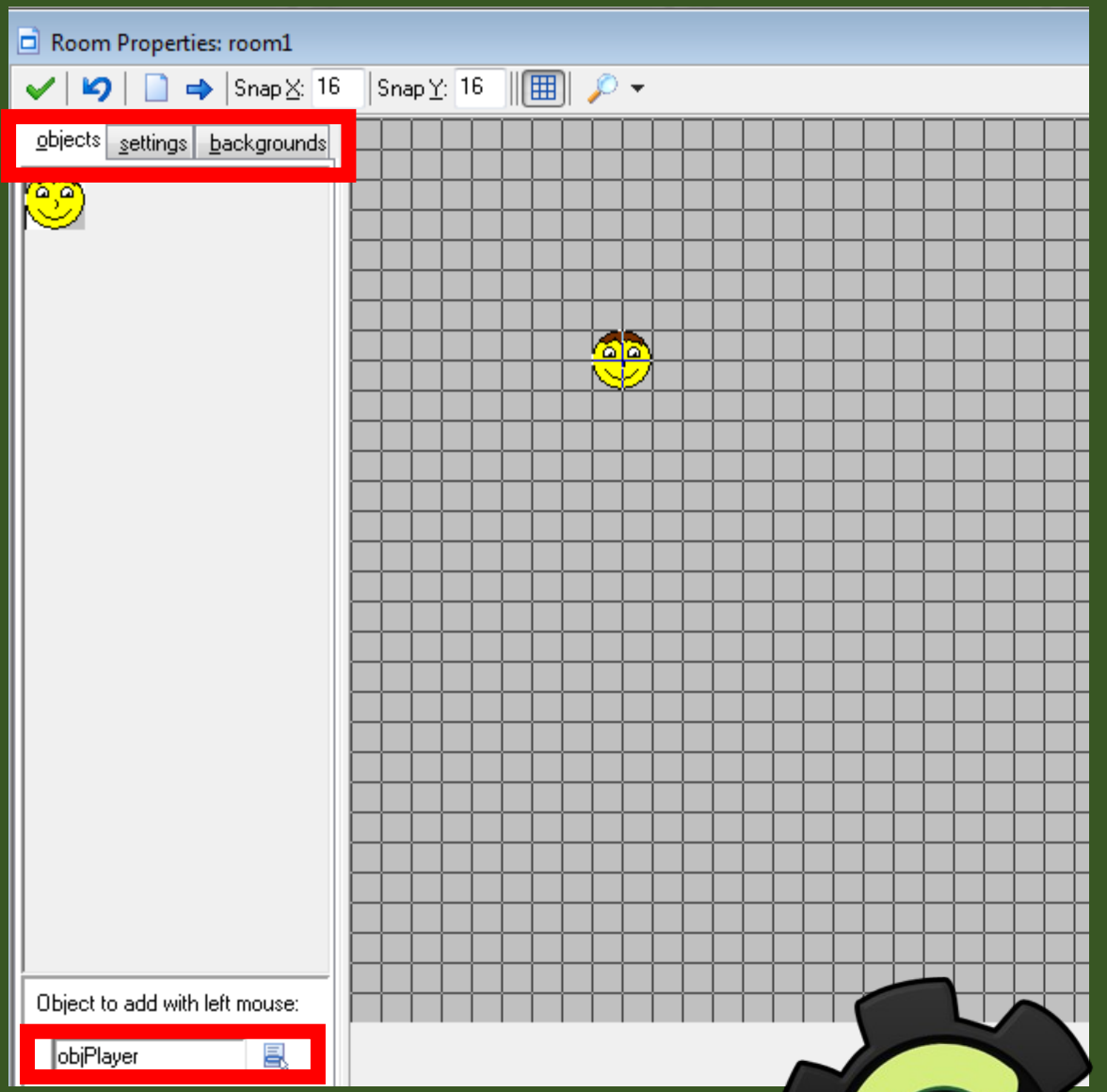
Once you have clicked on this, Go to the **settings tab. Give the room a** meaningful name. We will call this rmPractise.

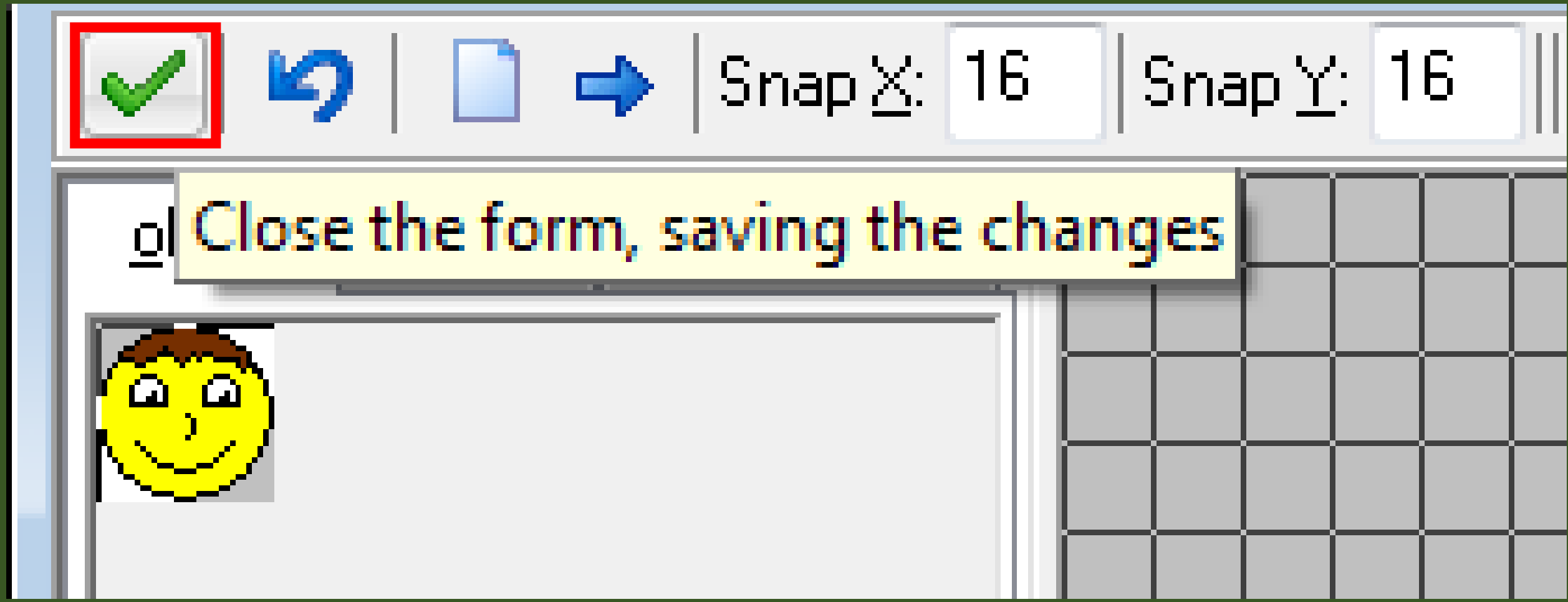


Further Task: Add a caption for the room.

Go to the objects tab. At the bottom, make sure objPlayer is selected.

Then click anywhere in the tiled area.





Click on the Green Tick to Save your Room

Theory: Rooms



A room in Game Maker is important to include in every game. This **is the area we use to add in objects**.

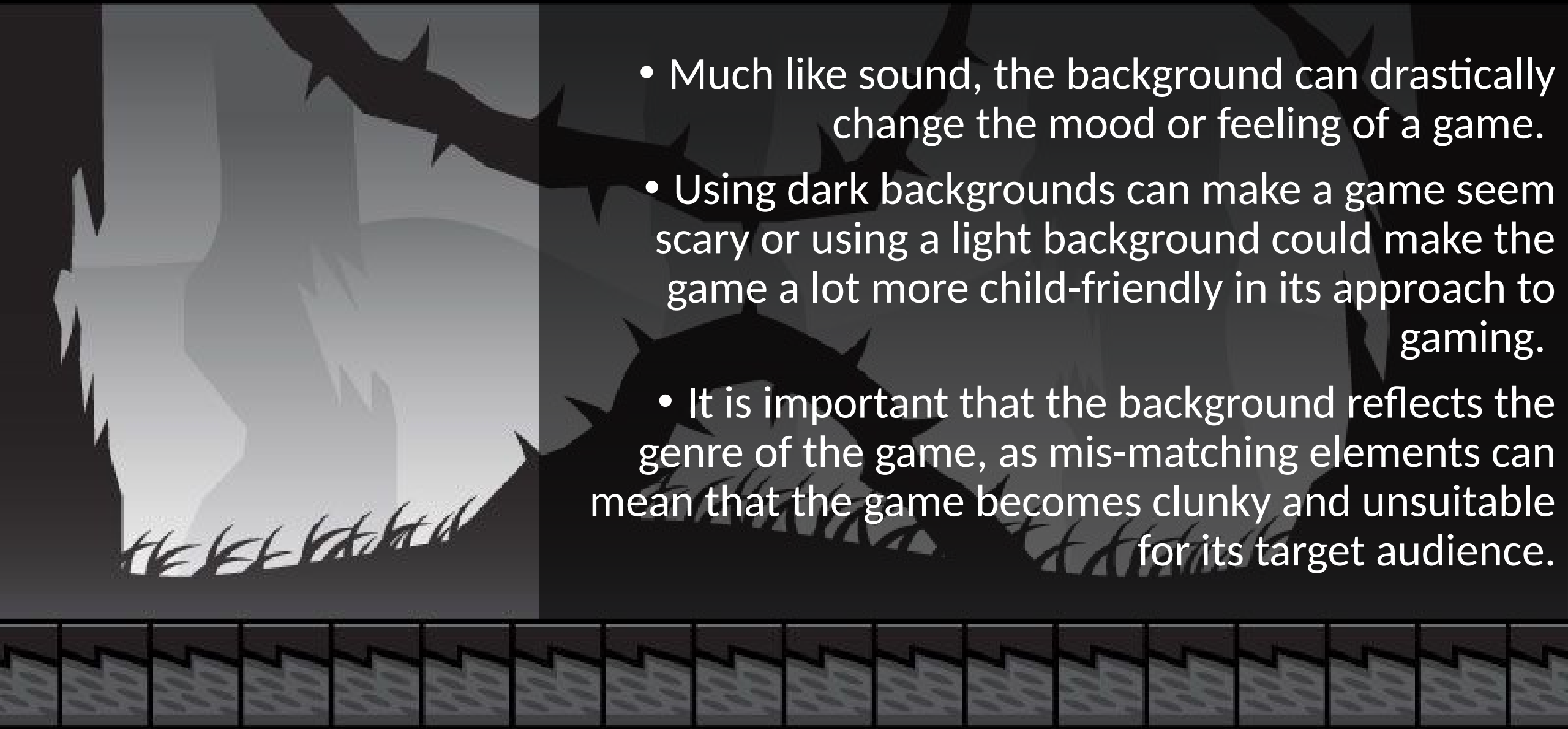
- Not only should you have a room for each level, but you should also have a room for the opening screen, help pages and any credits/end screens you will have.
- You must have **at least one** room in any game for it to run, but in general you will need many more.
- The room will be a design of **what the player will see** when they play a game. It is important that we create these to be very atheistically pleasing to ensure players are interesting in playing your game.



Theory: Sound

- Sound, or commonly known as audio, is an important part of creating an engaging game that keeps players interested in the gameplay and mechanics of a game.

- Two main aspects of audio in games:
  - Background music
  - SFX (aka. Special effects)



- Much like sound, the background can drastically change the mood or feeling of a game.
- Using dark backgrounds can make a game seem scary or using a light background could make the game a lot more child-friendly in its approach to gaming.
- It is important that the background reflects the genre of the game, as mis-matching elements can mean that the game becomes clunky and unsuitable for its target audience.

Theory: Backgrounds

Theory: Genres

A genre **identifies the style or category of a piece of work, like a book, poem or film**. Games also fall into different genres.

- Games are not usually defined by the actual content of the games, but rather by the challenges, story and gameplay that are contained within the game.



Genre: ACTION!!!!!!!!

- Action games emphasize physical challenges that require Hand-eye coordination and motor skill to overcome.
  - They center around the player, who is in control of most of the action.
  - Most of the earliest video games were considered action games; today, it is still a vast genre covering all games that involve physical challenges.
- Action games are classified by many subgenres.
  - Platform games and fighting games are among the best-known subgenres.



Genre: Adventure

- Unlike adventure films, adventure games are not defined by story or content.
  - Rather, adventure describes a manner of gameplay without reflex challenges or action.
- They normally require the player to solve various puzzles by interacting with people or the environment, most often in a non-confrontational way.
- It is considered a "purist" genre and tends to exclude anything which includes action elements beyond a mini game.



Genre: Role Playing (RPG)

- Most of these games cast the player in the role of one or more "adventurers" who specialize in specific skill sets (such as melee combat or casting magic spells) while progressing through a predetermined storyline.
- Many involve manoeuvring these character(s) through an overworld, usually populated with monsters, that allows access to more important game locations, such as towns, dungeons, and castles.



Genre: Are we in a Simulation?

- Simulation video games is a diverse super-category of games, generally designed to closely simulate aspects of a real or fictional reality.





Genre: Have you got a good strategy?



- Strategy video games focus on gameplay requiring careful and skilful thinking and planning in order to achieve victory and the action scales from world domination to squad-based tactics.

Genre:
Are you good at sports?



- Sports are video games that simulate sports.
- This opposing team(s) can be controlled by other real-life people or artificial intelligence.

Task: Genres



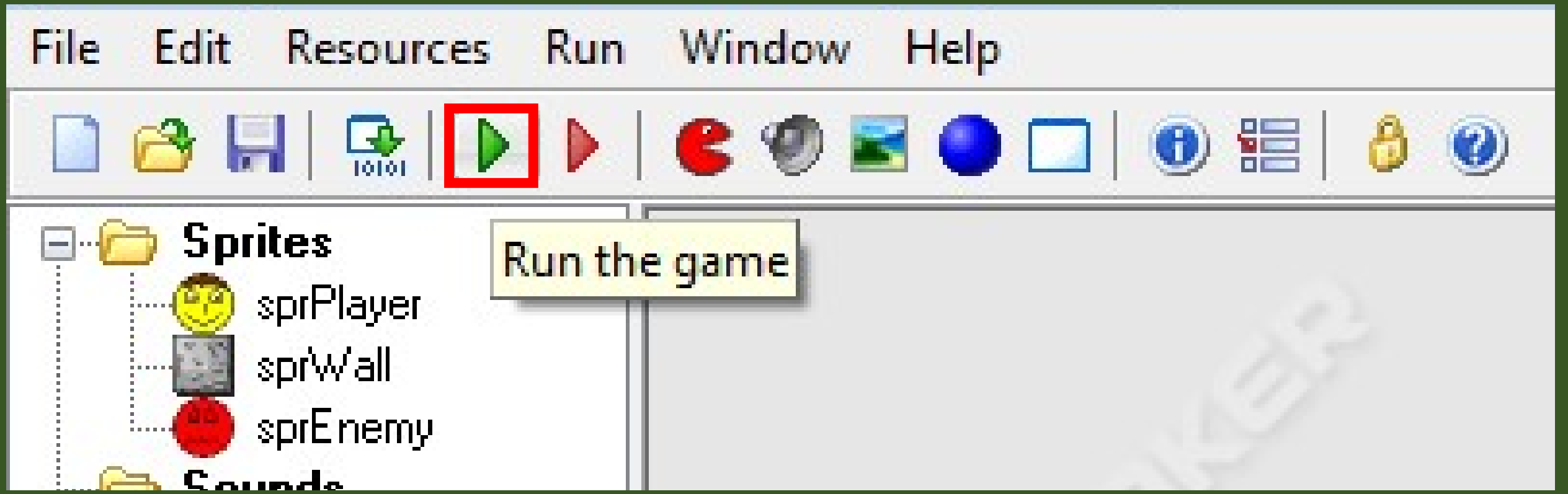
- Pick two different genres from the previous slides and research them.

- Produce an A4 poster describing both, with examples of real games or popular games as examples. Describe how the games fall into the genre(s).

- Record the Game Information into your booklet.

Let’s Create more of our Game

Tutorial Part 3



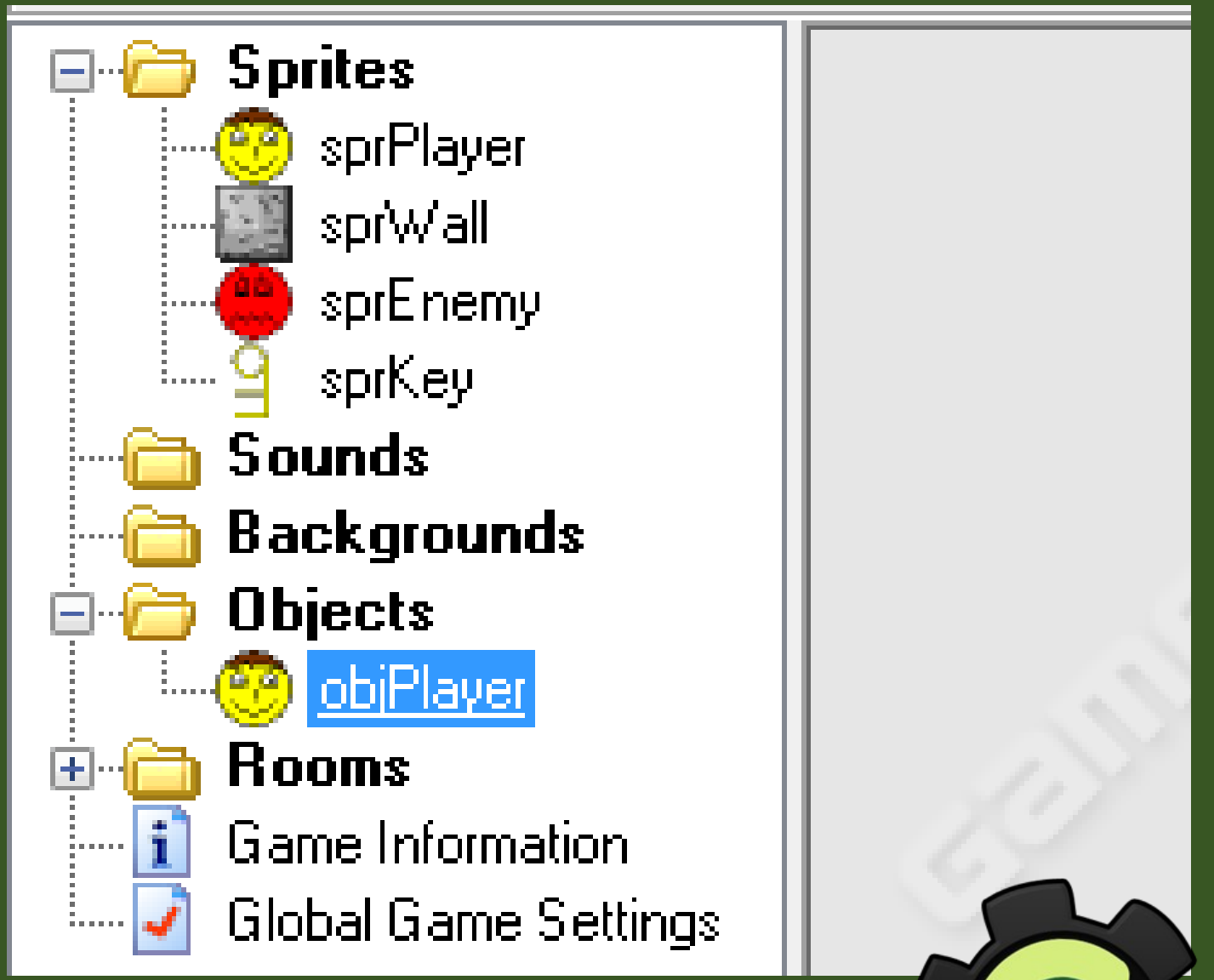
To test run your game, you can click on the green arrow / triangle at the top of GameMaker.

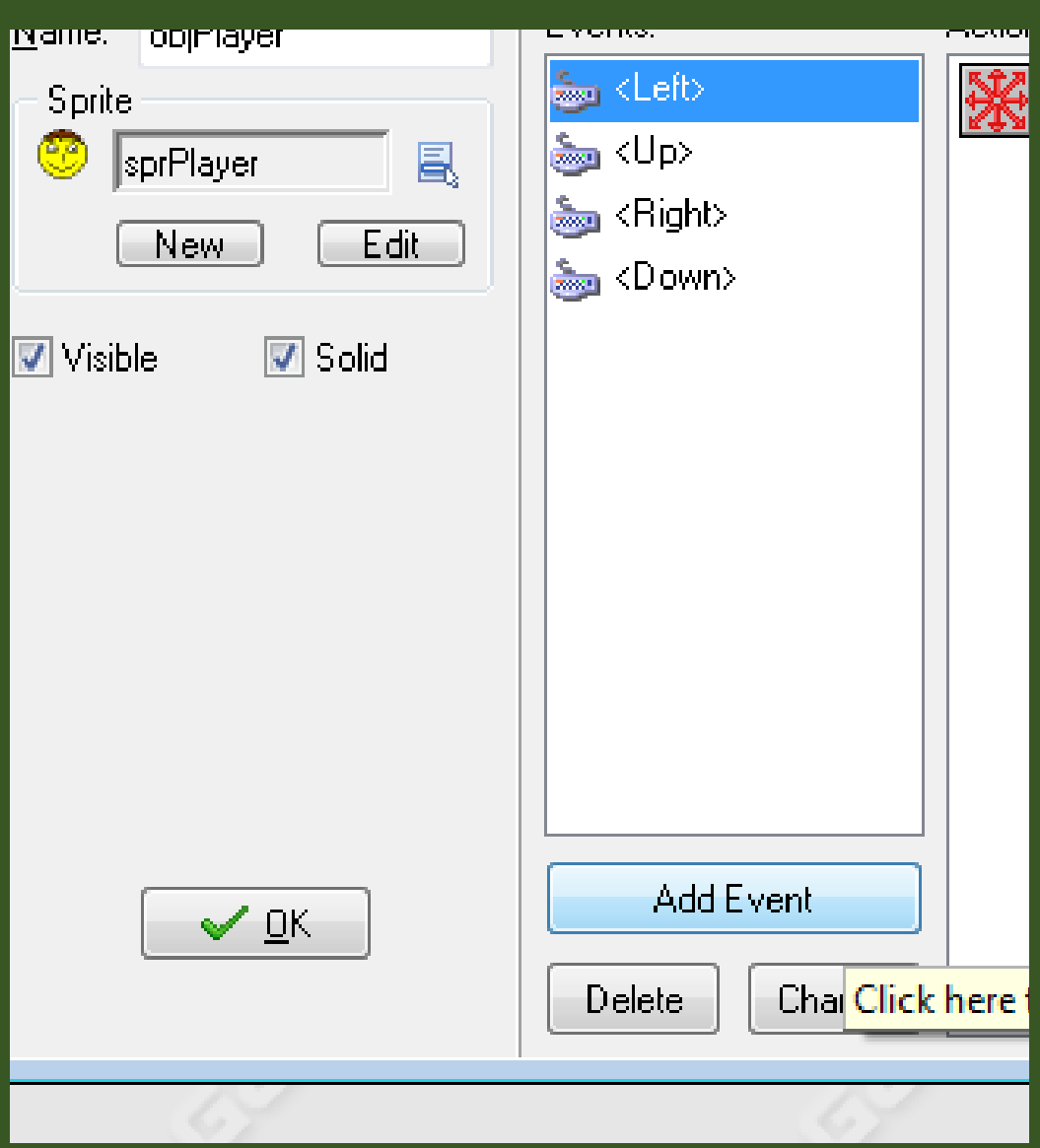
Click Escape to Exit the Game

What do you notice about the character. Does he stop when you press nothing?



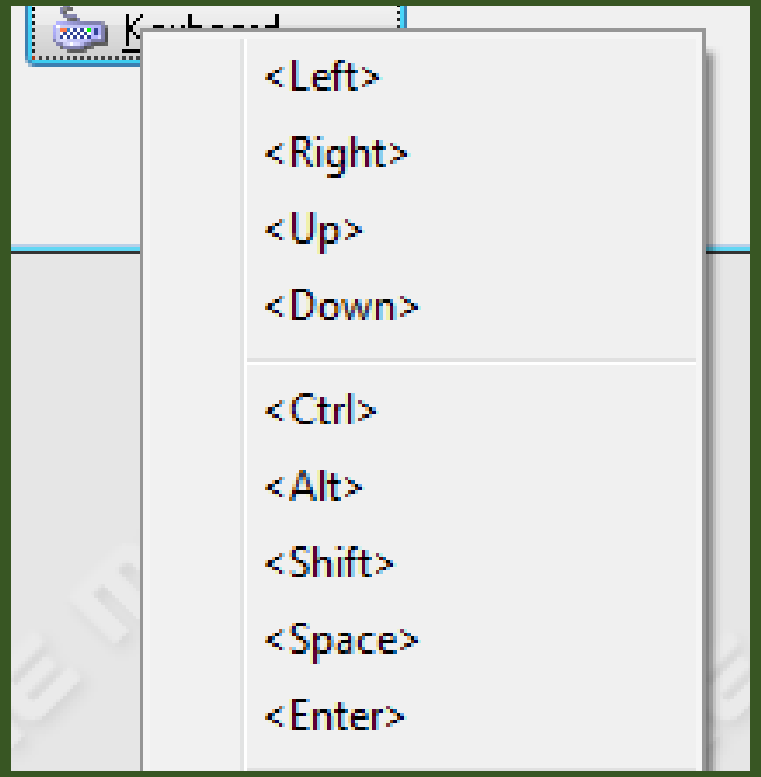
We will fix this now. Double click on your objPlayer.

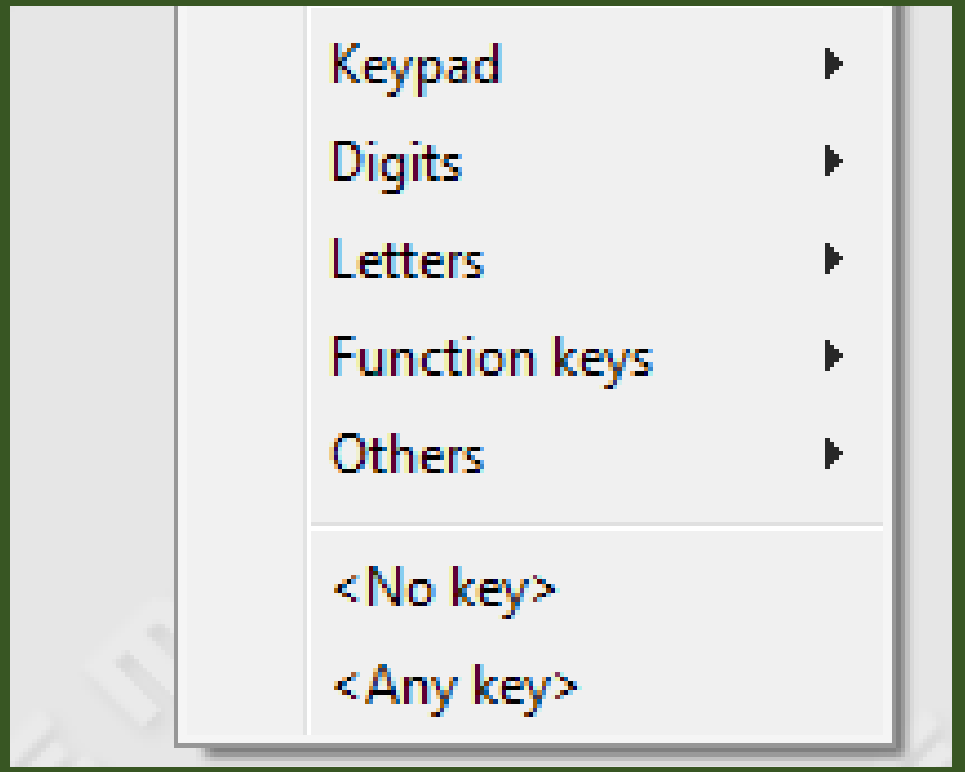




Much like the other events, we are going to add another keyboard event.

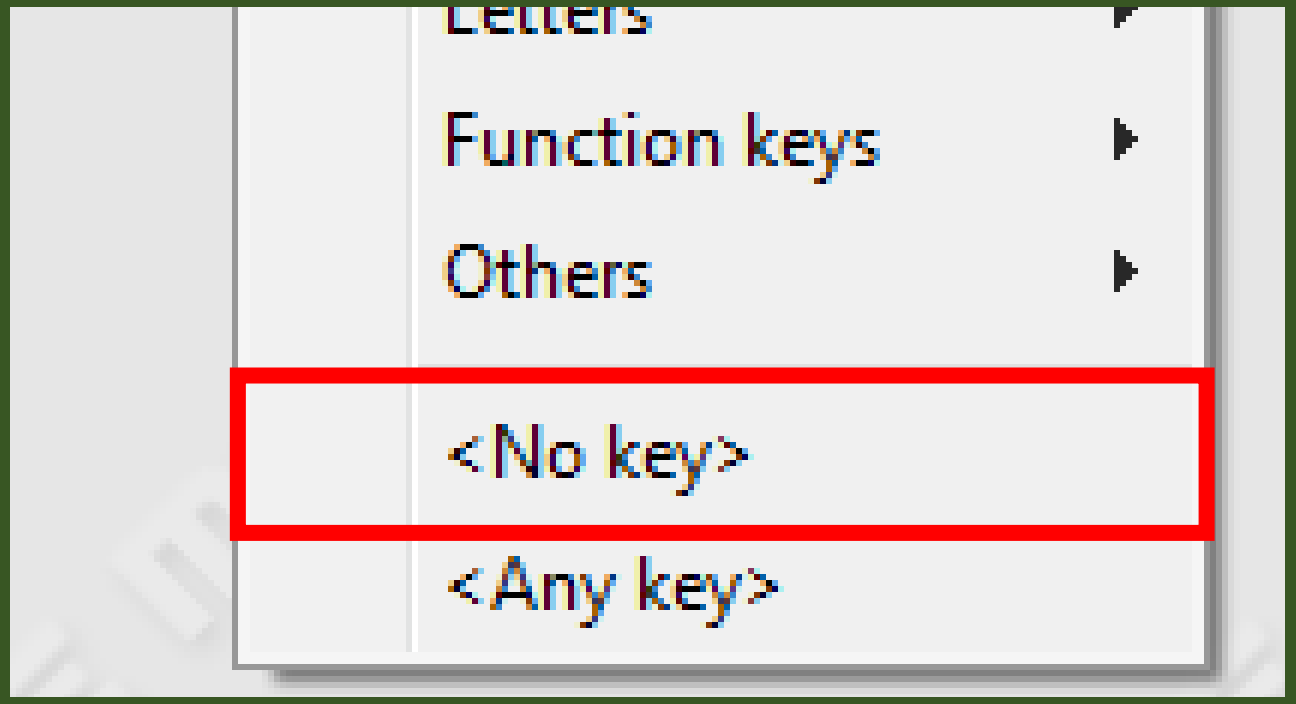
Click Add Event again and select Keyboard.



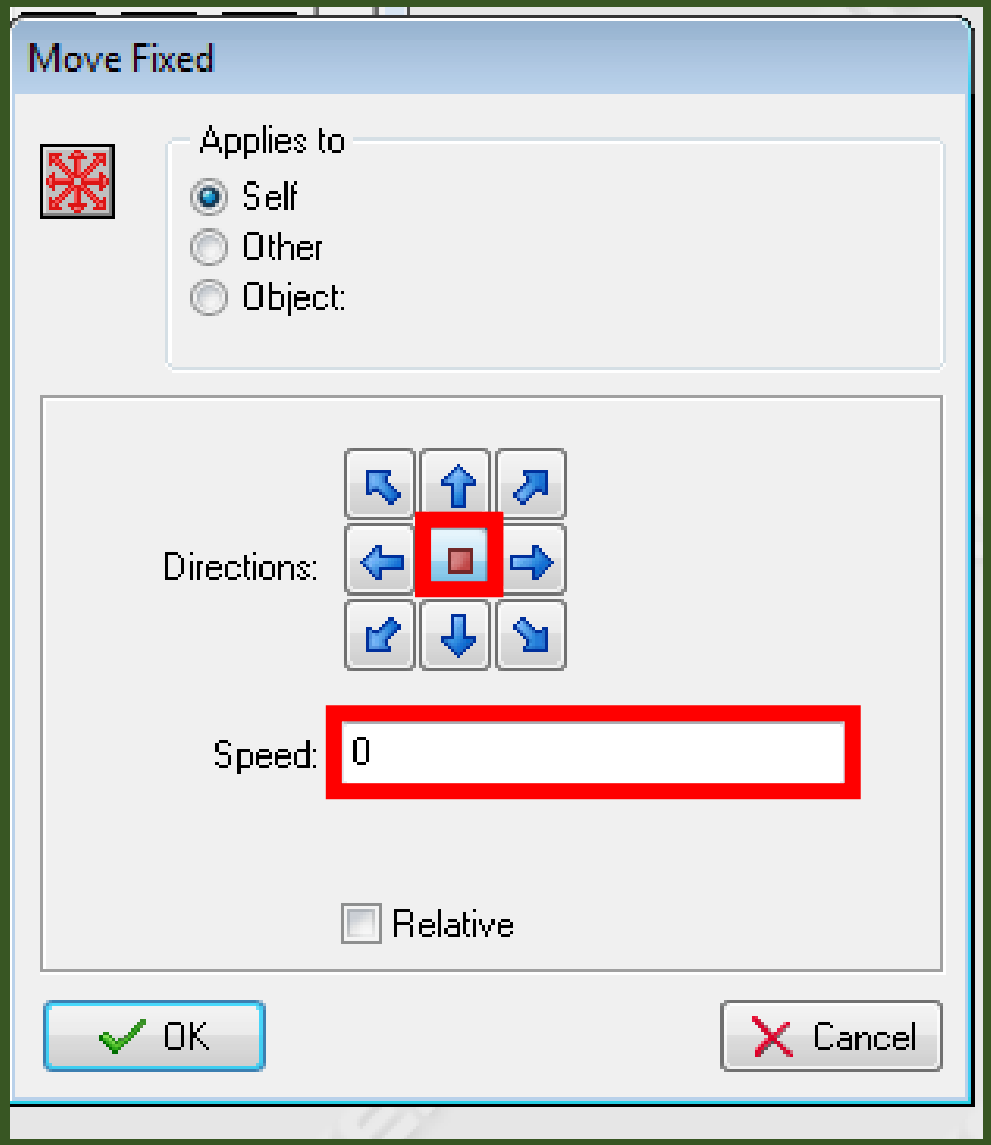


Which of these keyboard events do you think will allow us to check if we're not pressing a directional key?



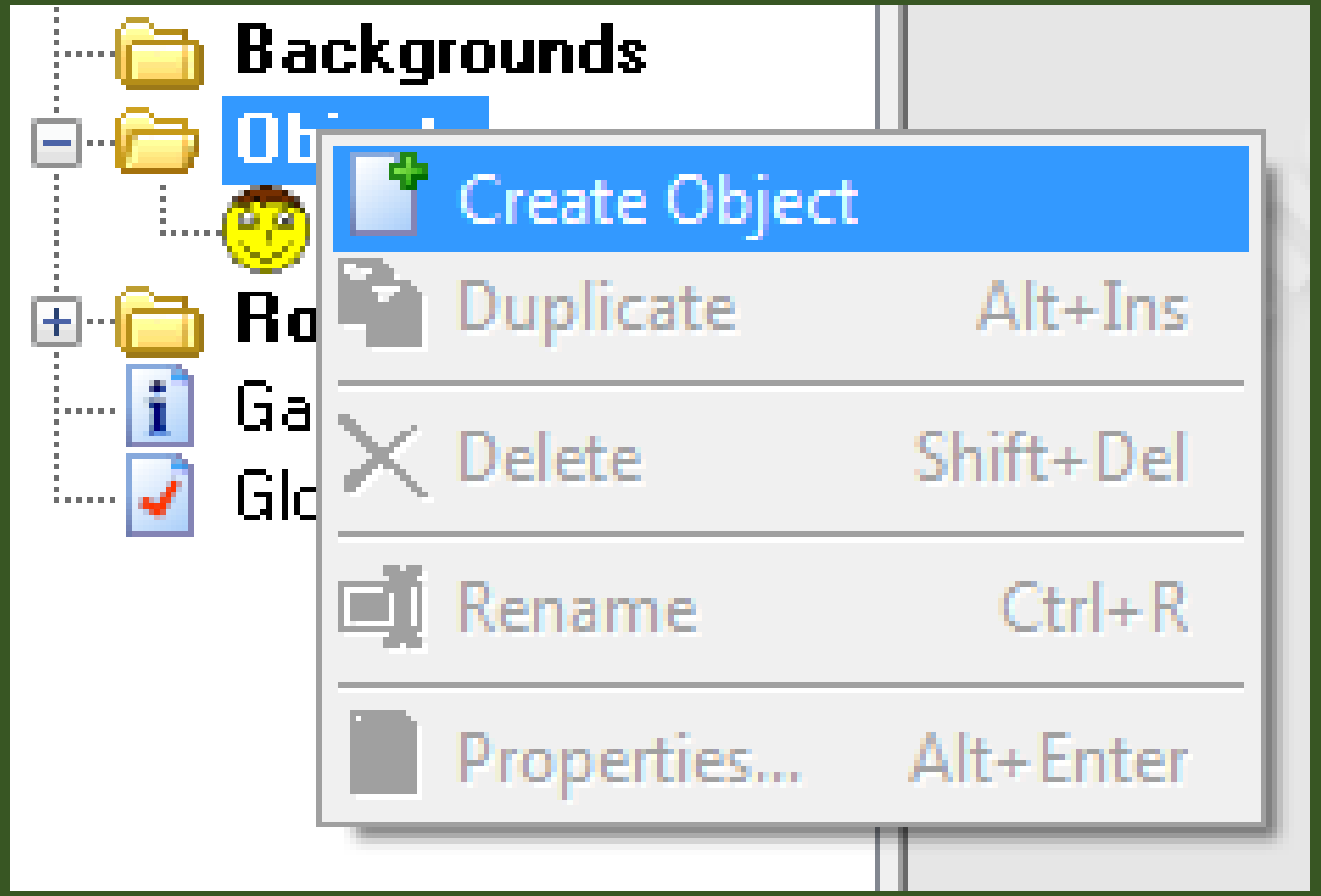


So, select <No key>. That way we can tell no key is being pressed.



We want the player to stop when no key is pressed, so we need to click on the square in the centre. Set a speed of 0

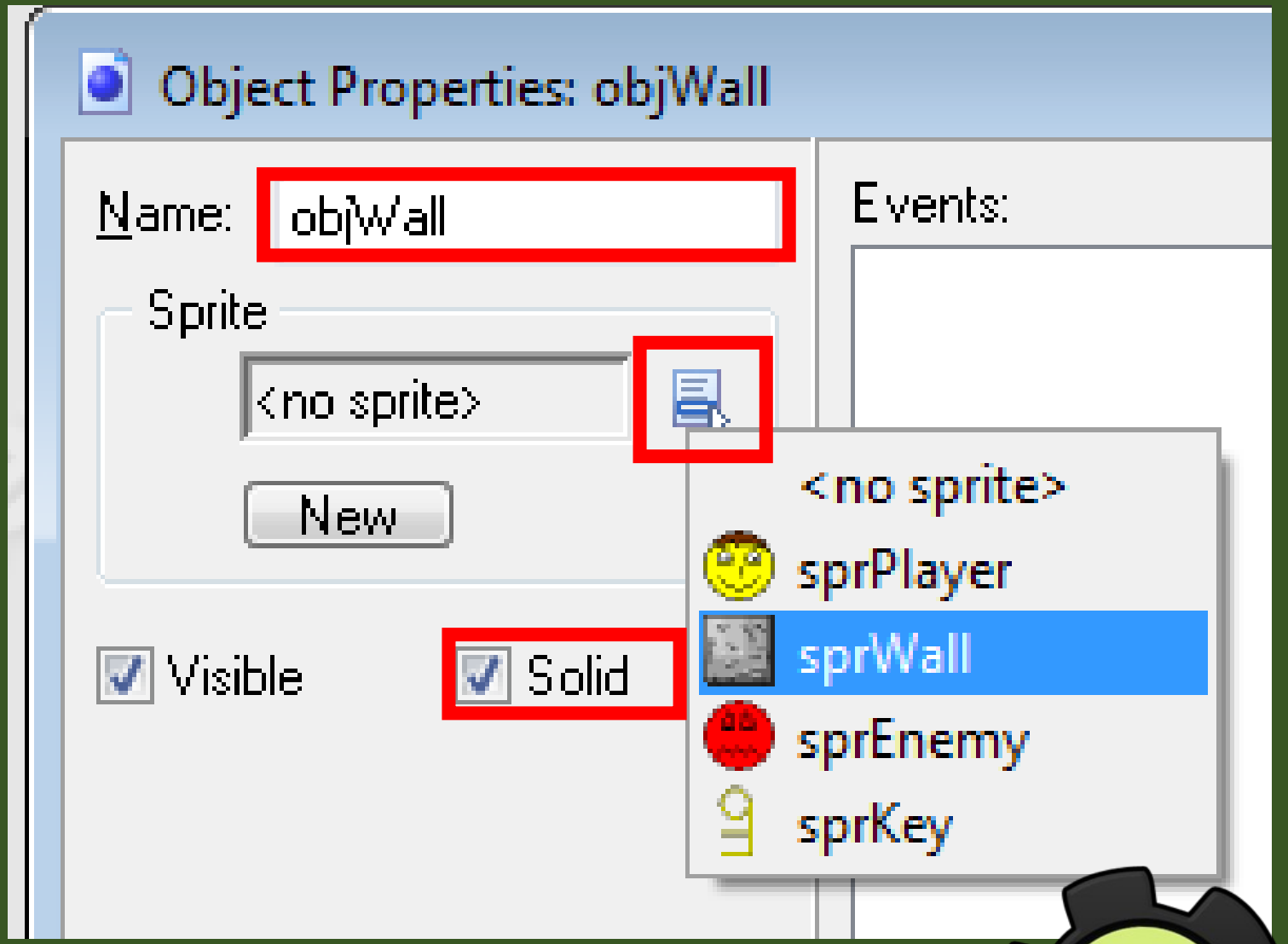
Run your game to test if your player will stop when no key is pressed



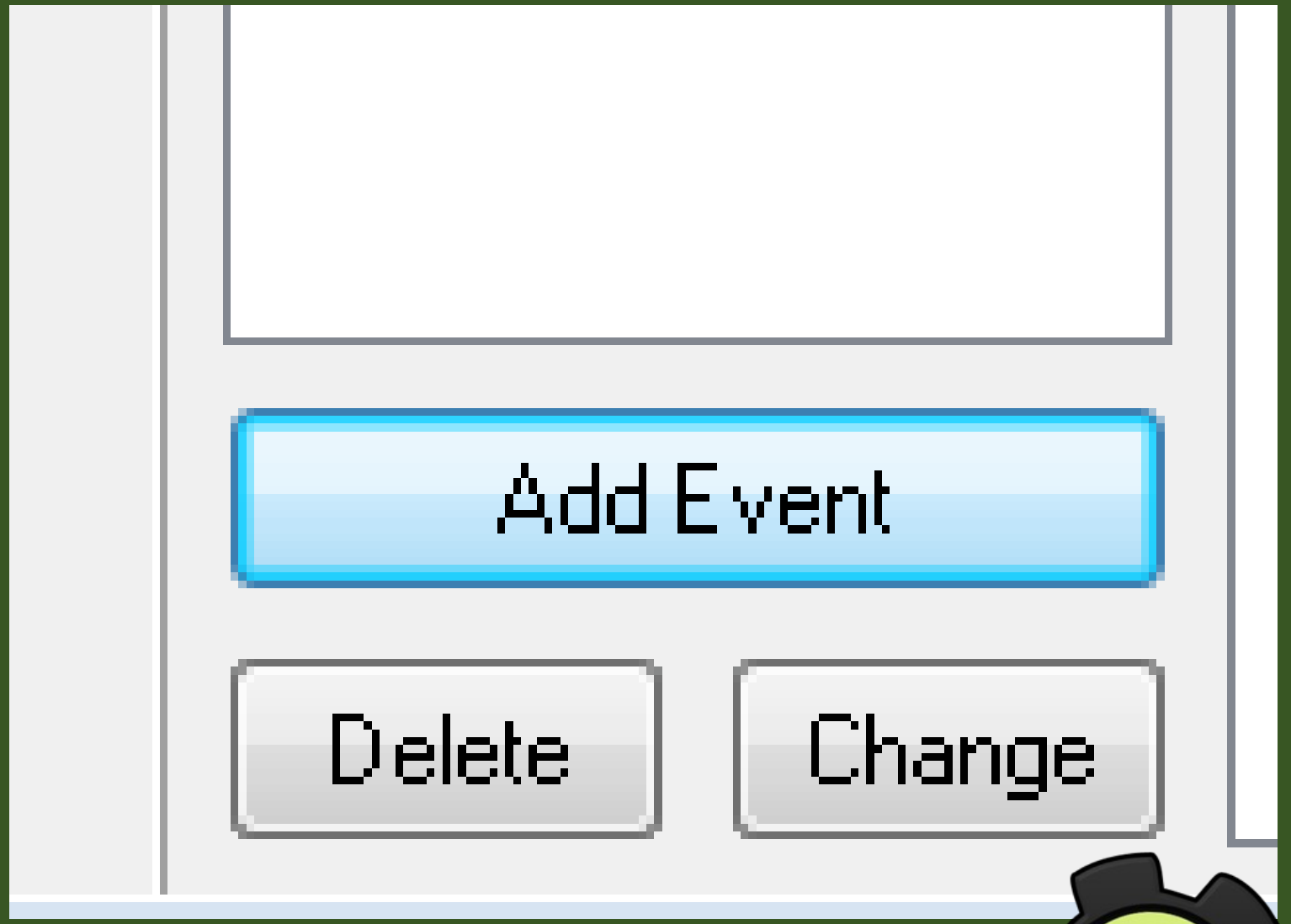
Let's now stop the player from being able to leave the room.

Right click on Objects and select Create Object again

Name this object objWall, the sprite should be set as sprWall and we need this to be solid



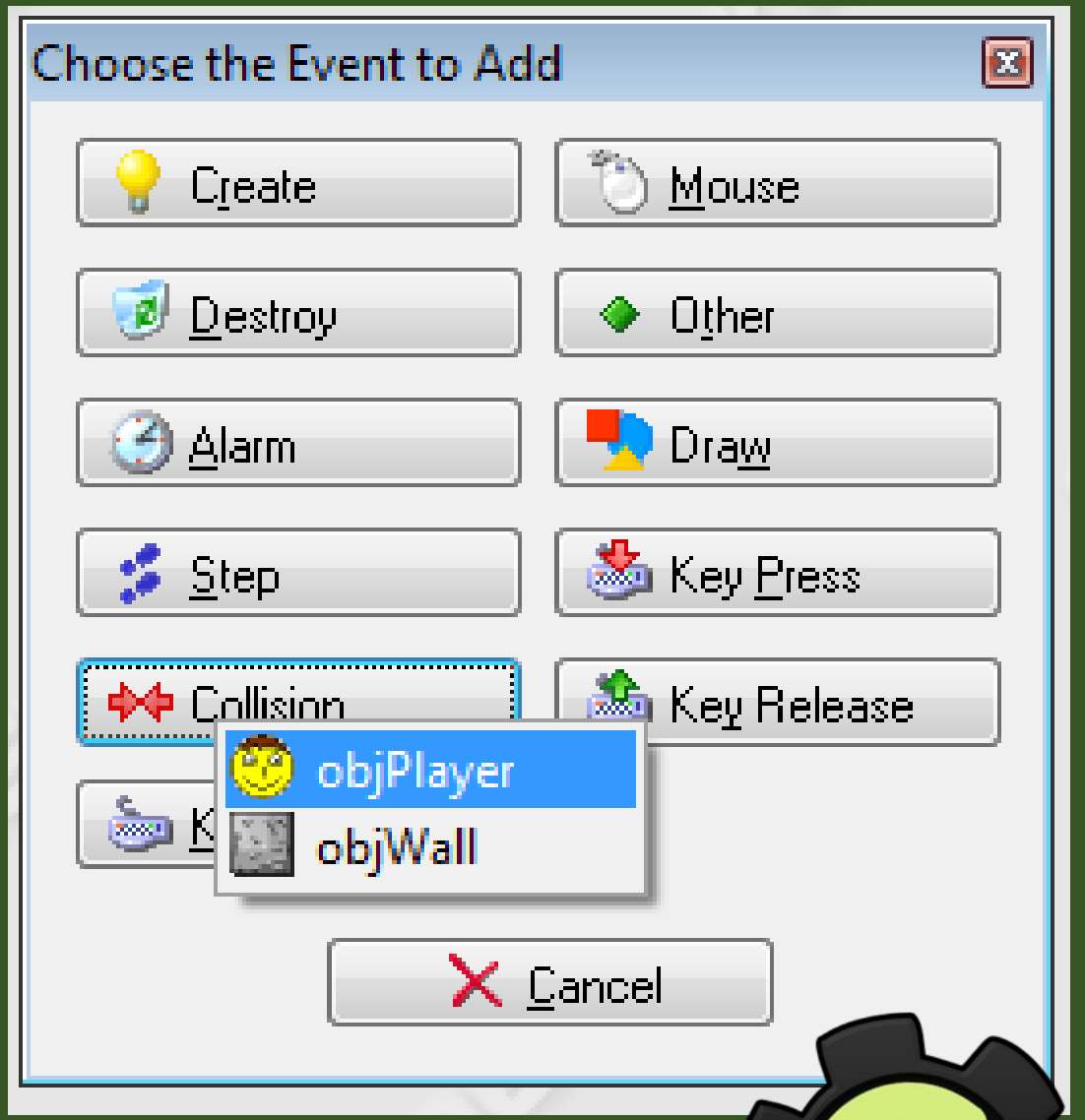
Whilst still on the objWall object, we need to check if the player crashes into the wall. So we need to Add Event.



Whilst still on the objWall object, we need to check if the player crashes into the wall.

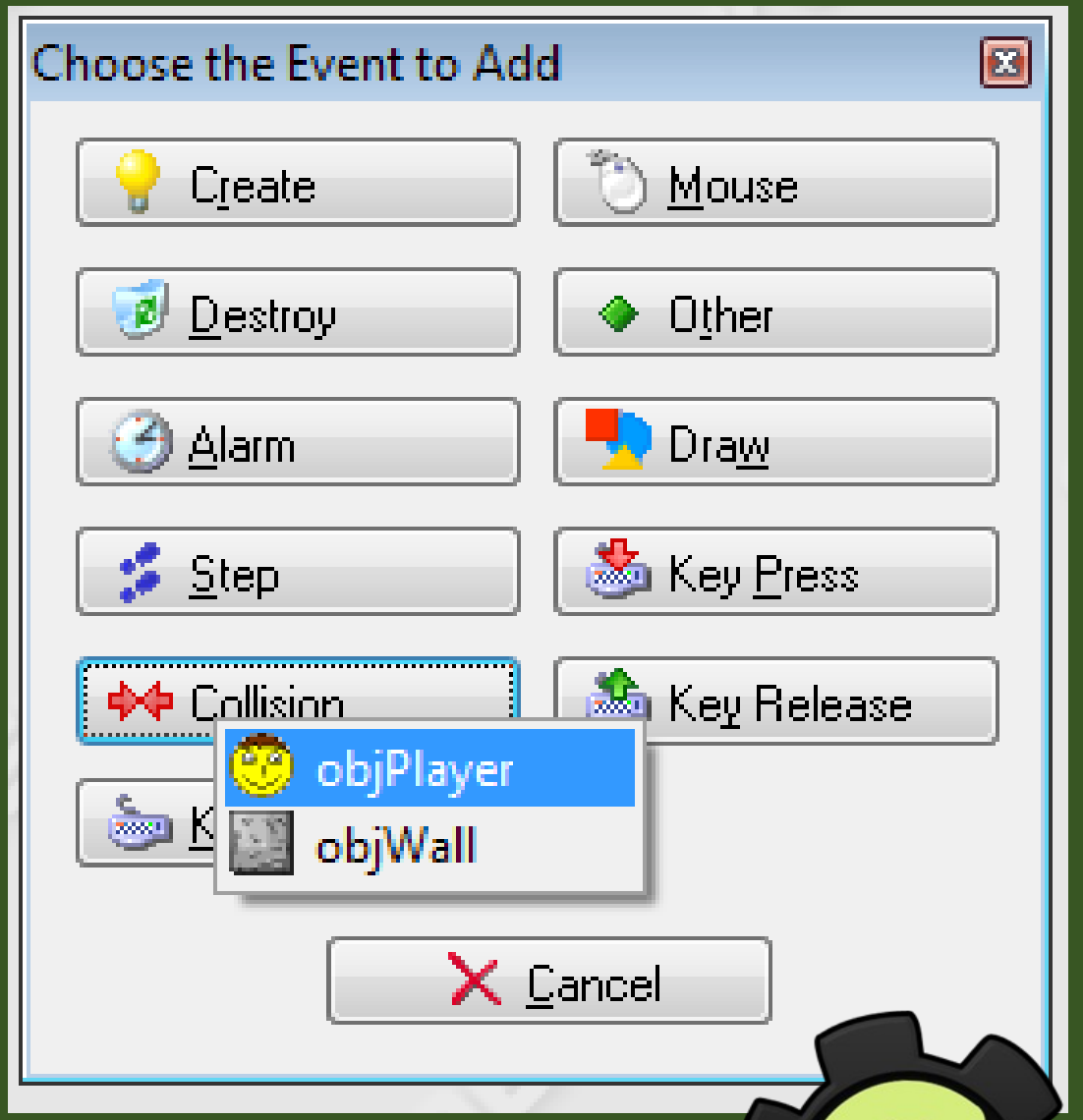
So we need to Add Event.

Select Collision.



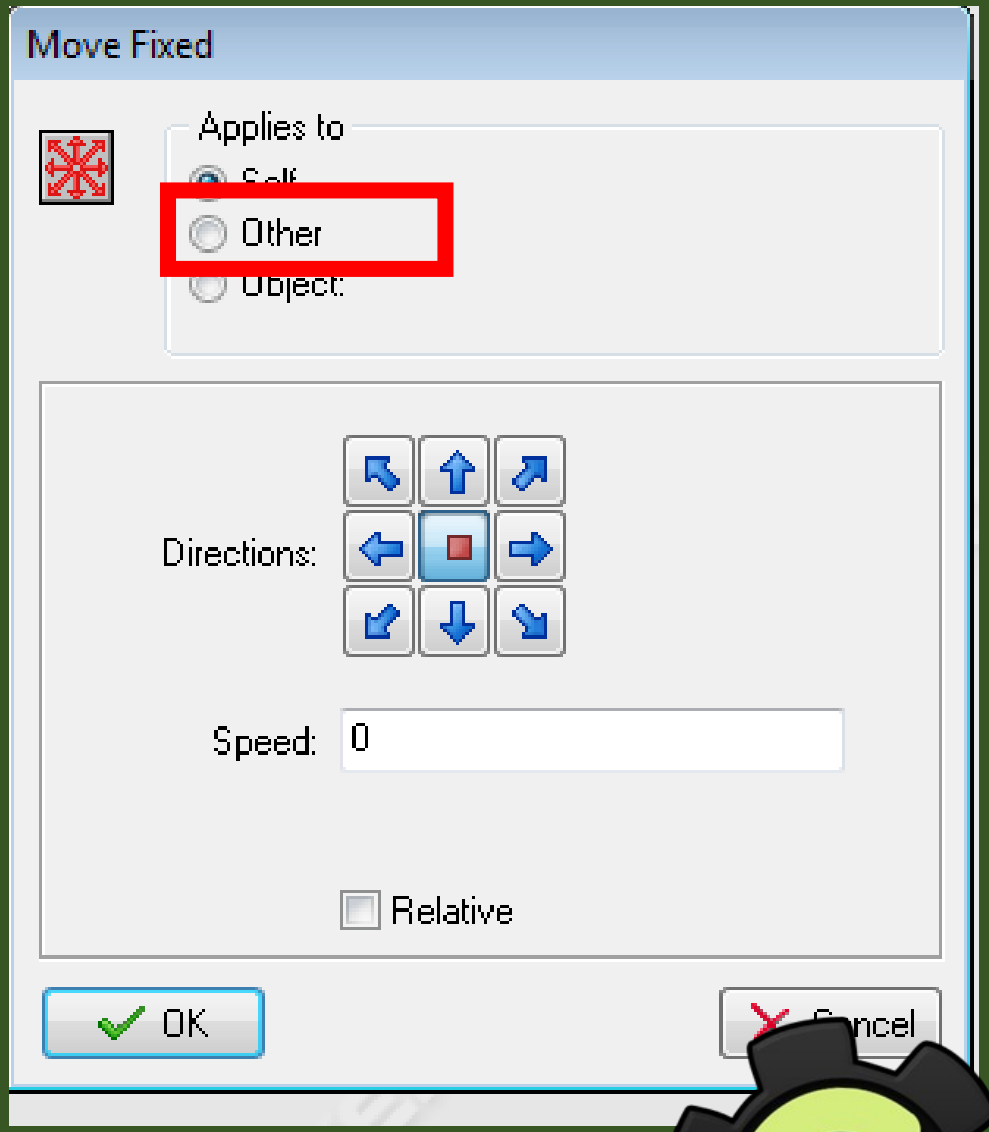
We are going to run when the objPlayer and objWall are colliding together.

To do this, we need to select objPlayer.

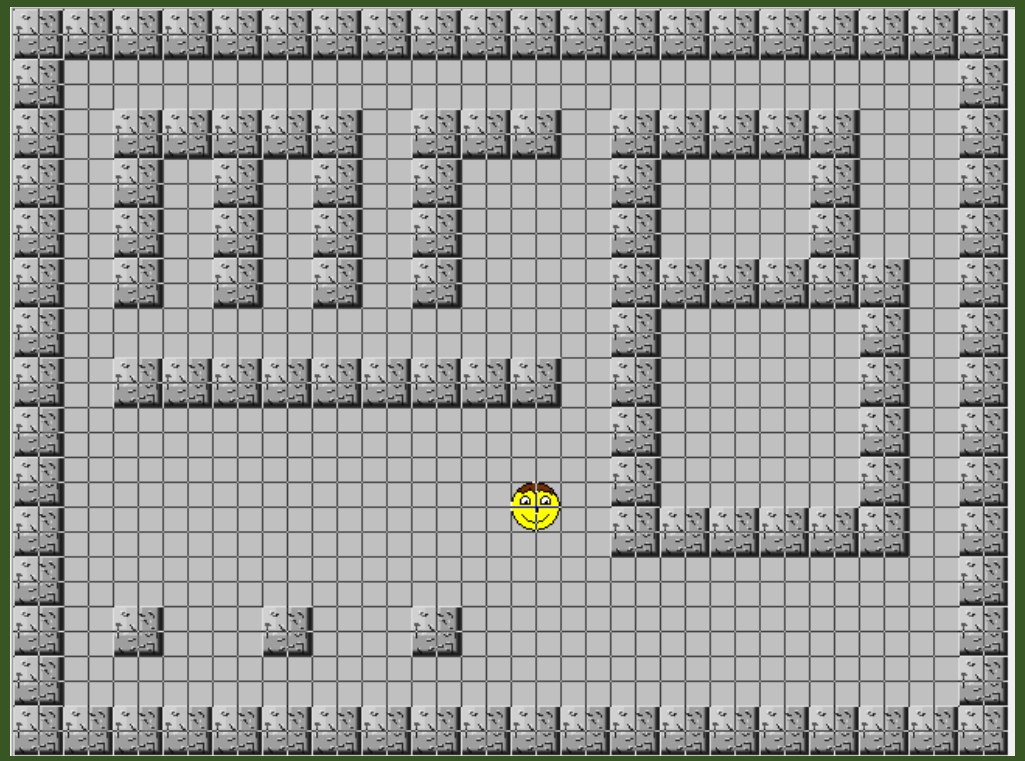


Much like we did for the ‘no key’ being pressed, we can use the ‘move fixed’ option.

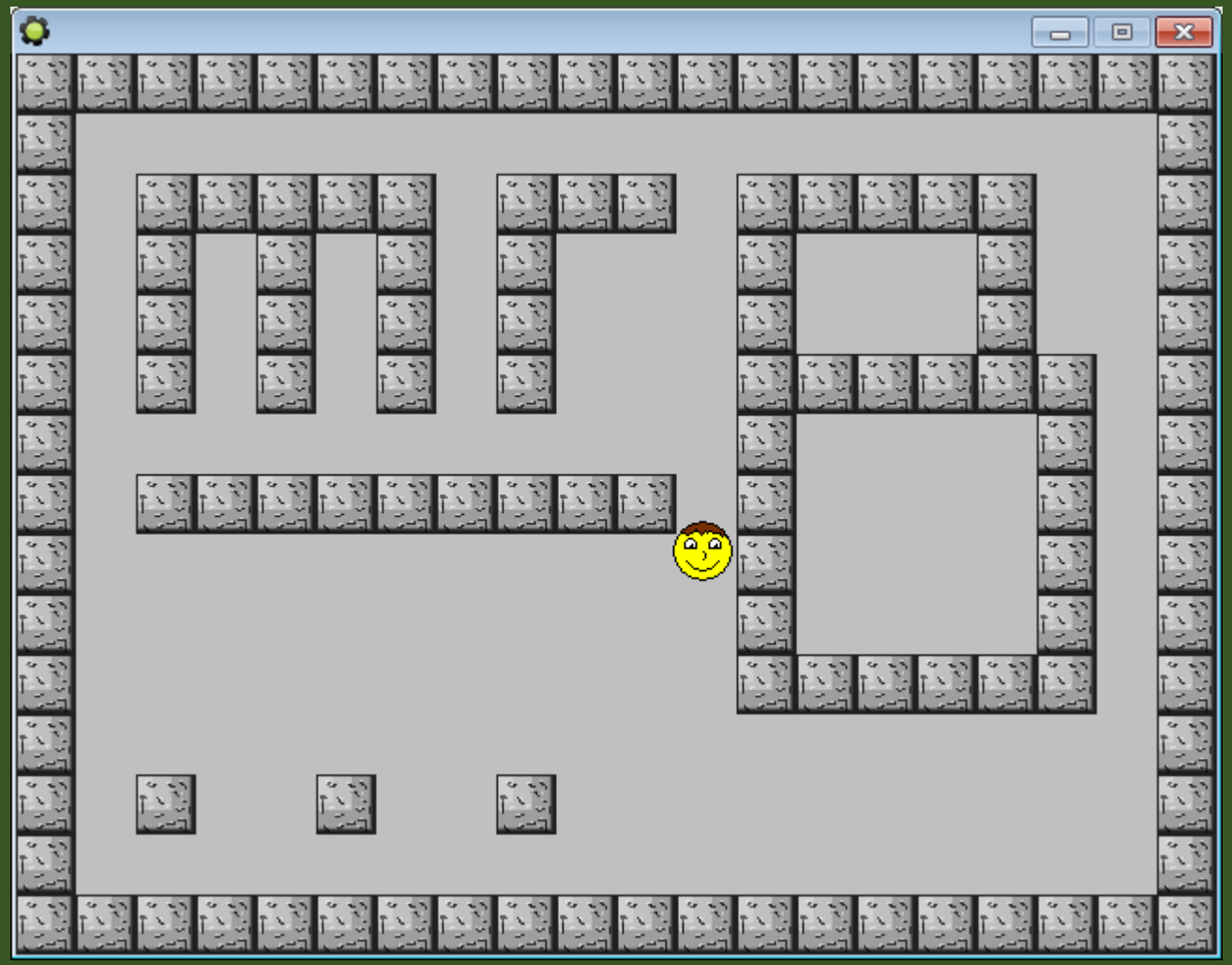
Except this time, we need to stop the Other object (the player).



Go into your PractiseRoom and add walls around the entire room. Add some in the middle too.



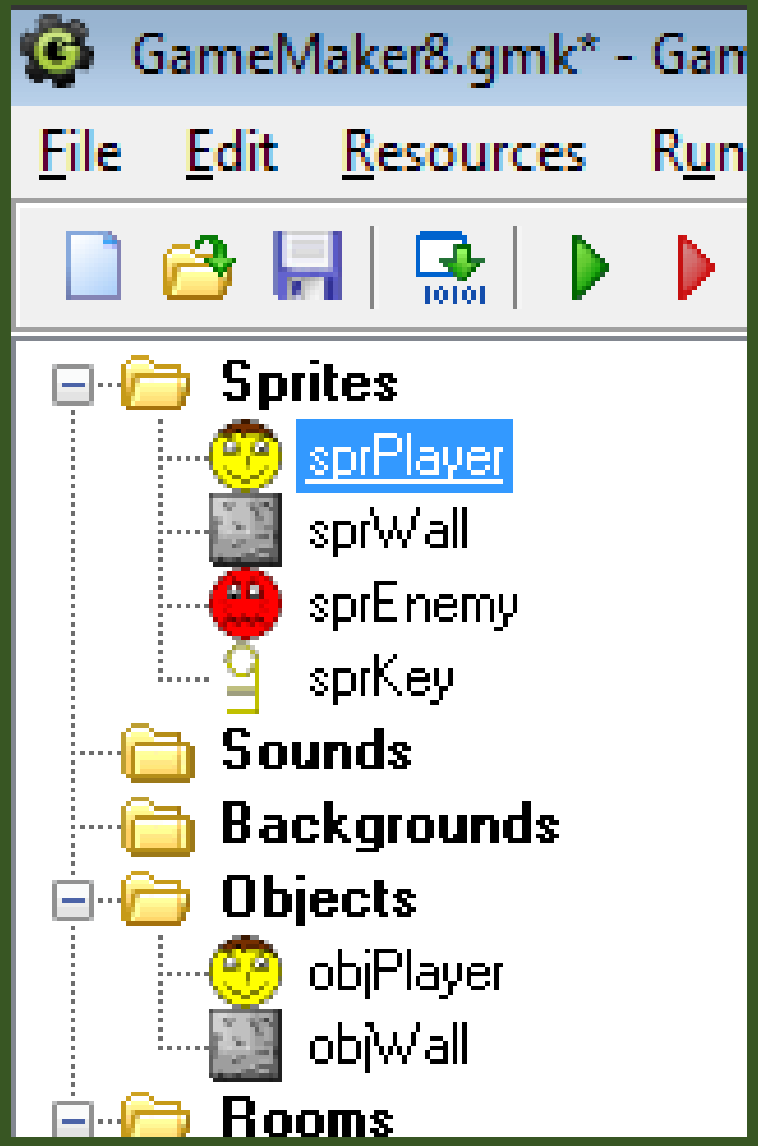
Run your game to see if the player can move through the wall.



Oh no!

The Character can't fit through the gap!

We need to make our character slightly smaller.

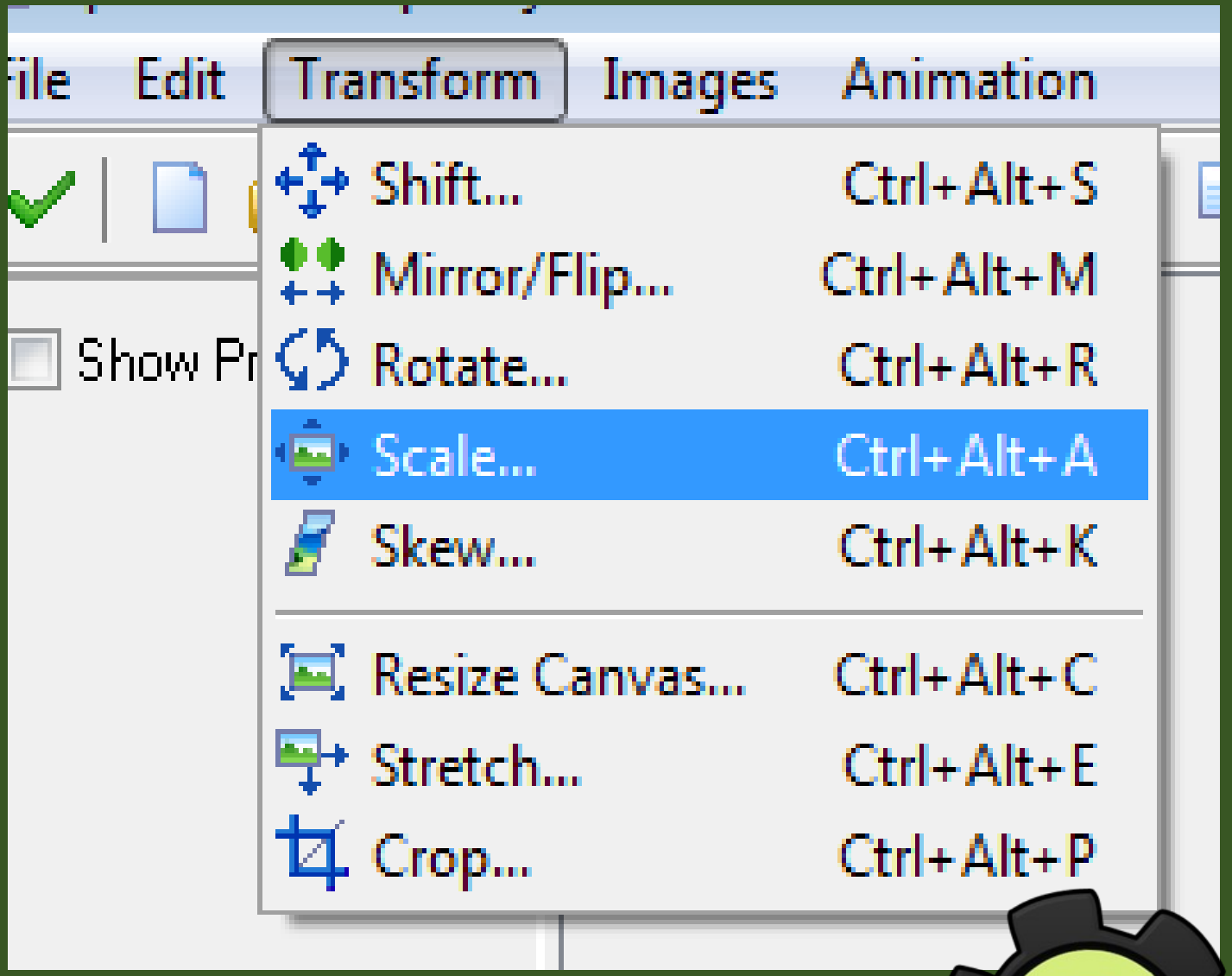


Exit the game by pressing Esc.

Double click on our sprPlayer.

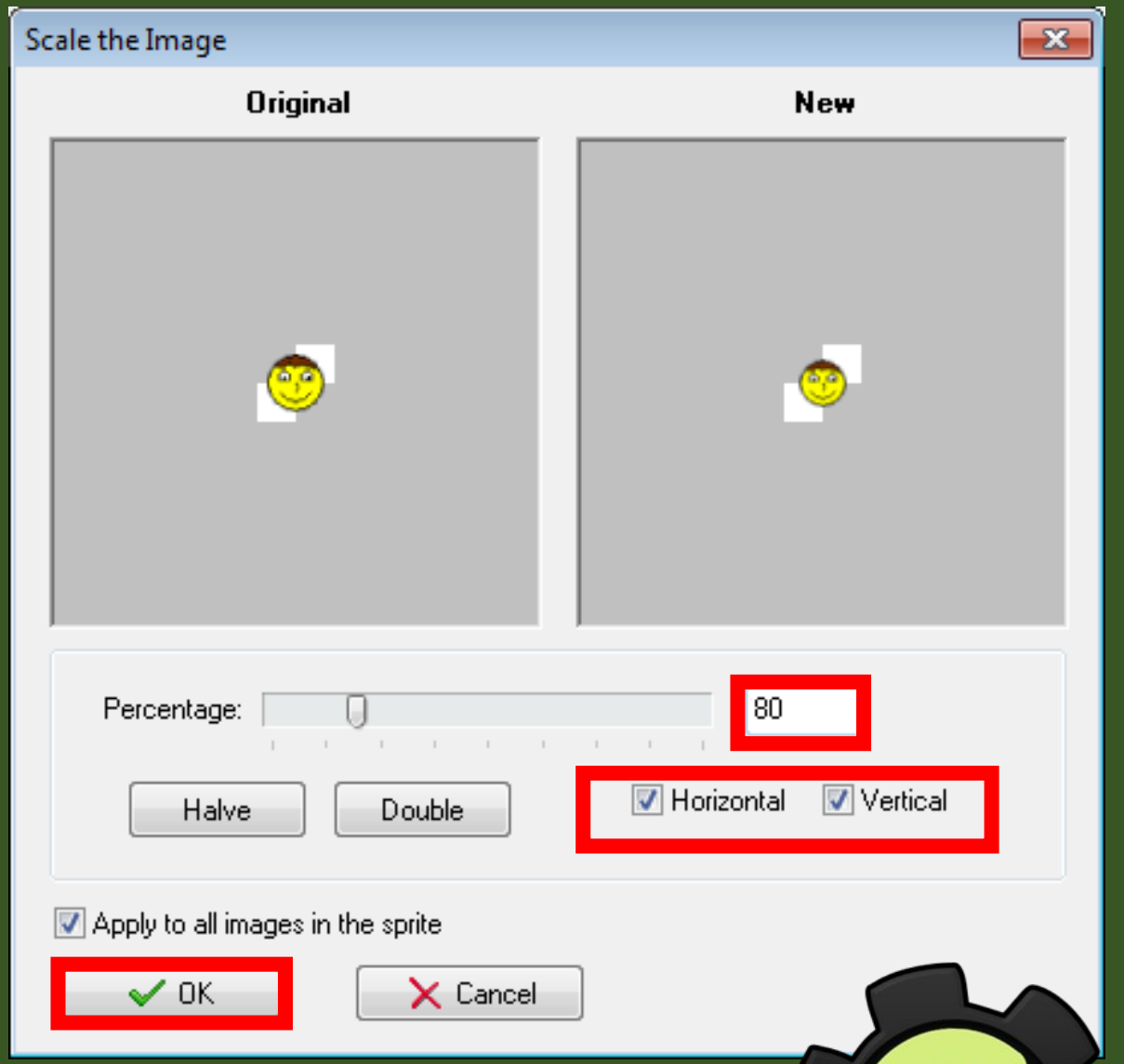
Now click Edit Sprite

On the menu bar there is a dropdown for Transform. Select Scale...



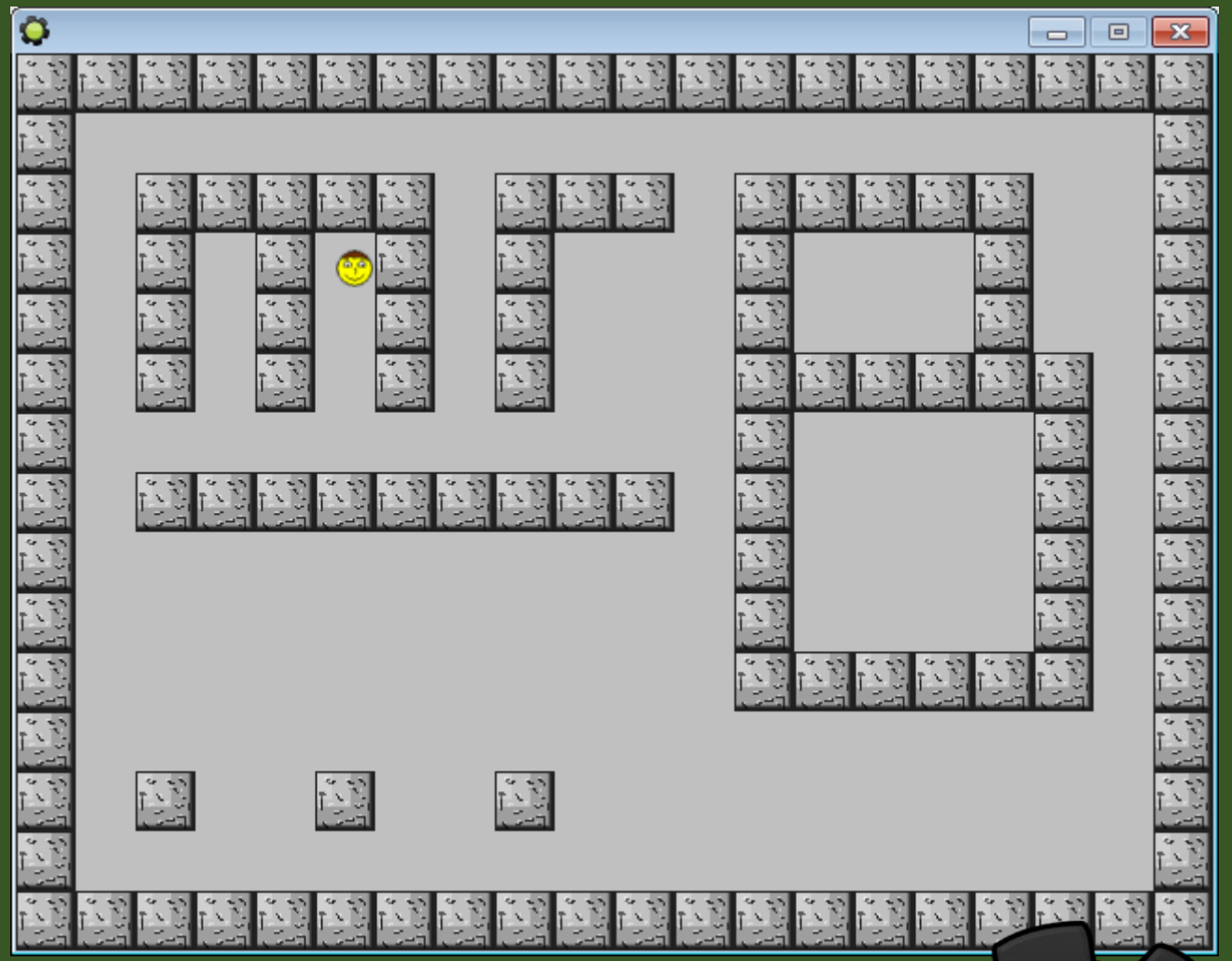
In the Scale popup, set the Percentage to 80

Make sure both Horizontal and Vertical is checked, otherwise you'll stretch your sprite.



Run your game now. You character should be able to fit through the gaps!

Repeat this for your sprEnemy



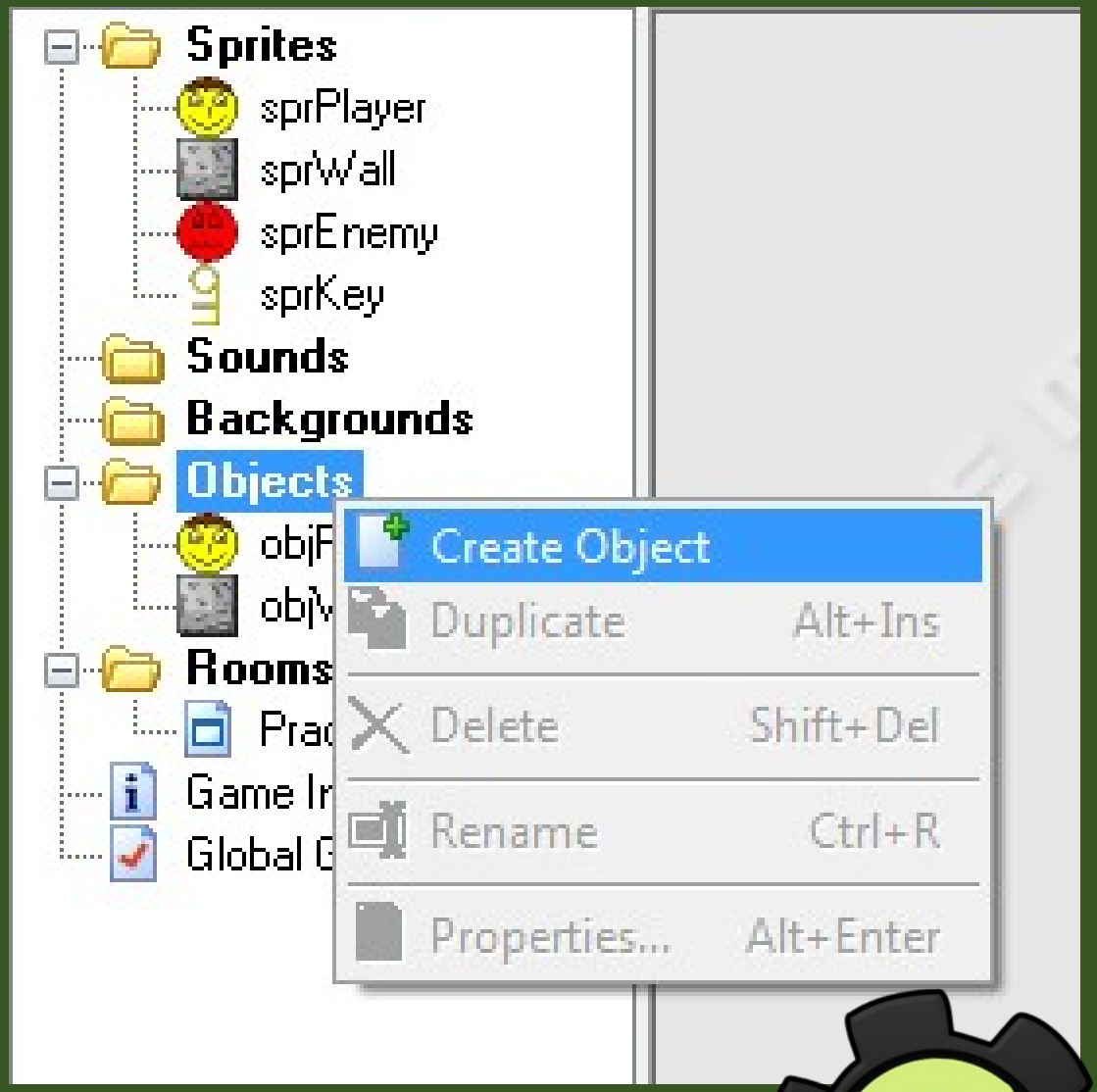
In our game, if the key is collected, we're going to progress to the next level.

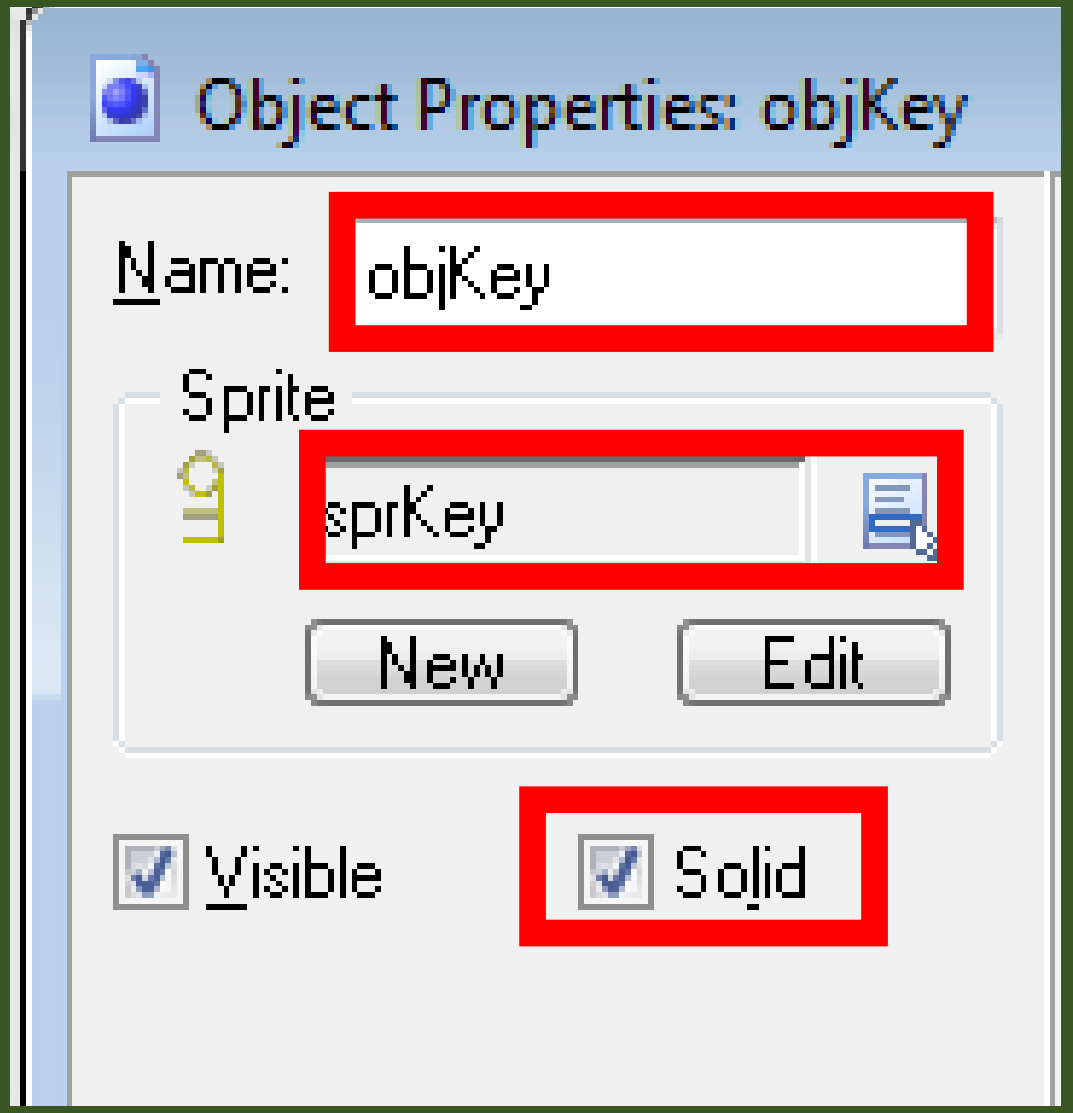


Exit the game’s preview. We’re now going to make the key object.



Right Click on Objects and click Create Object.

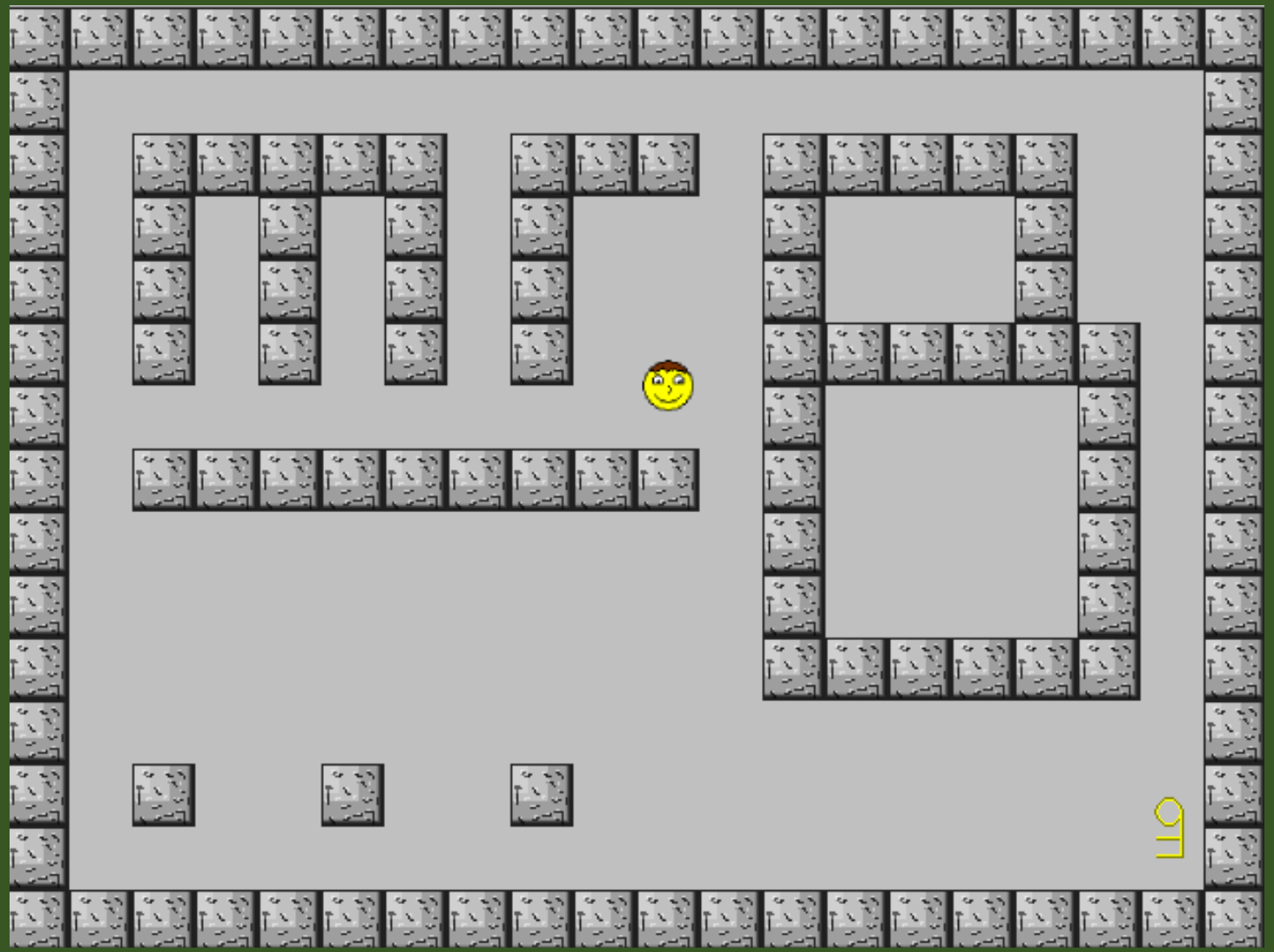




Keep the naming convention the same.

Set the Sprite.

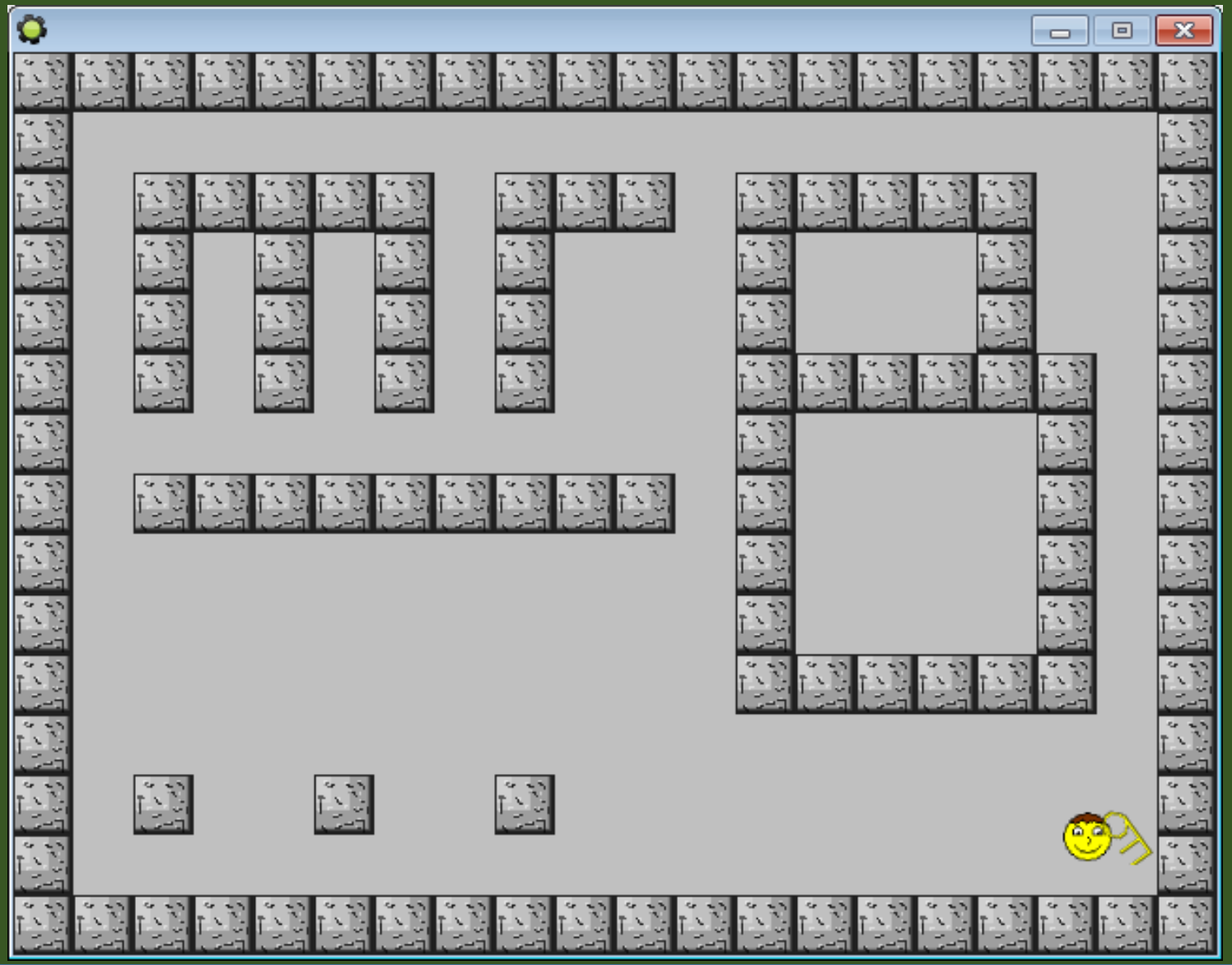
Set this to be solid.



Add the objKey into your room.

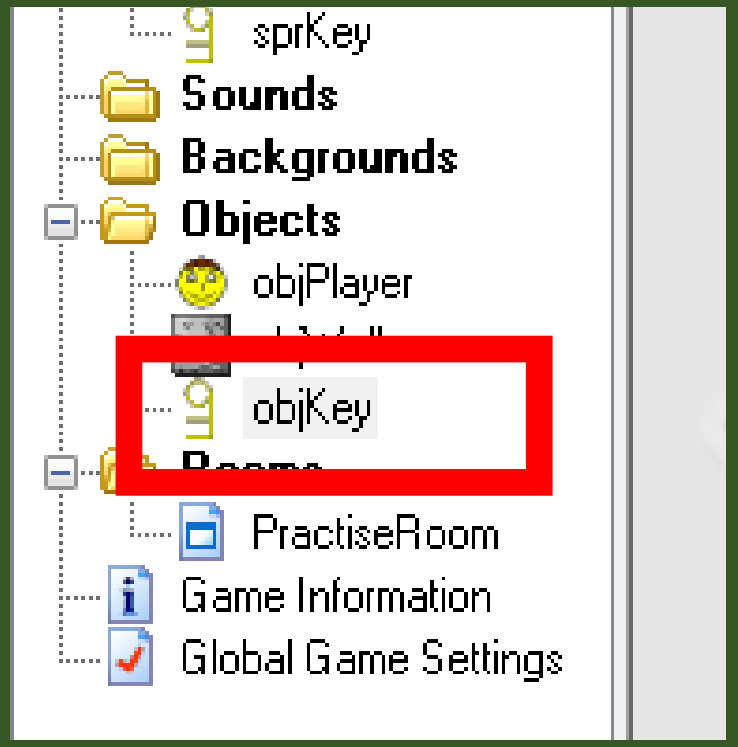
Run your game.

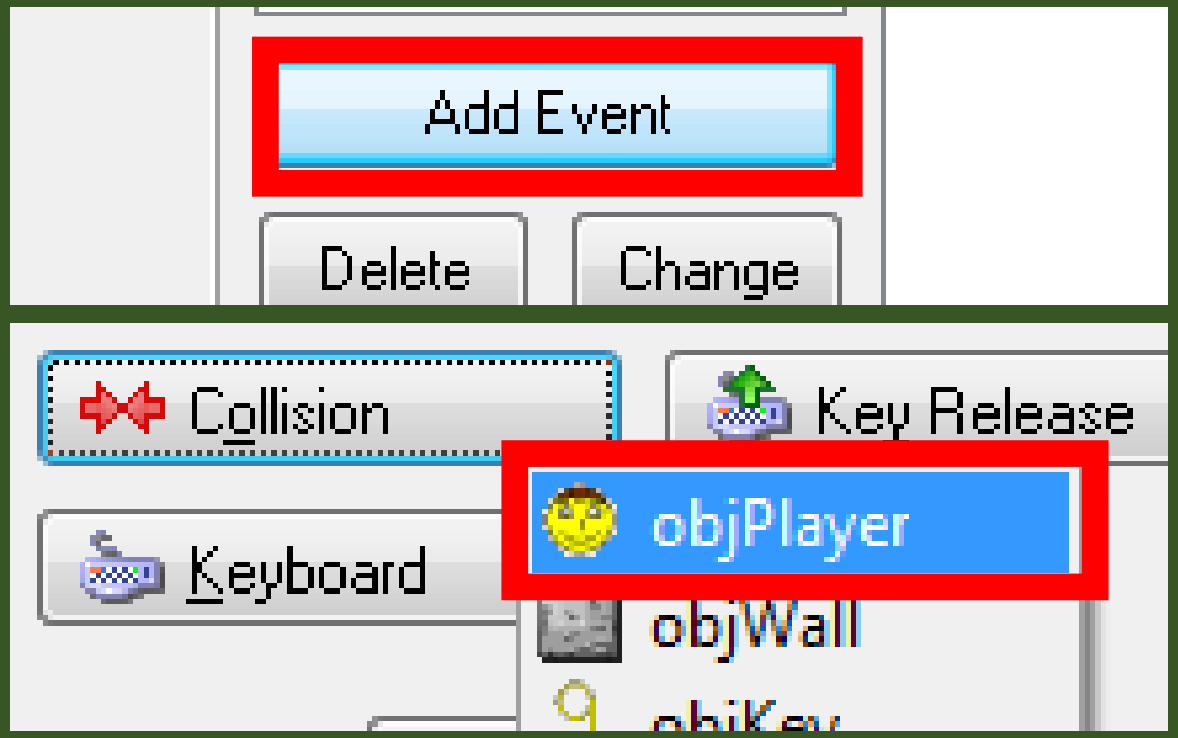
What happens when you hit the key?



We can run over the key. Why is this?





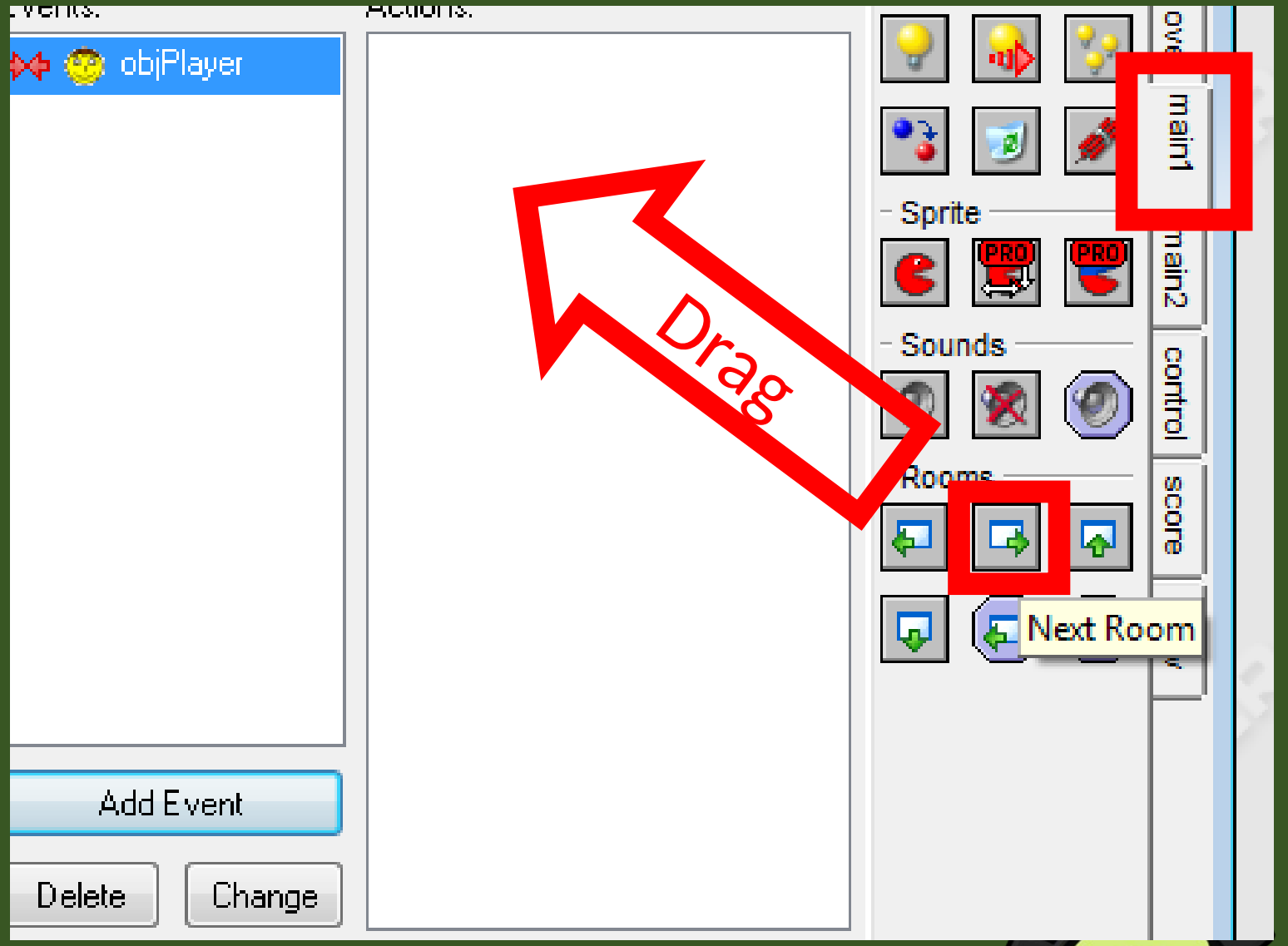


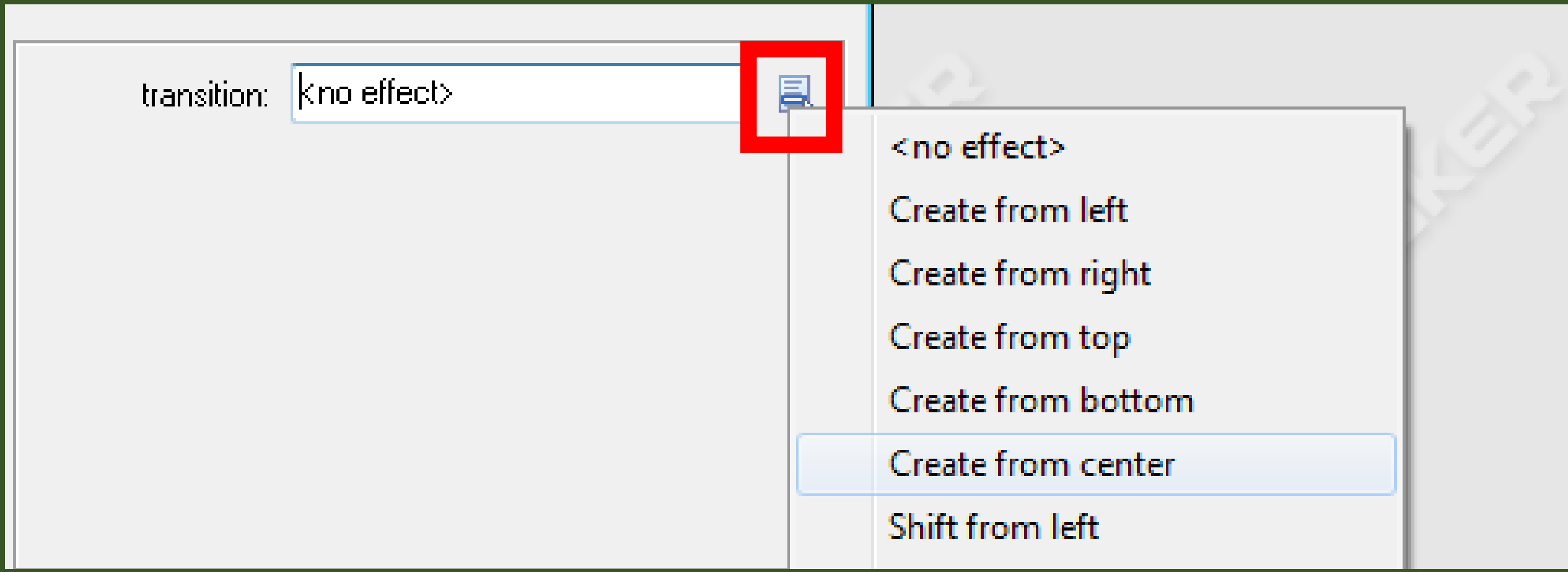
Double click on the objKey.
We're going to Add Event > Collision > objPlayer

Click on the main1 tab.

Under rooms, there is an option for Next Room.

Drag this into the Actions Section.

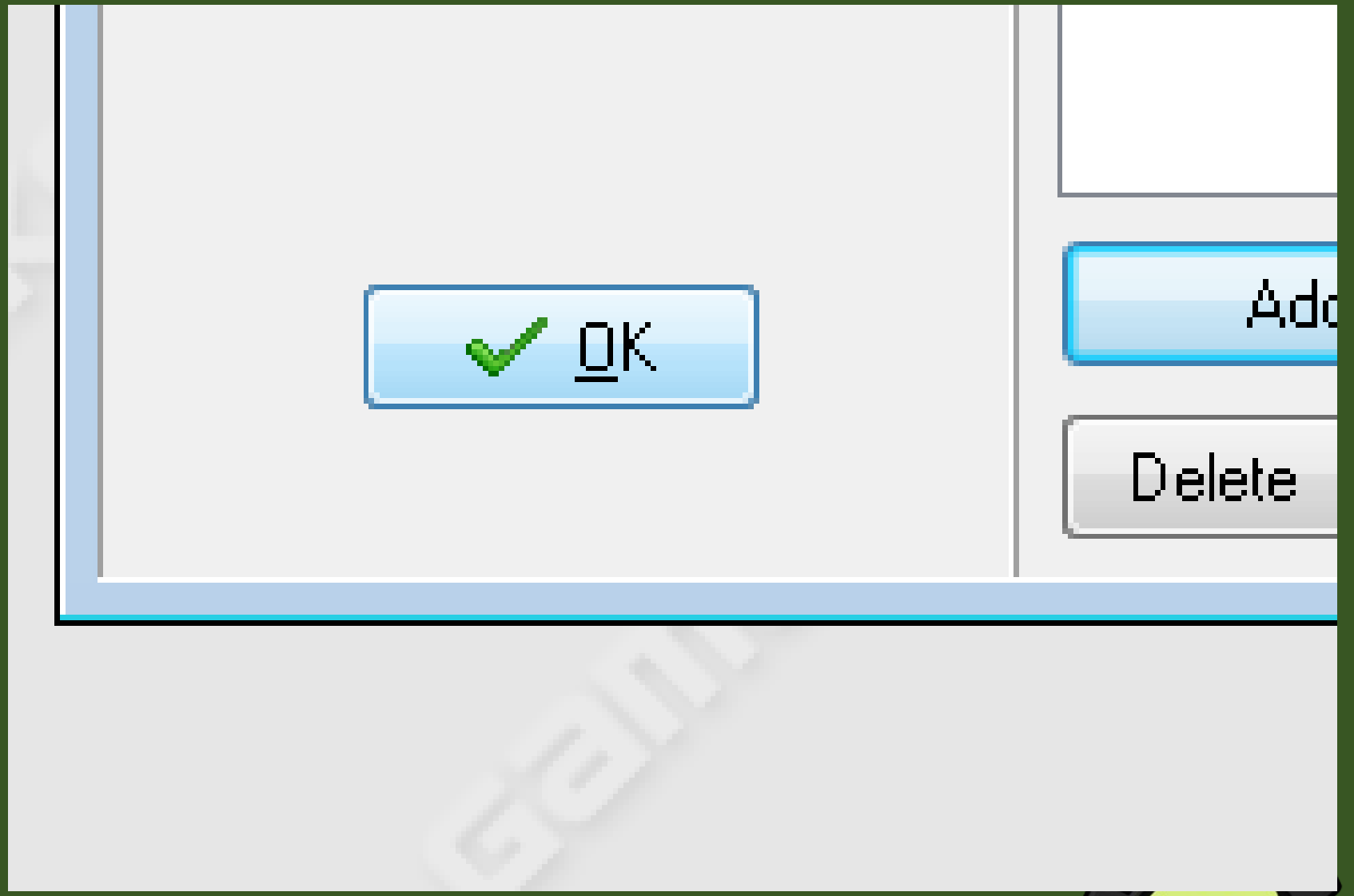




We can add a transition effect to add between levels.

Hit Ok and test your level.

What happens now when you hit your key?



Error Messages

```
------------------------------------------
FATAL ERROR in
action number 1
of Collision Event with object objPlayer
for object objKey:

Moving to next room after the last room.
```

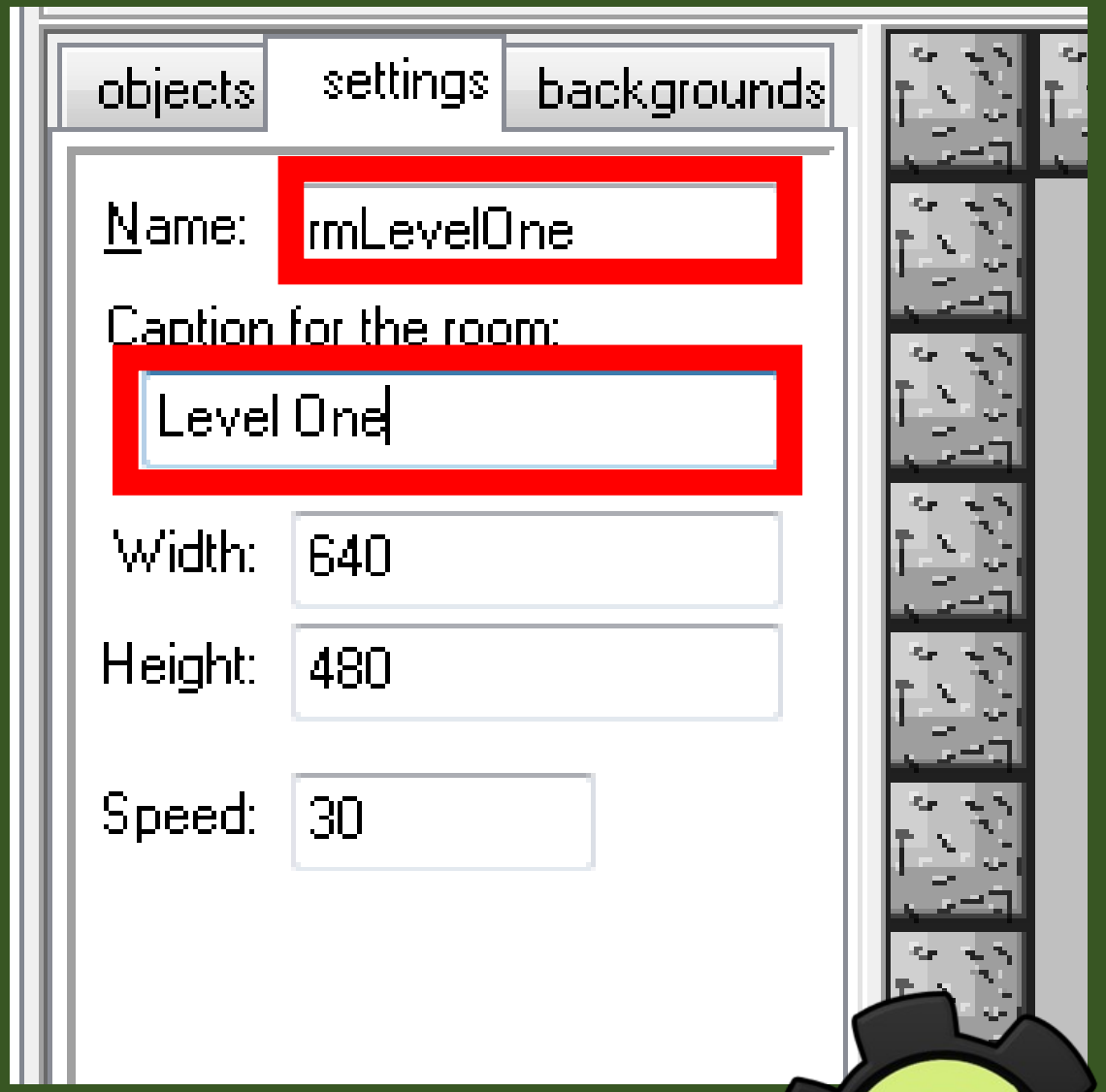
FATAL ERROR

Why? It was going so well?



This was caused by not having another room to go to.

We'll add another room now.
We'll call it rmlevelOne



Make sure you add an objKey as well as a player to your new level!

Then hit the green tick to save



Run your game. When you hit the first key, you should progress to ‘level One’.



Gaming Platforms

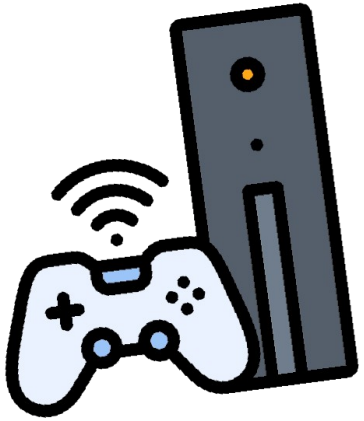


How do we play games?

What do we use?

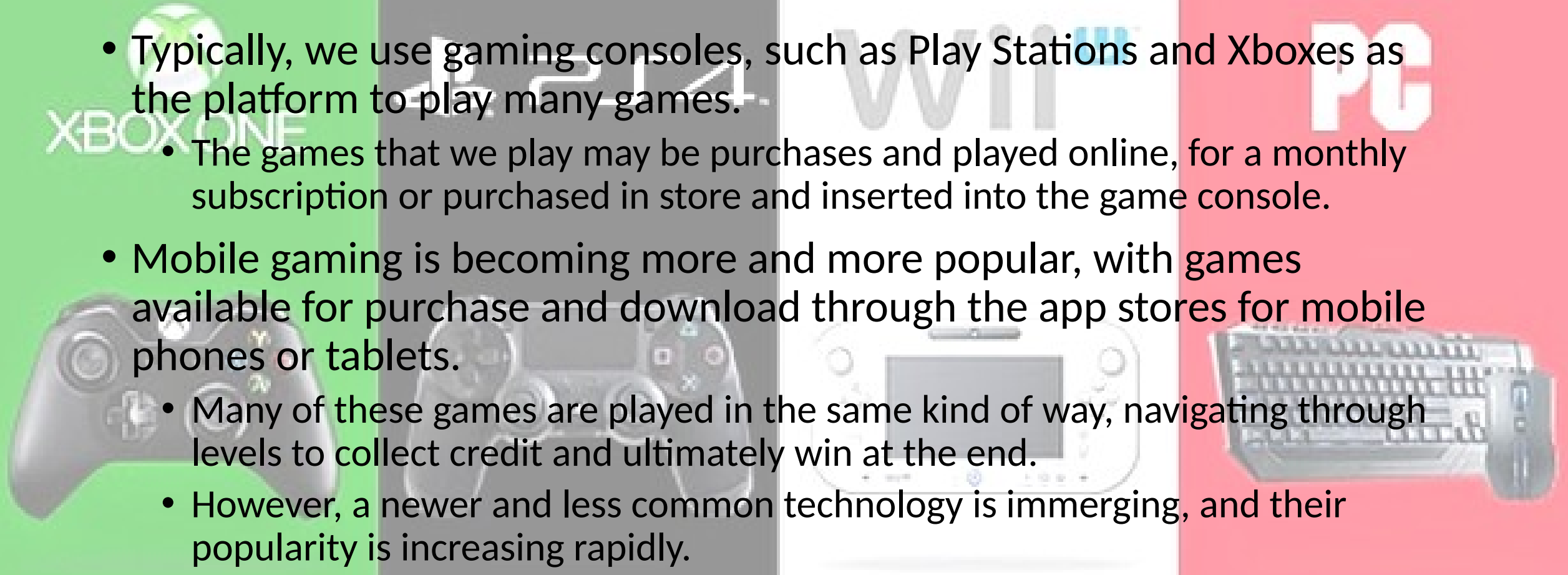
Where are we?

Hardware/Software?



Game Platforms

- Typically, we use gaming consoles, such as Play Stations and Xboxes as the platform to play many games.
  - The games that we play may be purchases and played online, for a monthly subscription or purchased in store and inserted into the game console.
- Mobile gaming is becoming more and more popular, with games available for purchase and download through the app stores for mobile phones or tablets.
  - Many of these games are played in the same kind of way, navigating through levels to collect credit and ultimately win at the end.
  - However, a newer and less common technology is immerging, and their popularity is increasing rapidly.
- These technologies are called **VR** and **AR**.



VR



- stands for Virtual Reality
- It’s a world that doesn’t exist.
  - This virtual reality has been created by a computer, which allows users to experience and interact with a 3D world
  - uses a VR headset
  - Additional technologies can be incorporated to add sound and movement.
- The most well-known example of VR is Minecraft. Minecraft allows users to build a variety of different blocks in a 3D procedurally generated world, using the players creativity.

AR



- stands for Augmented Reality
  - This means adding computer generated images into real life
  - For example, some furniture shops have an app or additional functionality on their websites, which allow users to view pieces of furniture in their own rooms.
  - This incorporated the use of device cameras and their internet connection.
- Biggest example is Pokemon Go

Let’s Create more of our Game

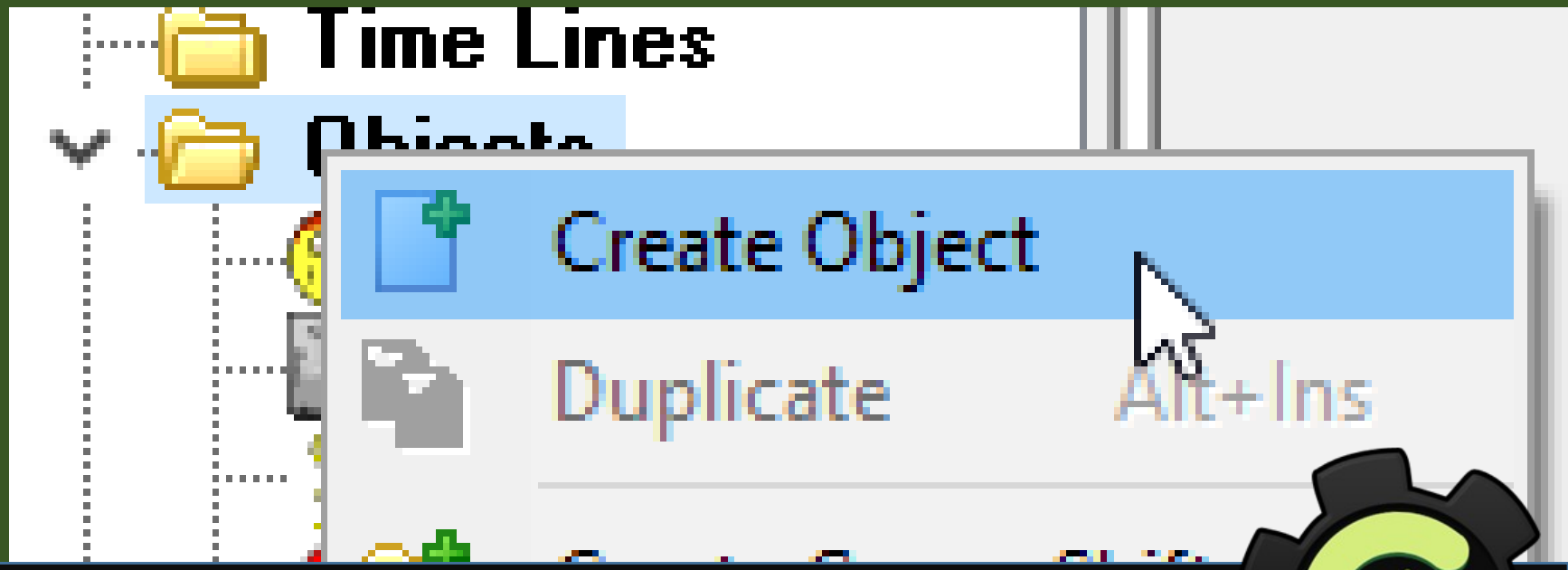
Tutorial Part 4

We’re now going to add another object to our game.

We’ll name this one objEnemy.



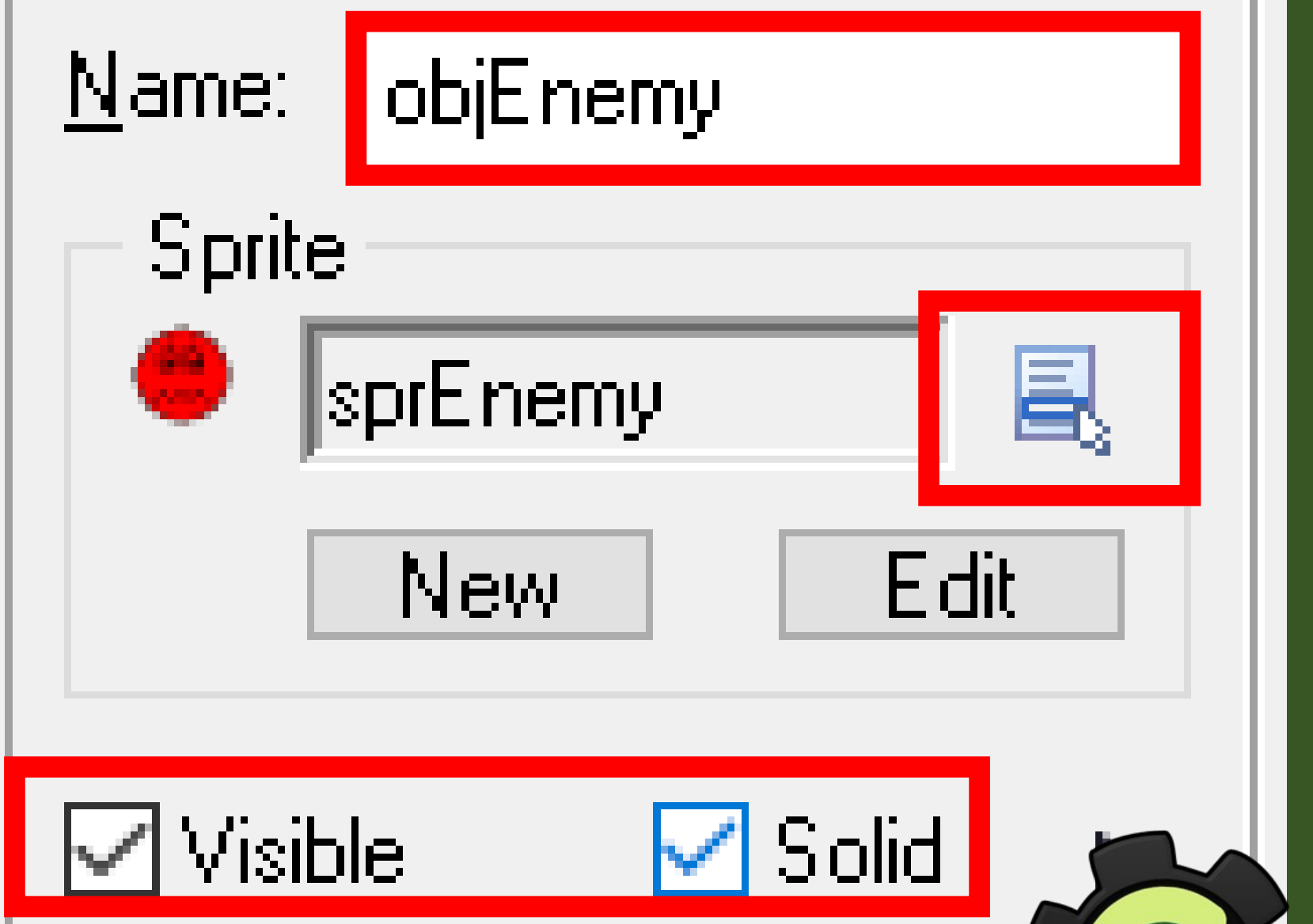
Right Click on Objects >
Create Object



objEnemy

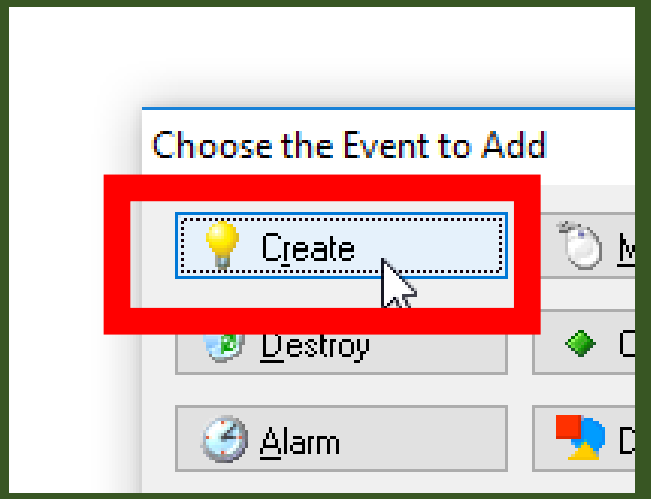
Set The Sprite

Visible AND Solid

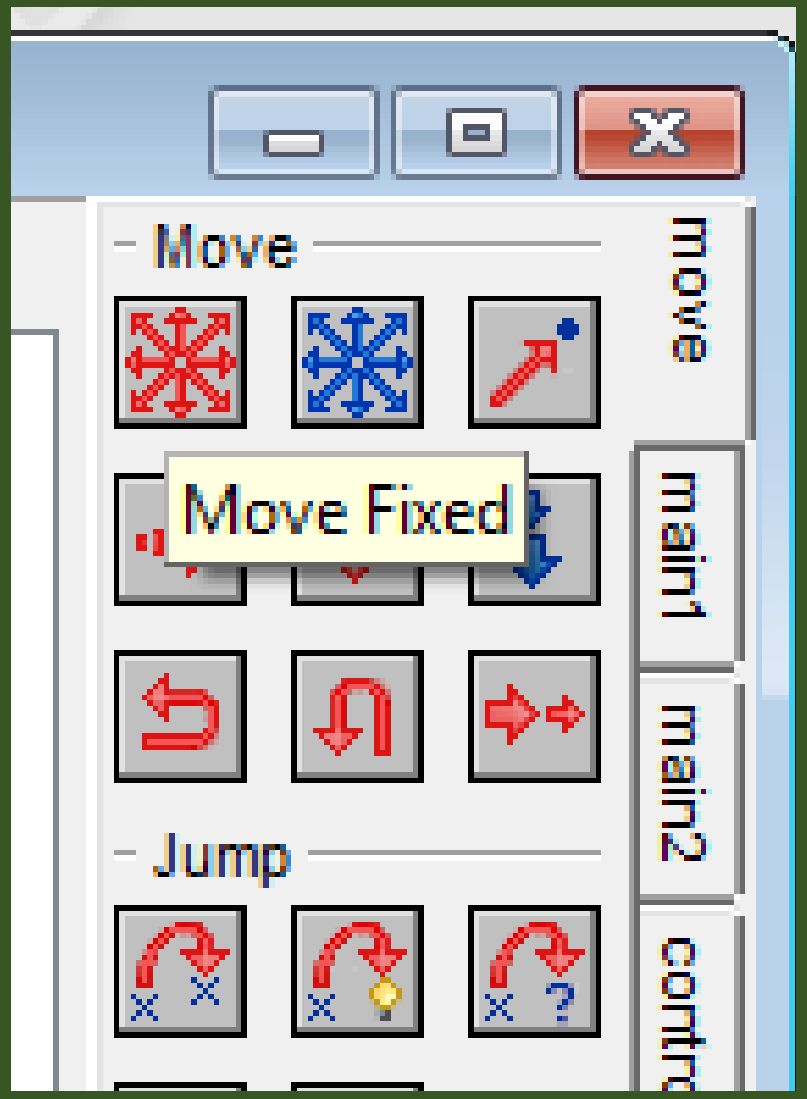




Go To Add Event > Create



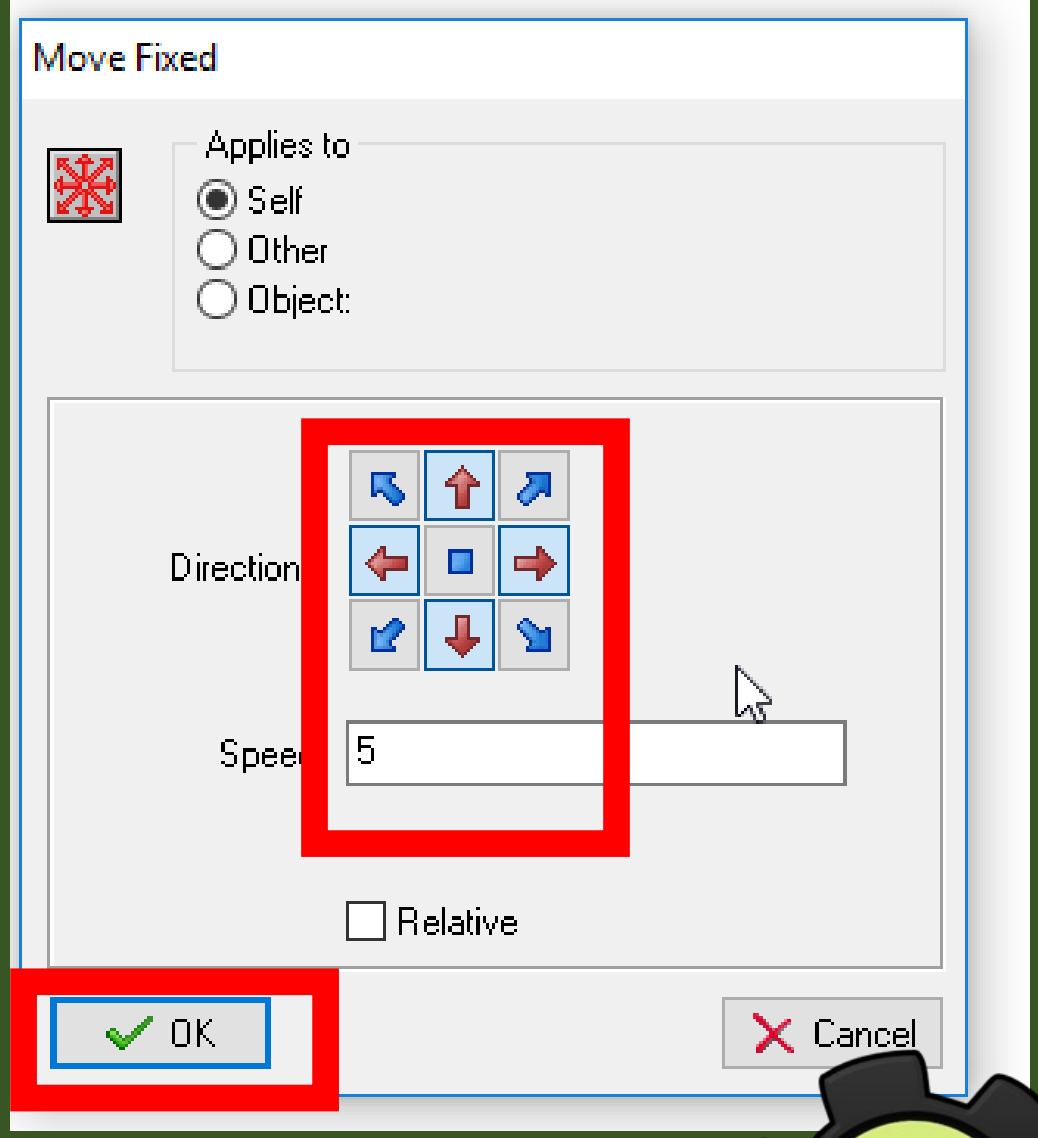




In the Move tab, we want to use the red arrows icon. The tooltip reads ‘Move Fixed’

Set The Direction to Up down left and right at a speed of 5.

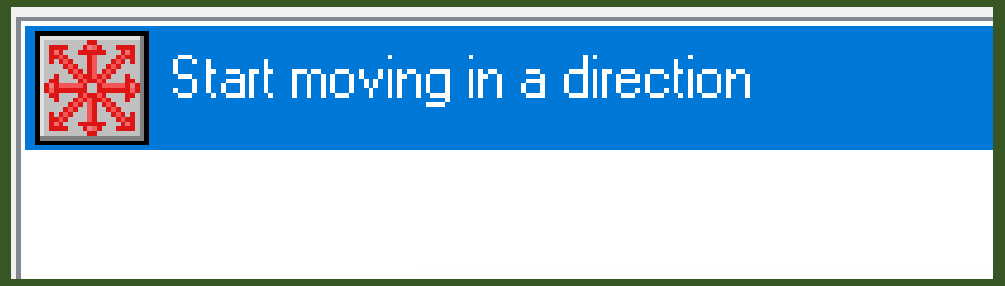
Then Hit OK



Add another Event

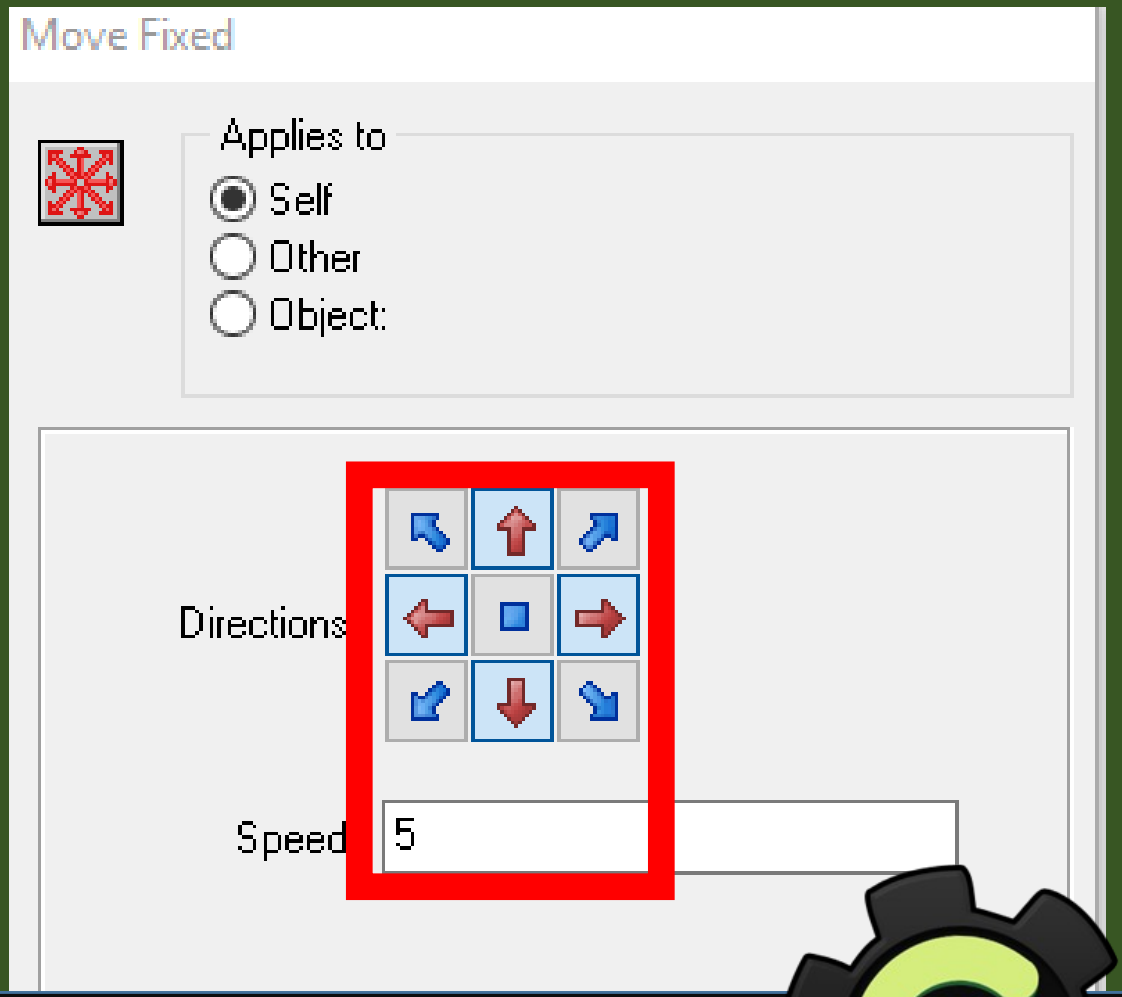
This time we are going to Collision > objWall





Add a Move Fixed block

Again, add the up down left and right with a speed of 5



Get adding!

Add some enemies into your game (not too many!!!)

Then test your game to see if there's anything wrong

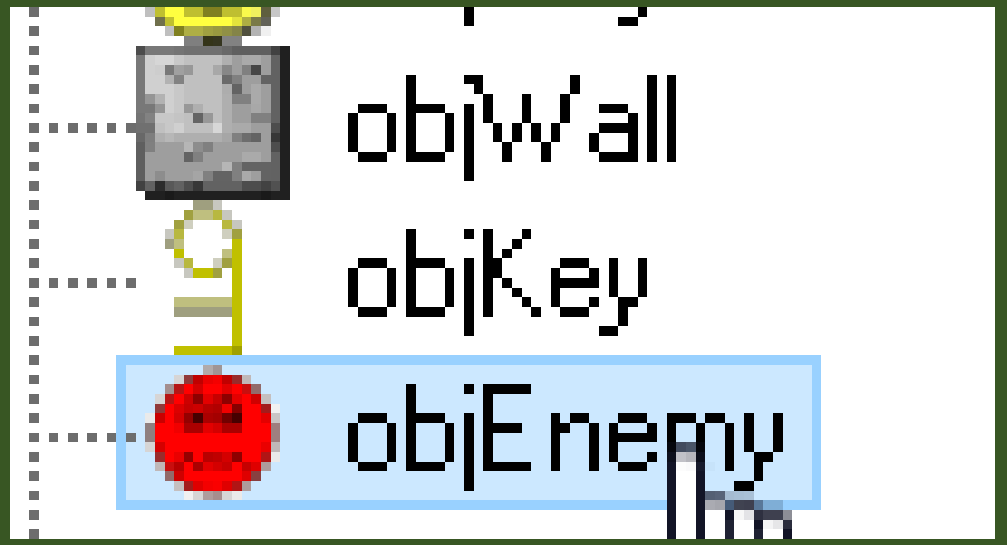


Q) What happens if the enemy hits you?

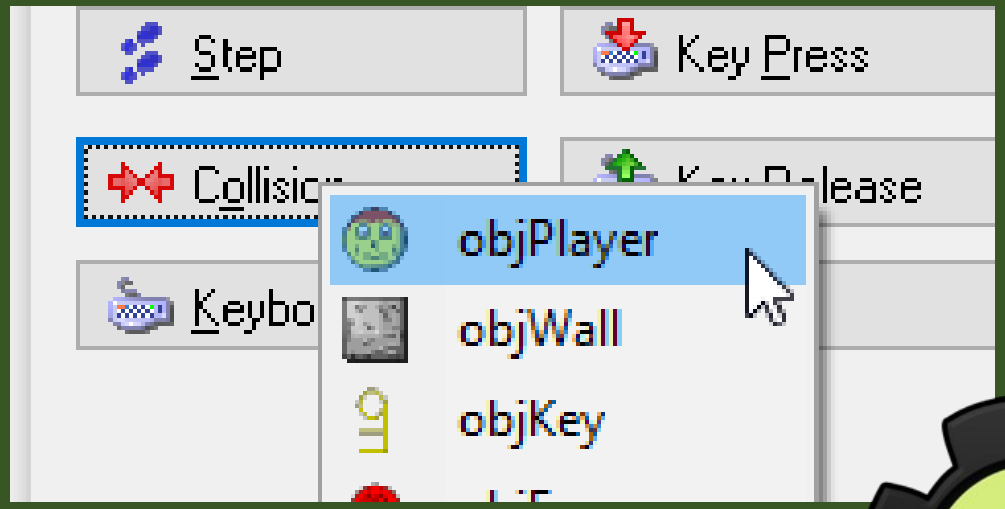
We need to add another collision event!



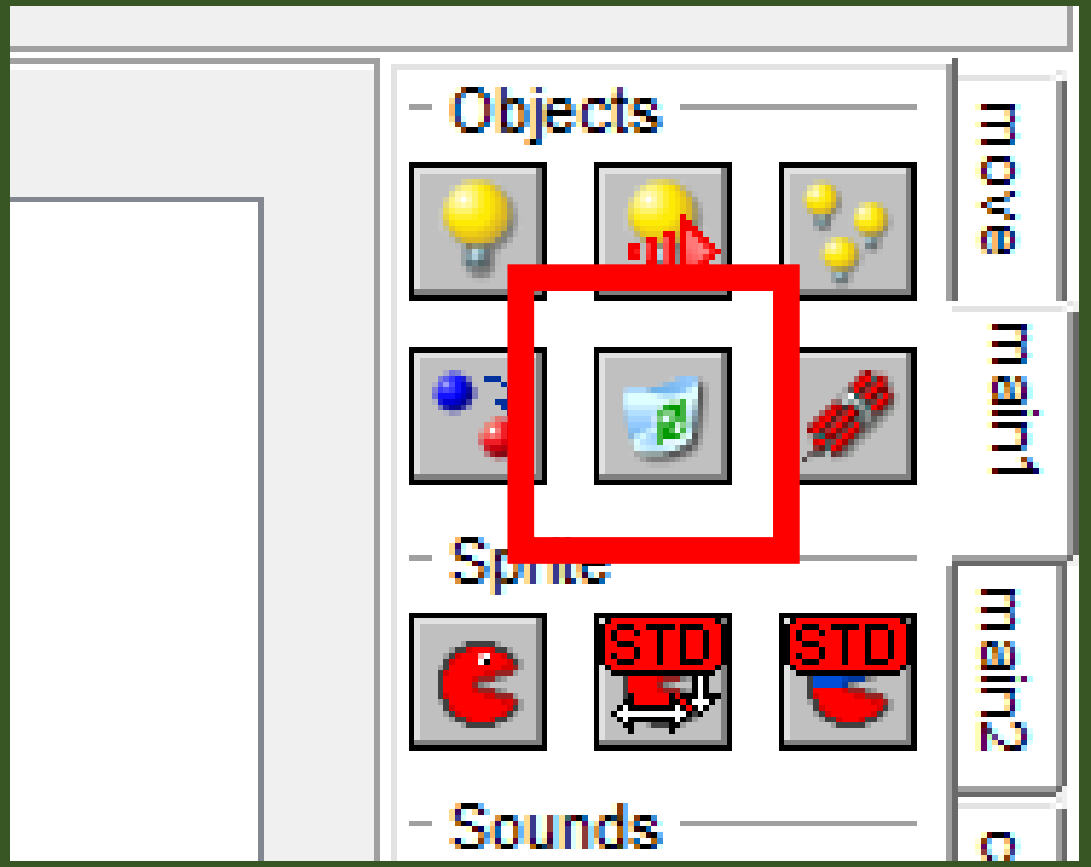
Go to objEnemy.



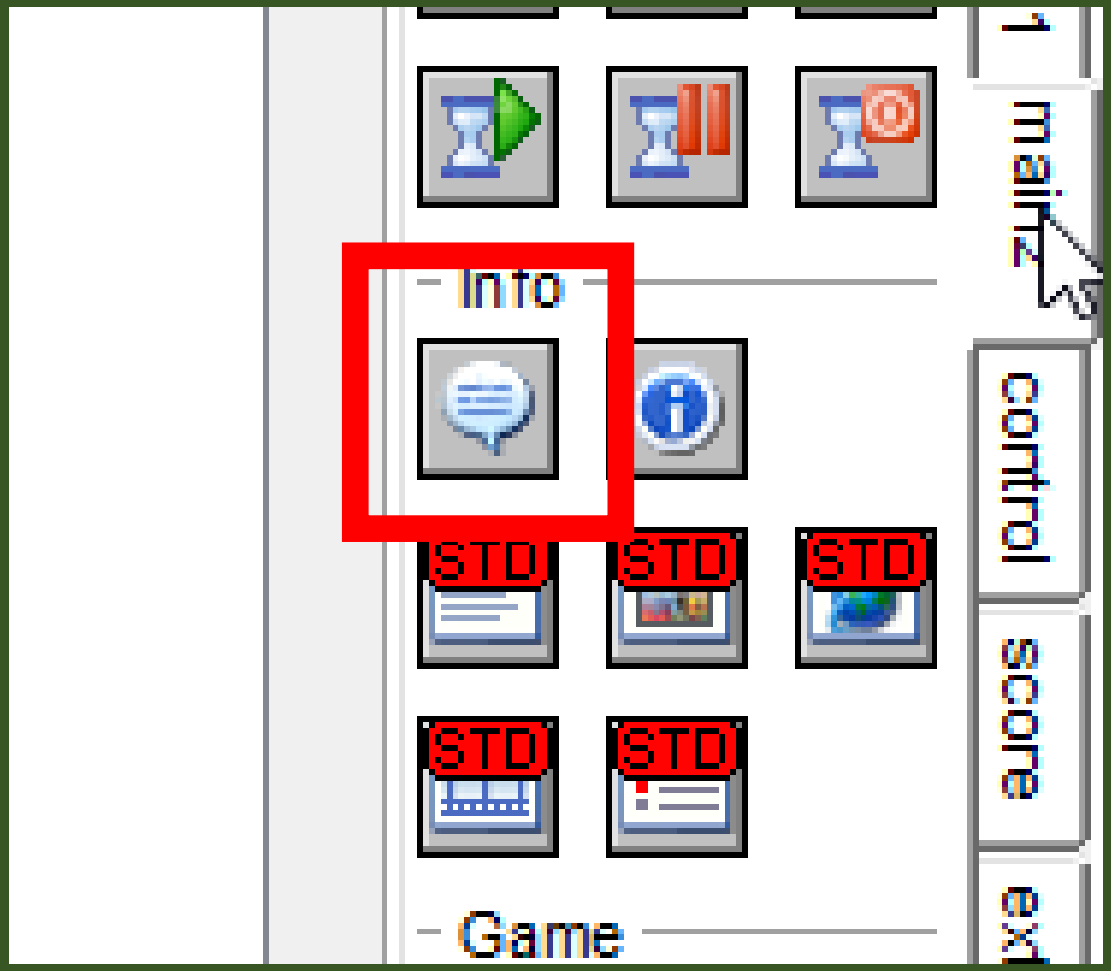
Add Event > Collision > objPlayer

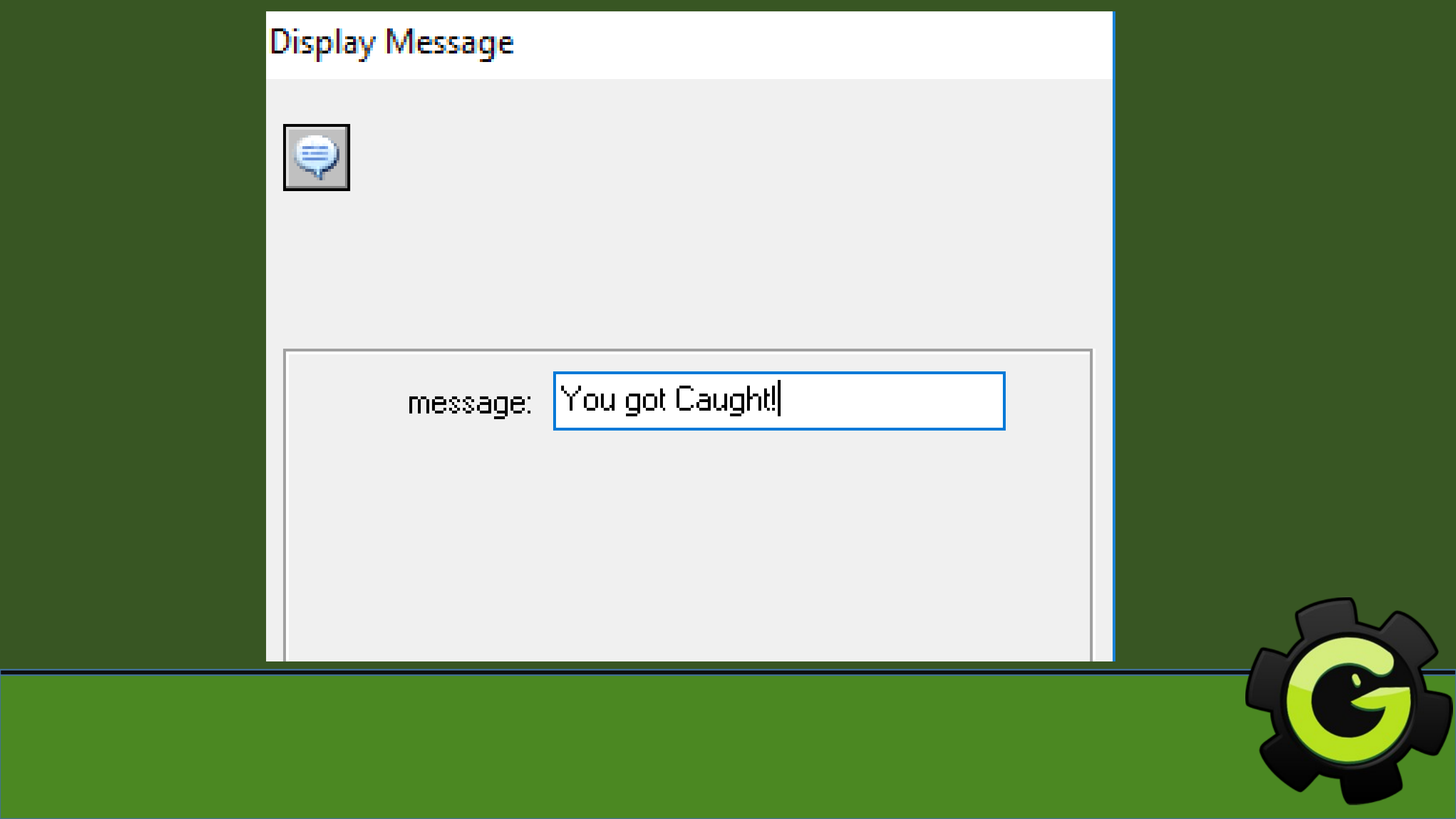


Drag the Destroy the Instance into your Actions section

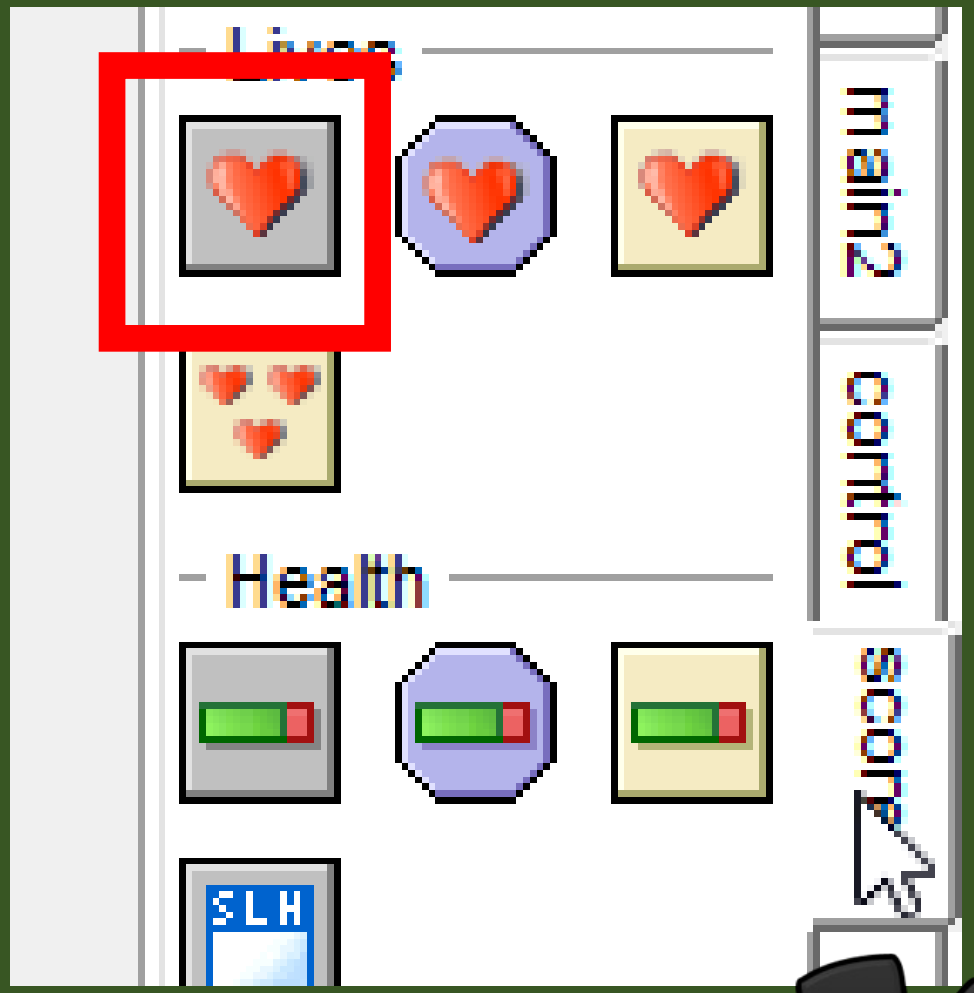


Now under main2 add a Display a Message Option




Display Message
message:
You got Caught!

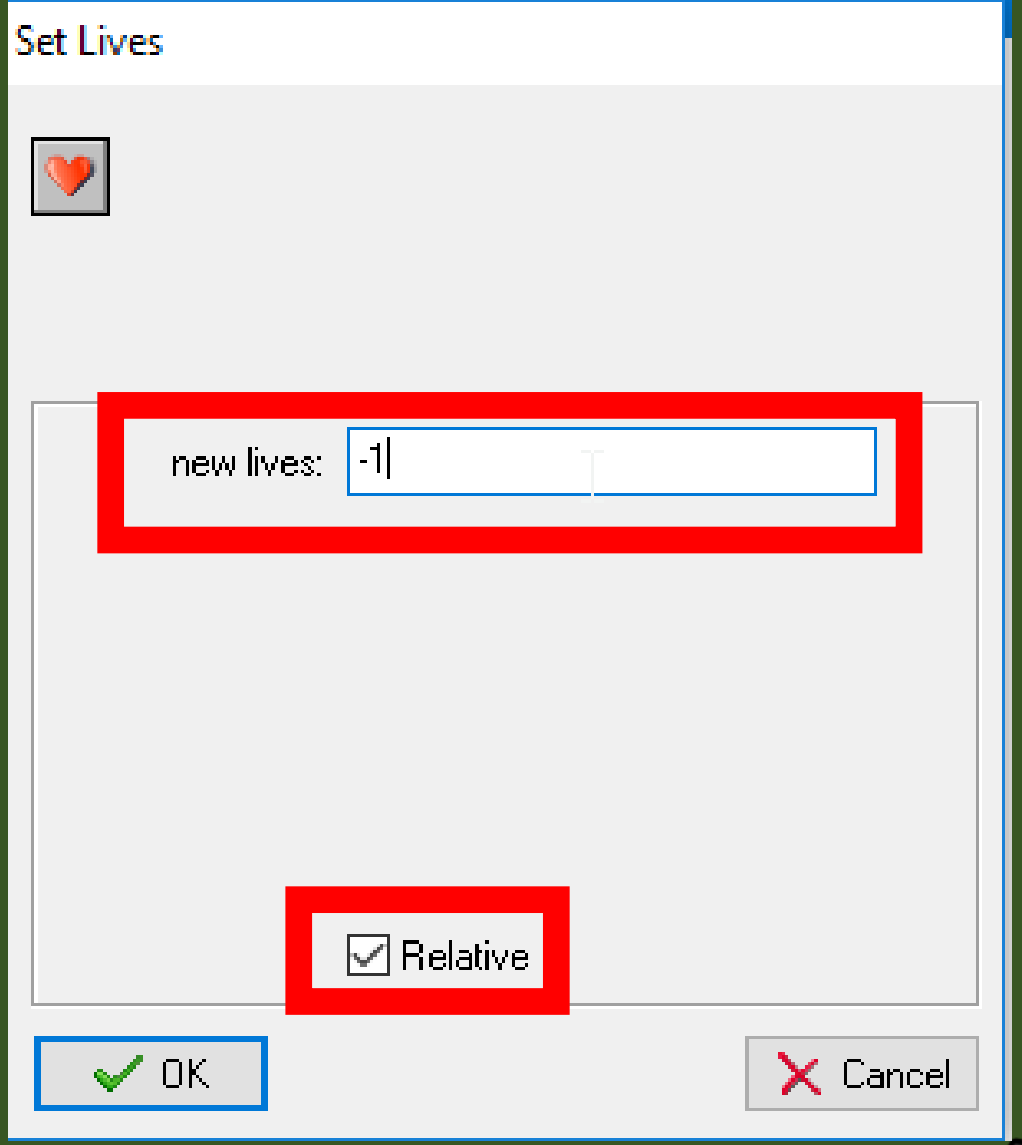
Under Score
Drag the Set Lives into your actions.

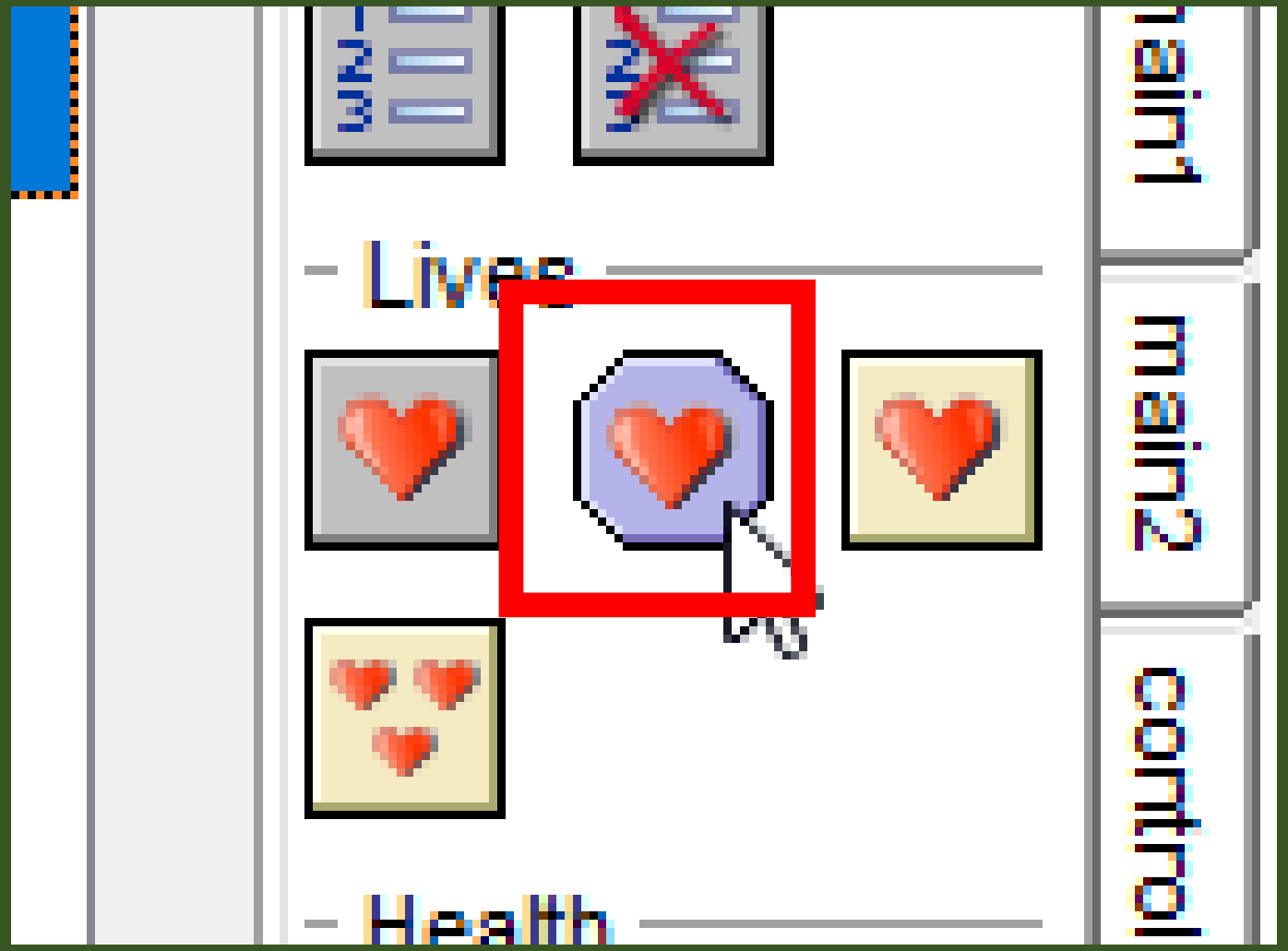


Set the new lives to -1

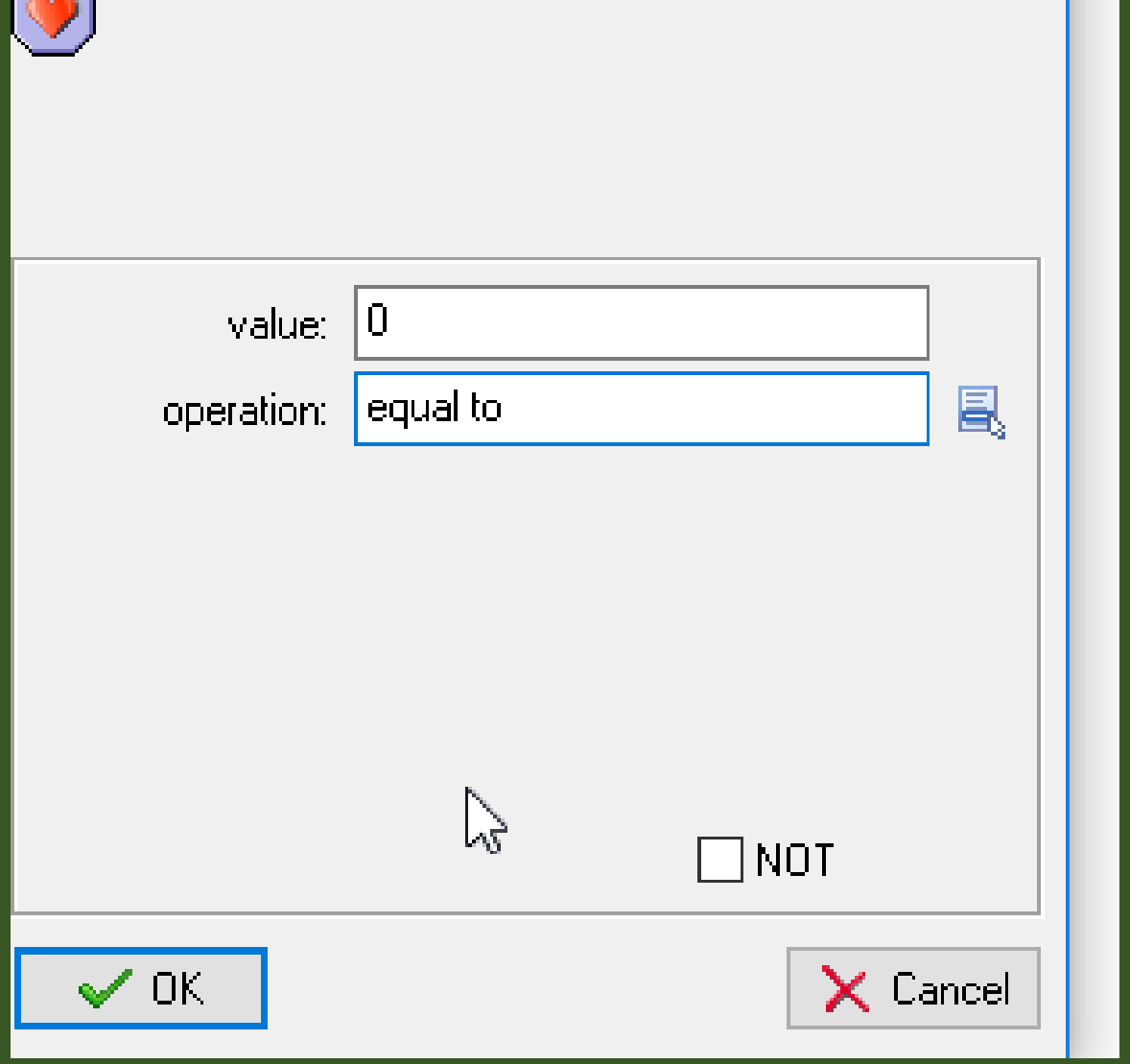
Check it is relative.

Now hit OK.



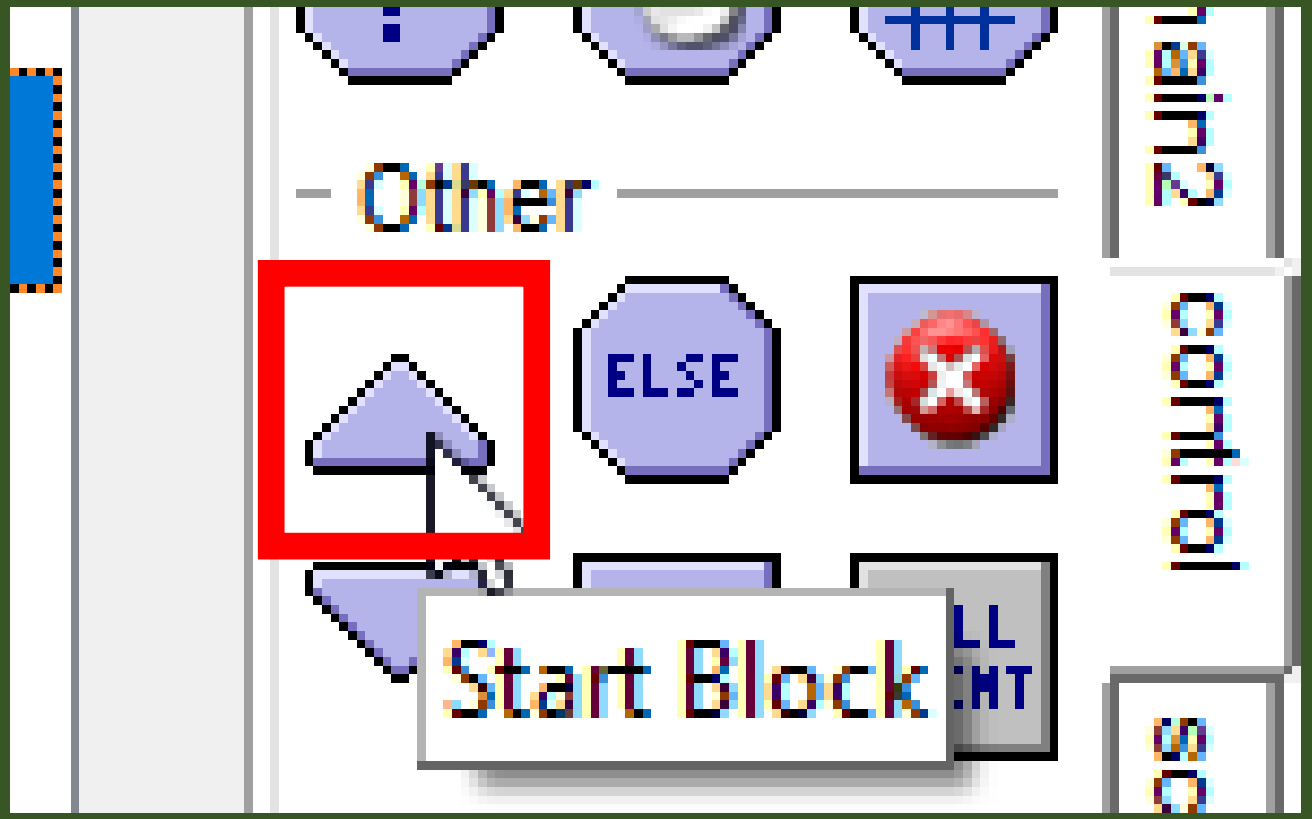


Now we need to use the Test Lives option



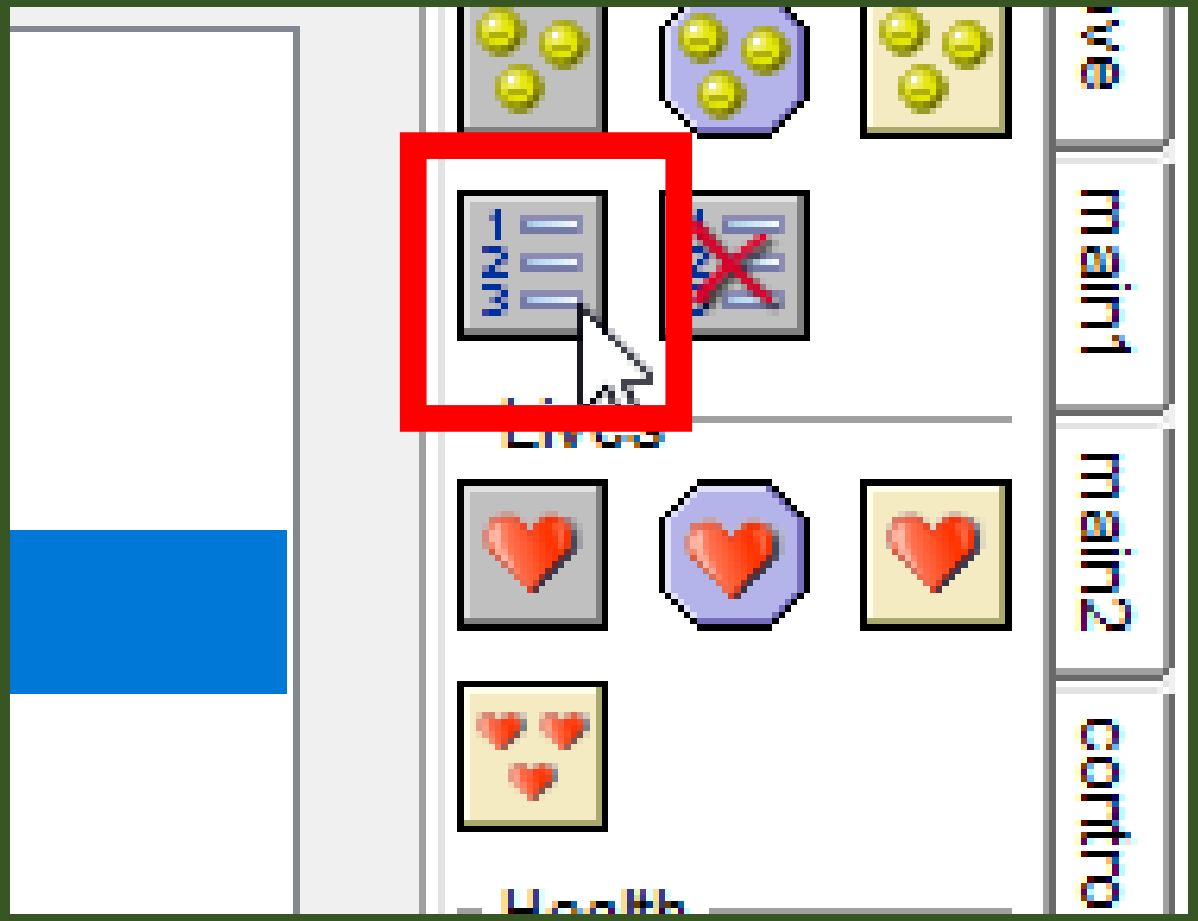
Set the Value to 0.
Set the operation to equal to.

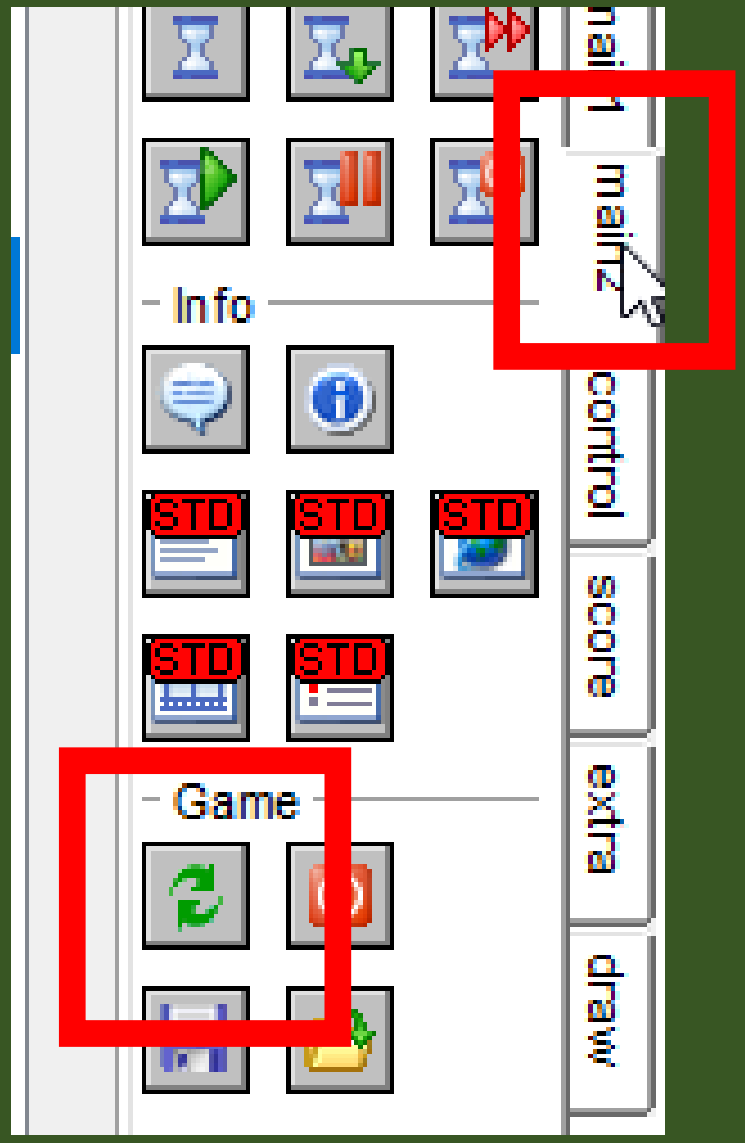
Then press OK



Under the Control tab, use the Start Block.

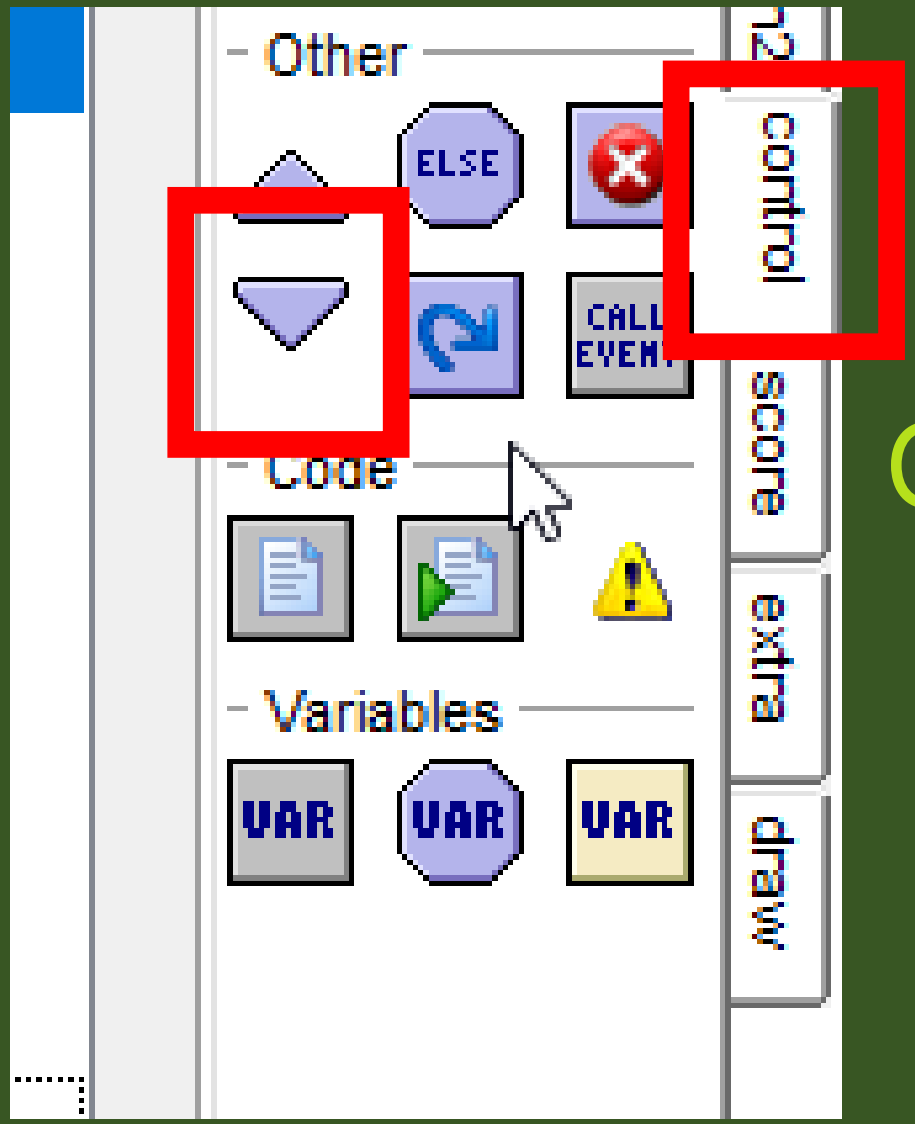
Now go to the score tab and drag the Show Highscore option



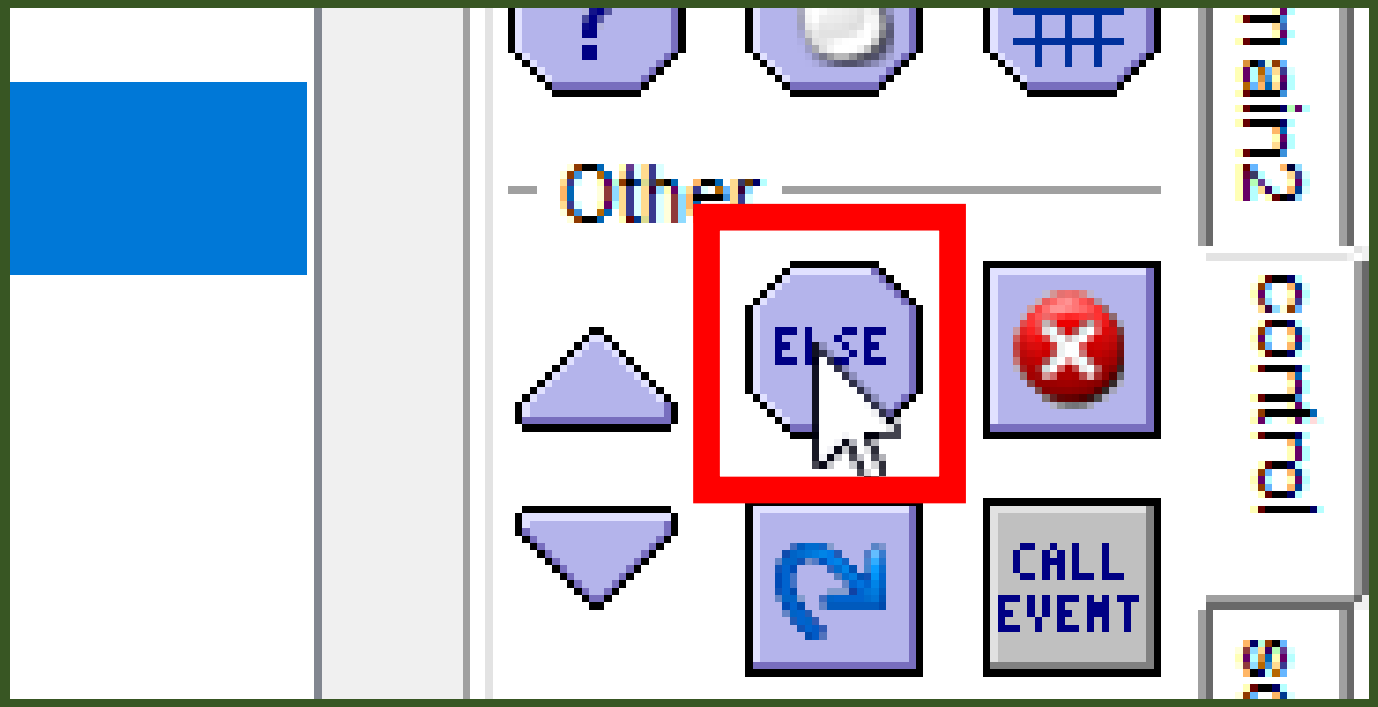


Go to the main2 tab.

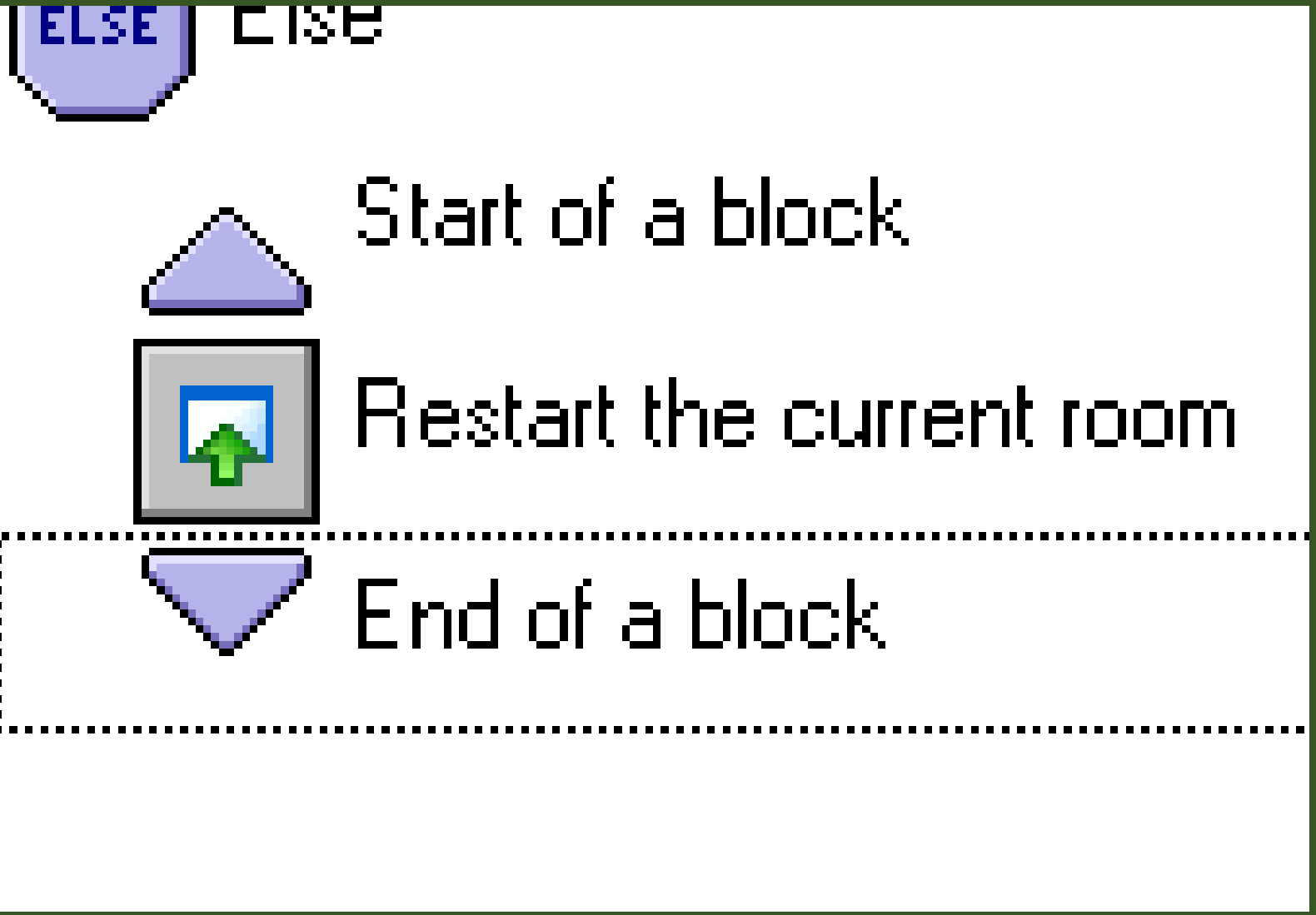
Under the Game section, use the restart game option.



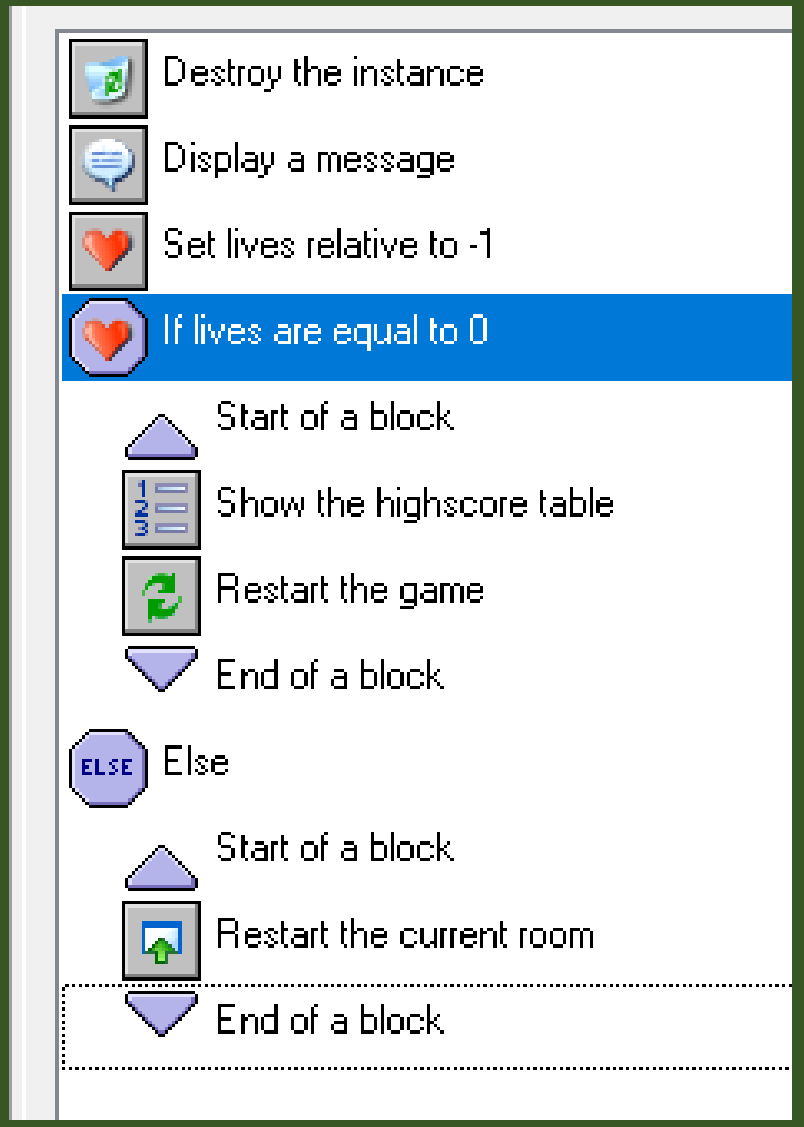
Go to the control tab. Drag a closing block underneath your actions.



Now add an Else block.



See if you can find these blocks now to finish our code.



This should be the completed code.

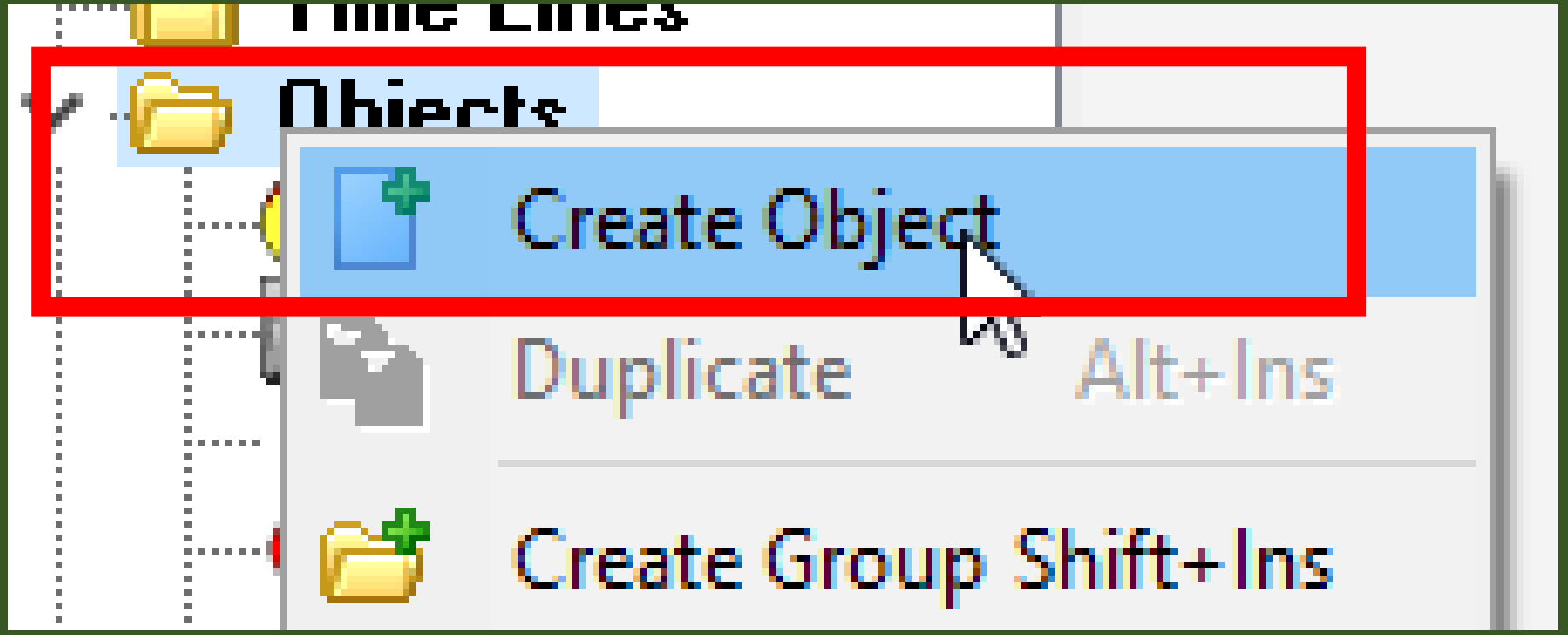
Check yours is correct otherwise it won't work!

Click OK and run your game. What happens if we collide with our enemy now?

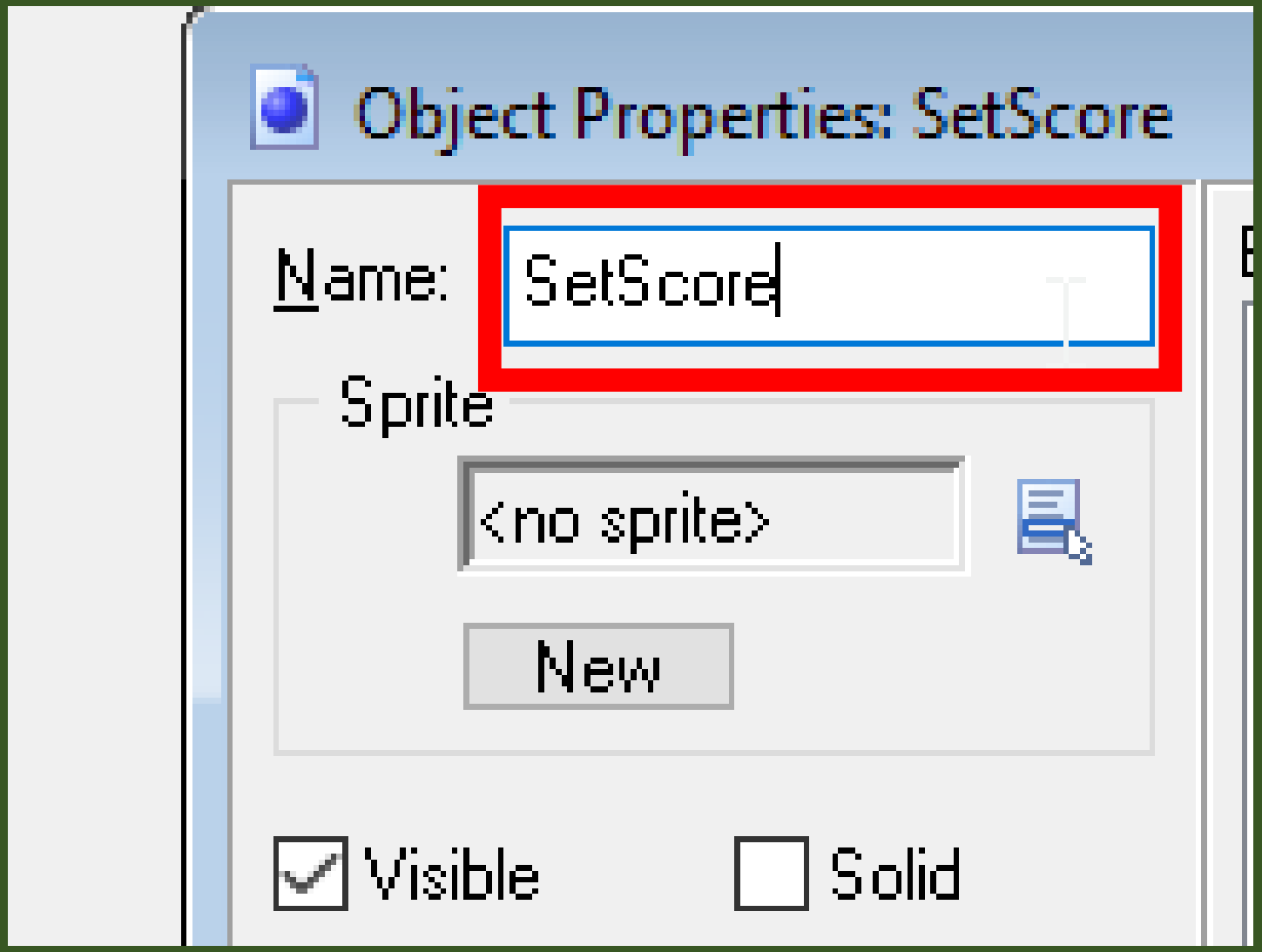




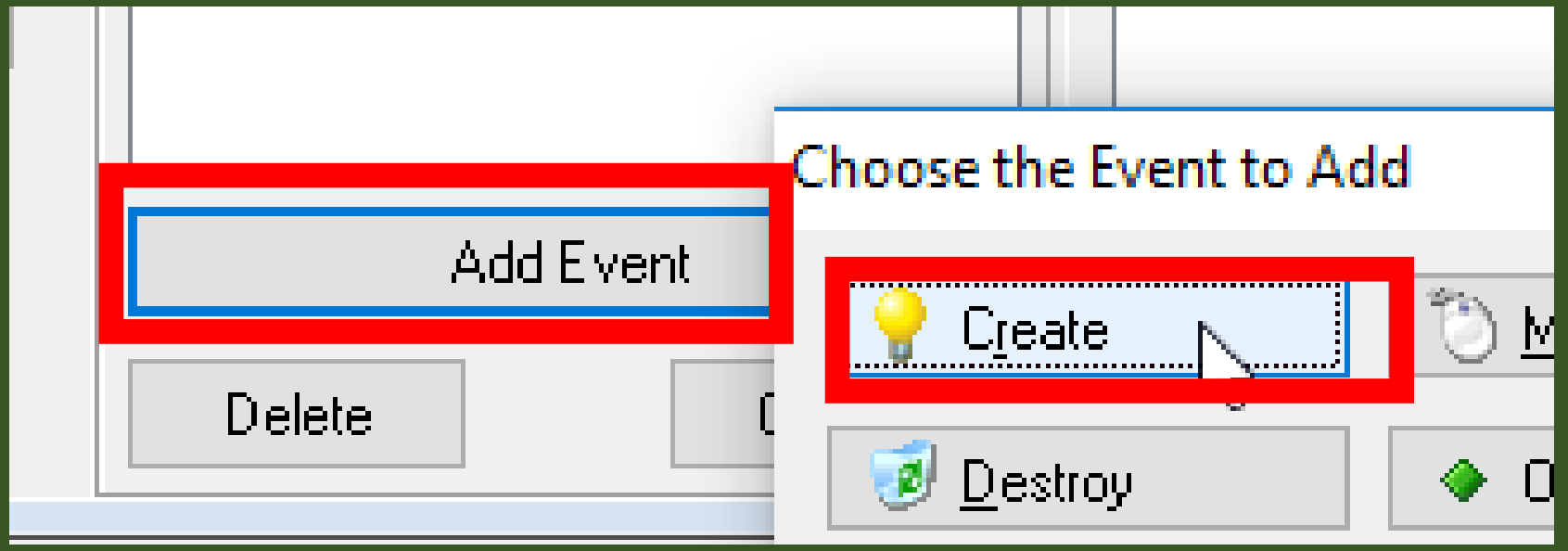
Let’s set up how many lives we should have



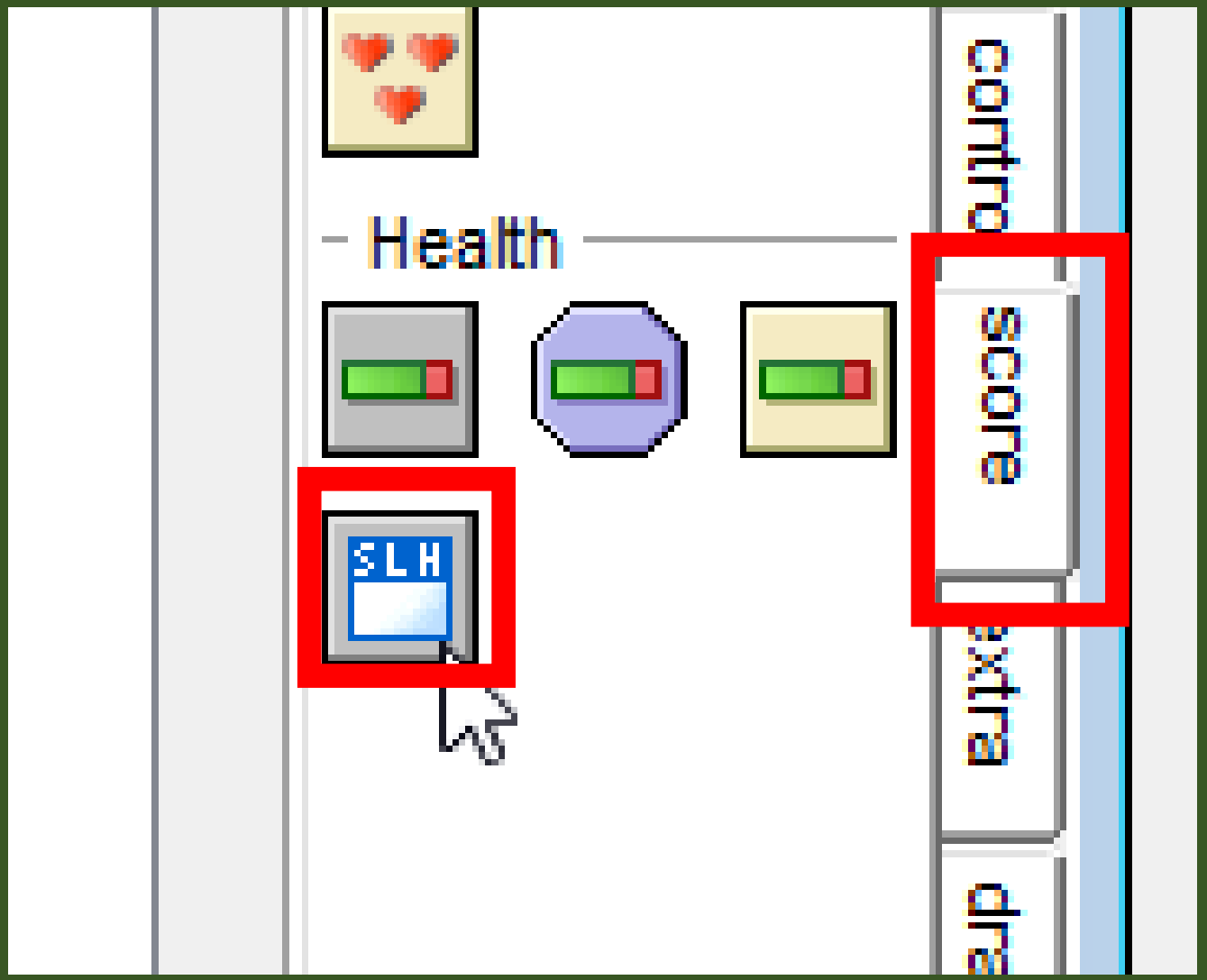
Right Click Objects > Create Object



Give this object a Name: SetScore

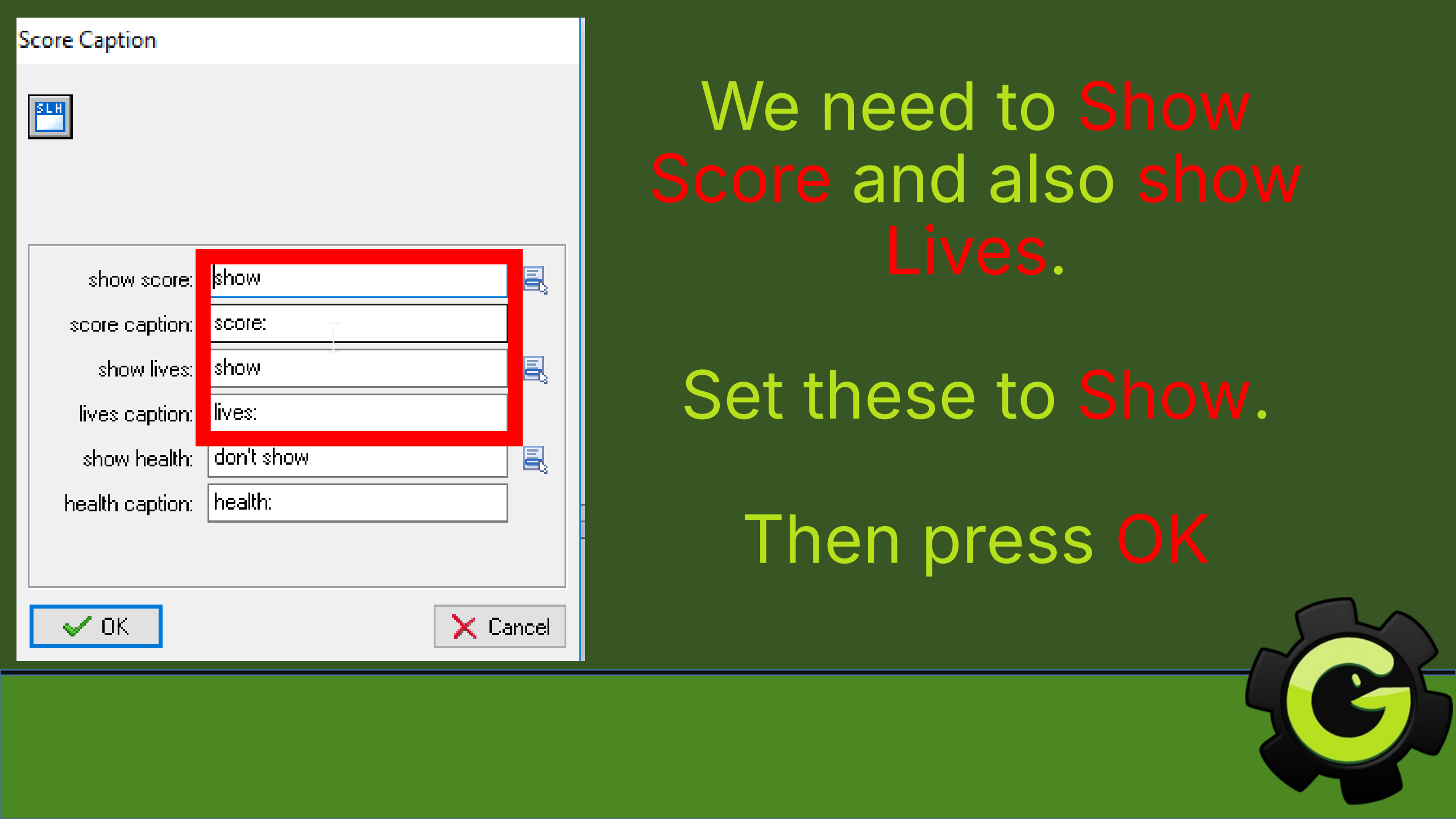


Click on Add Event > Create

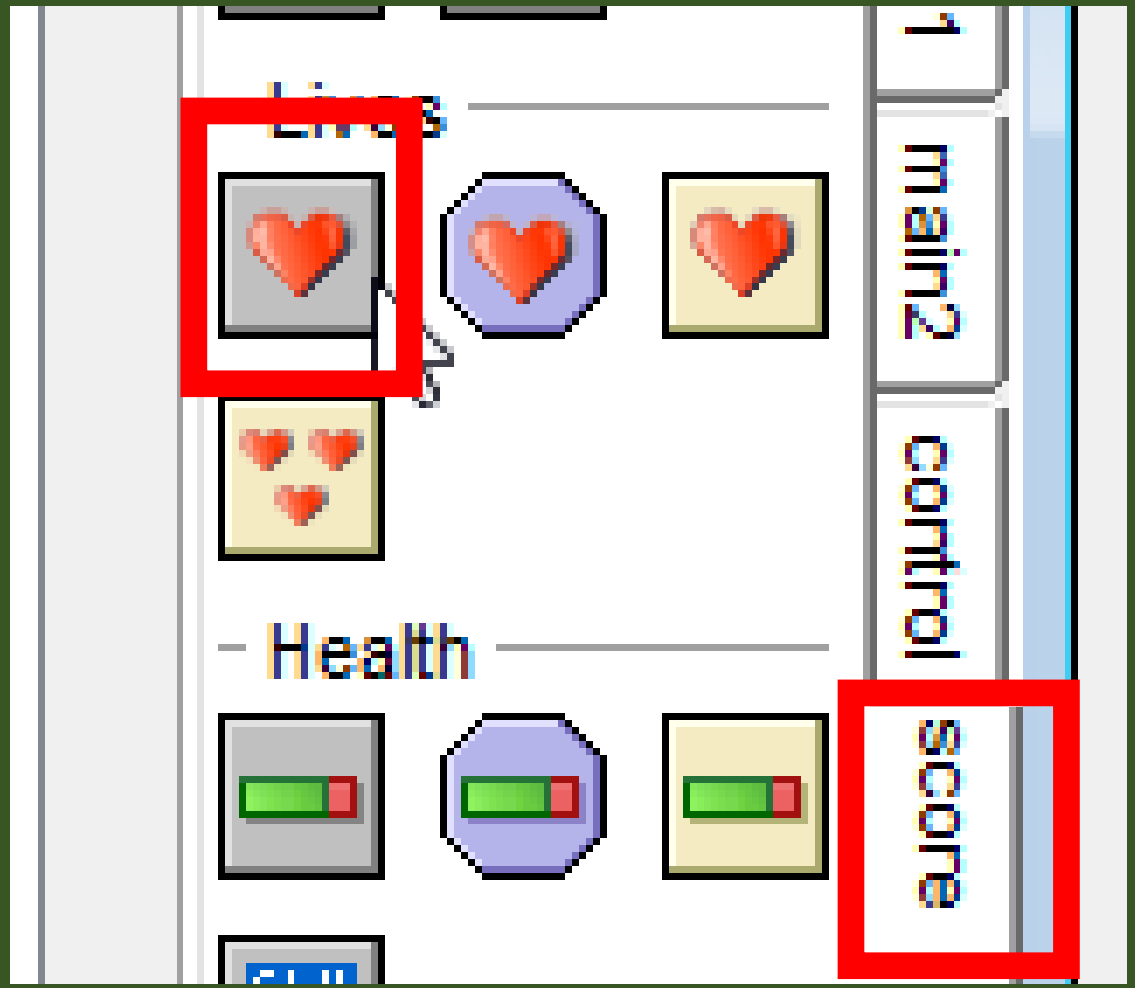


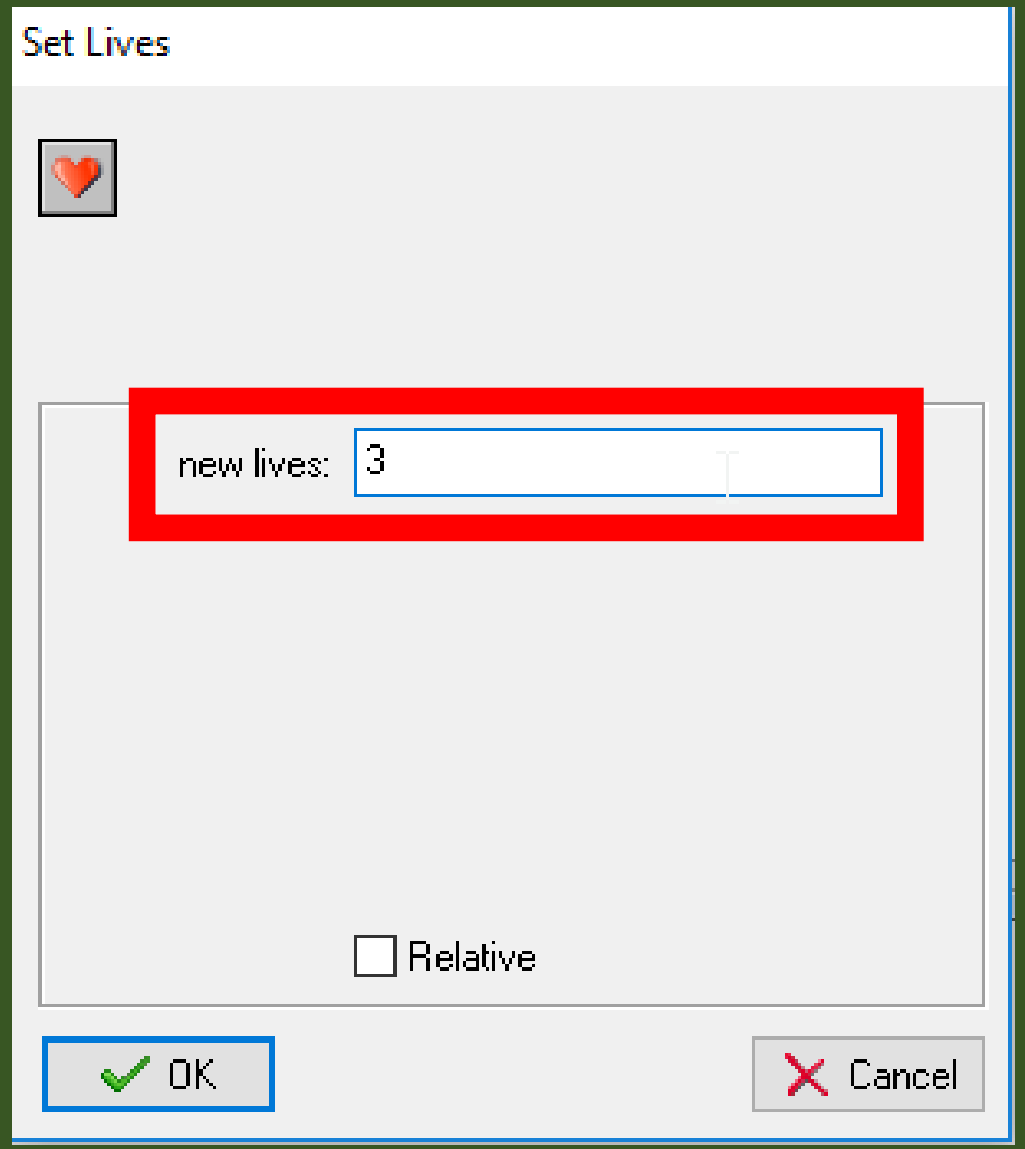
Under the Score tab, select SLH

Drag this onto your actions section.


Score Caption
show score:
show
score caption:
score:
show lives:
show
lives caption:
lives:
show health:
don't show
health caption:
health:
OK
Cancel
We need to Show Score and also show Lives.
Set these to Show.
Then press OK

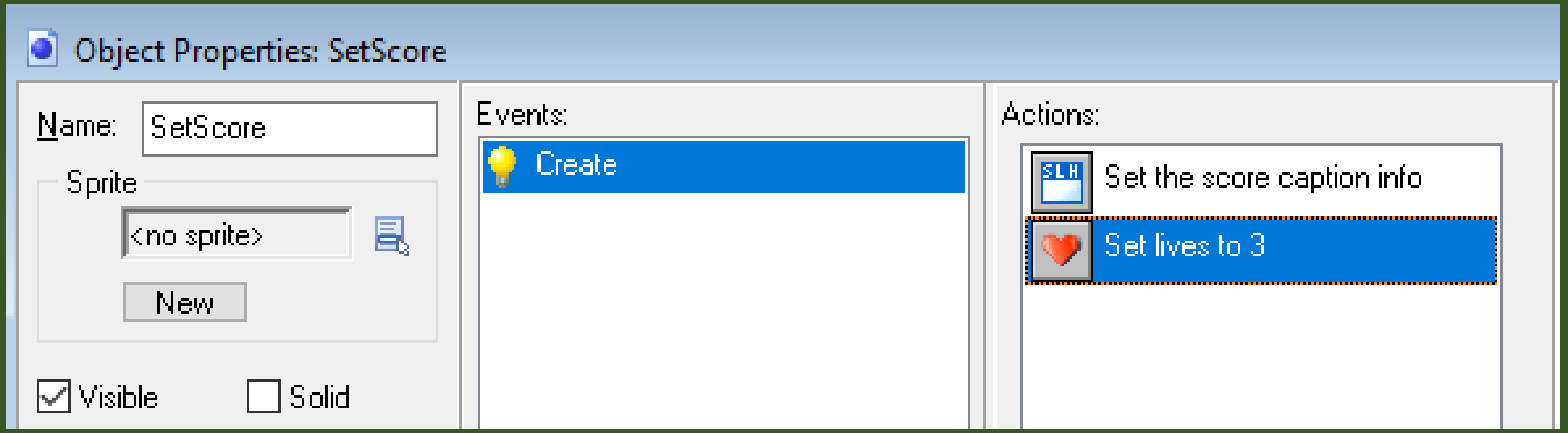
Still under the score tab, you need to drag the set lives option into your actions list.





Set your lives to 3.

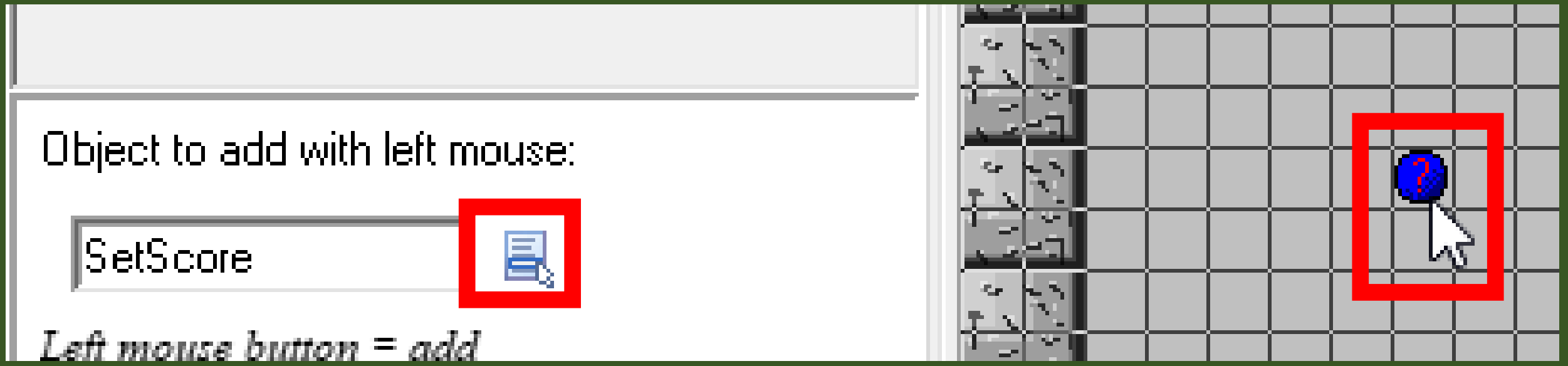
Then press OK



This is what your SetScore object should look like.



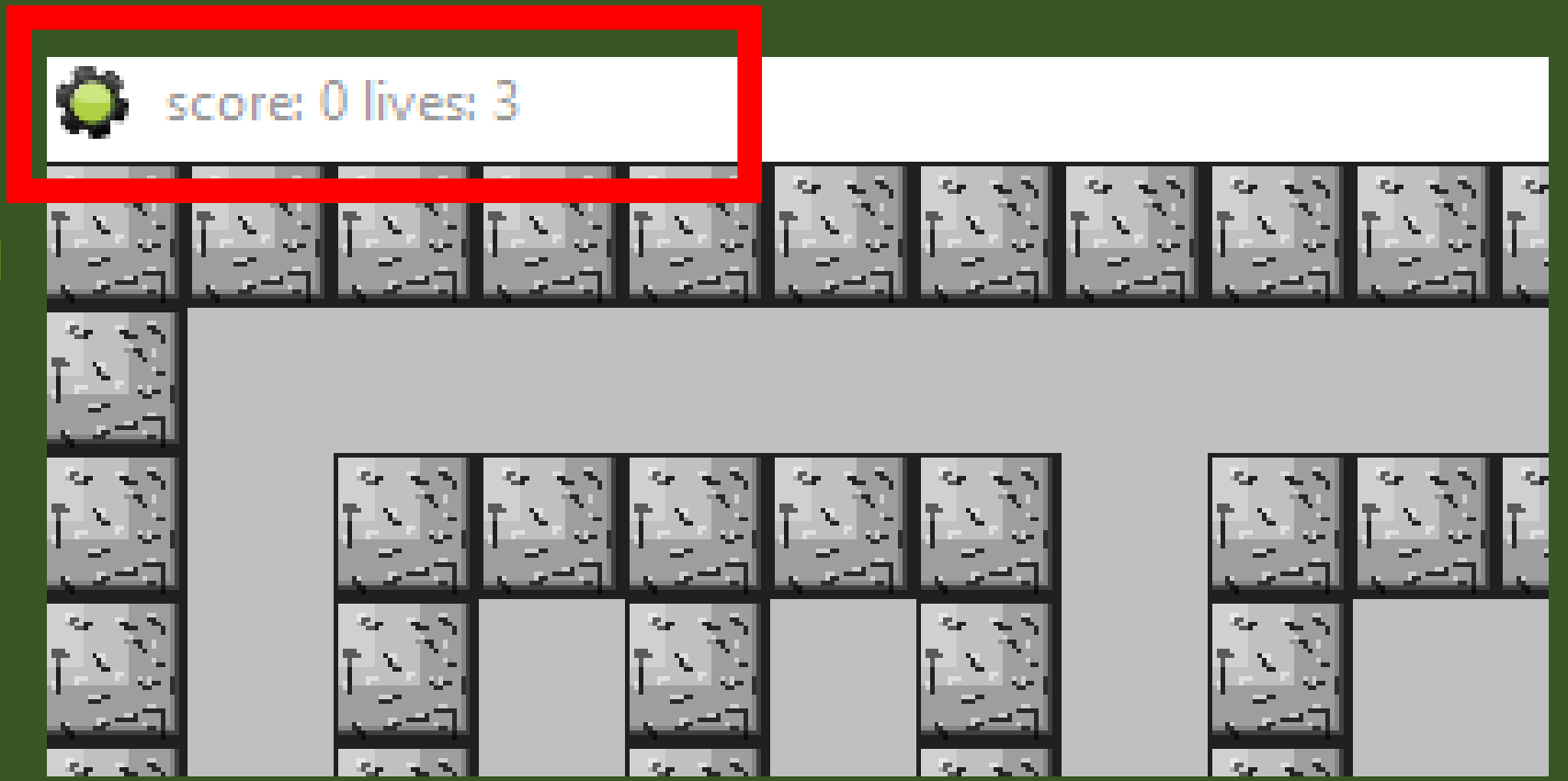
Double Click on the practiseRoom or rmPractise (the first room) to open its editor



Go back to your objects tab. Select SetScore and click once anywhere in your room.

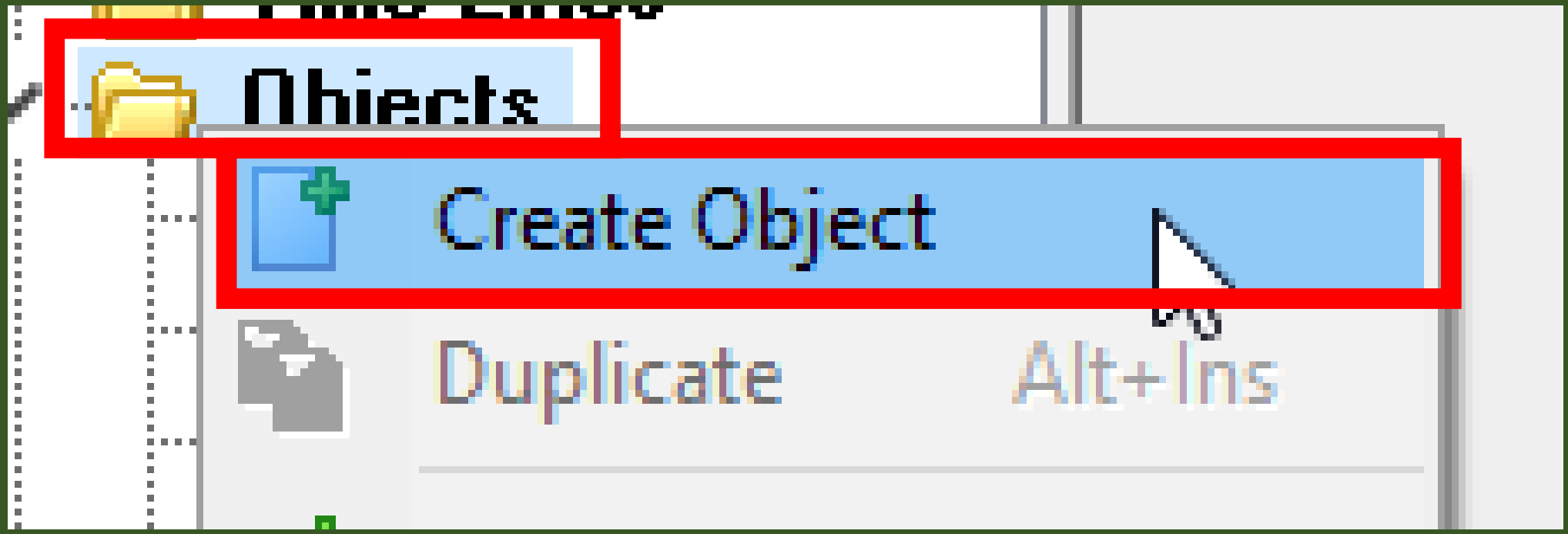
It should appear as a ? As it does not have a sprite to display.

Press the green tick to save and then run your game. Your game should display the score and lives.

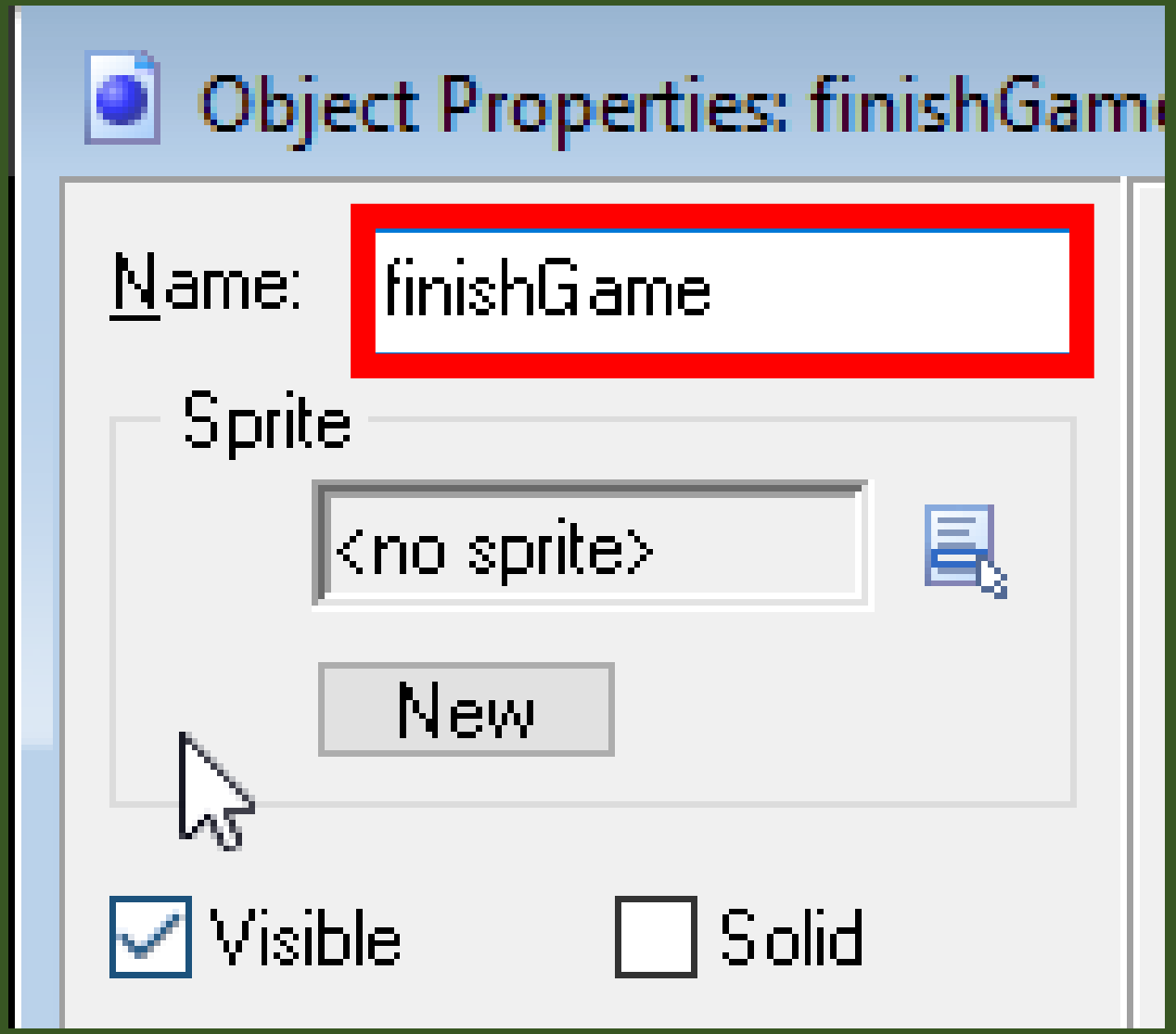


Let’s now add a ‘end game’ or ‘game finished’ screen.

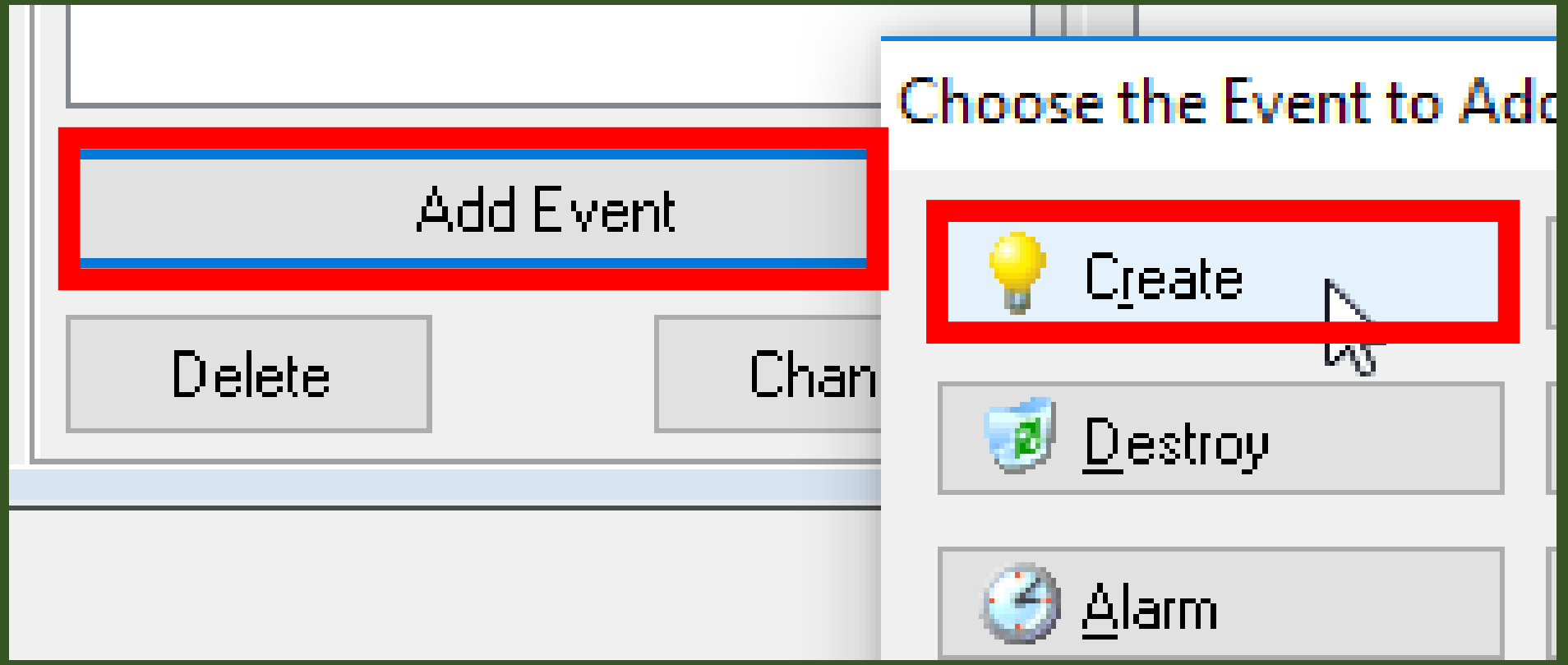




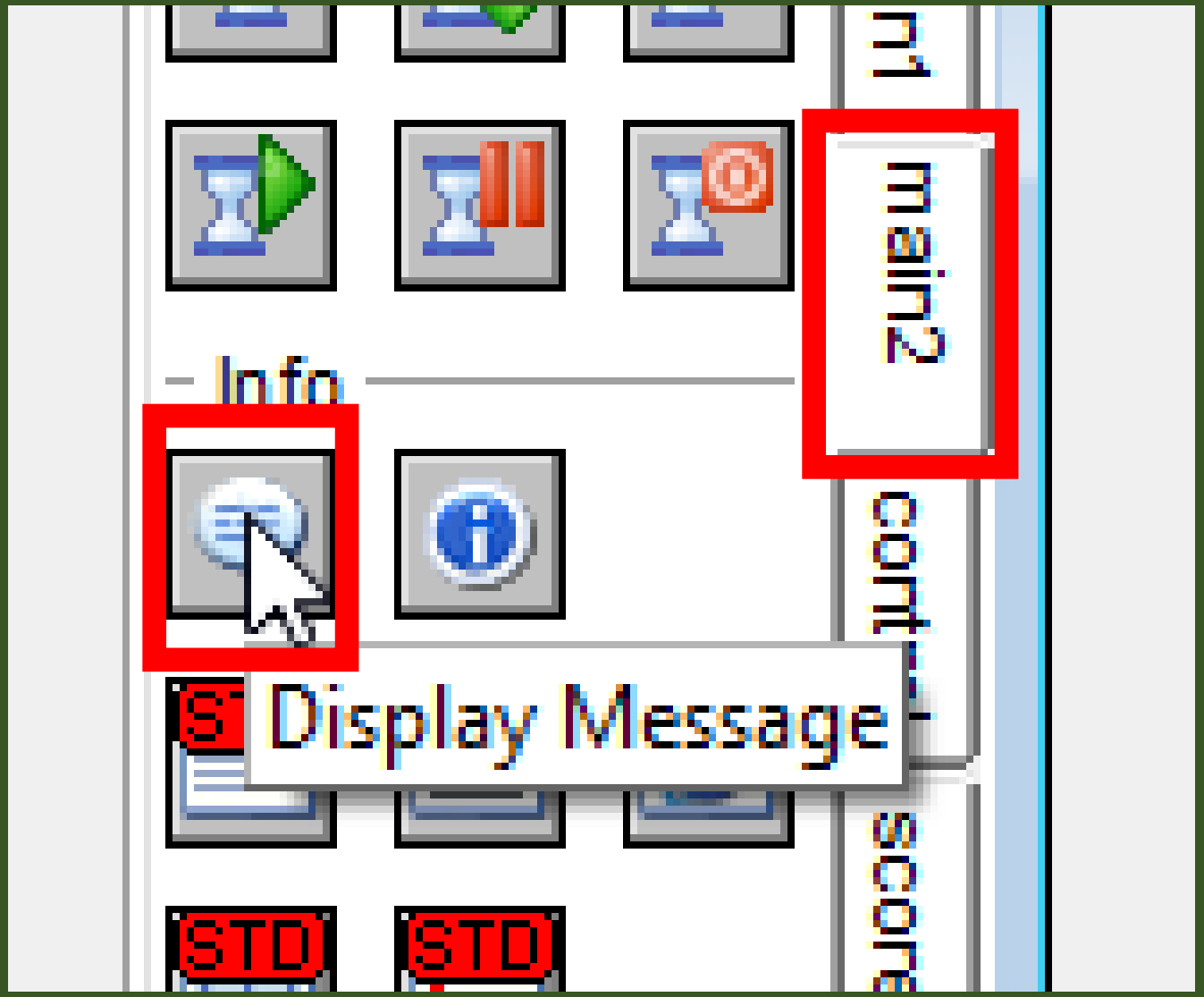
Right Click Objects > Create Object



Name this object finishGame. It does not require a sprite.



Add Event > Create



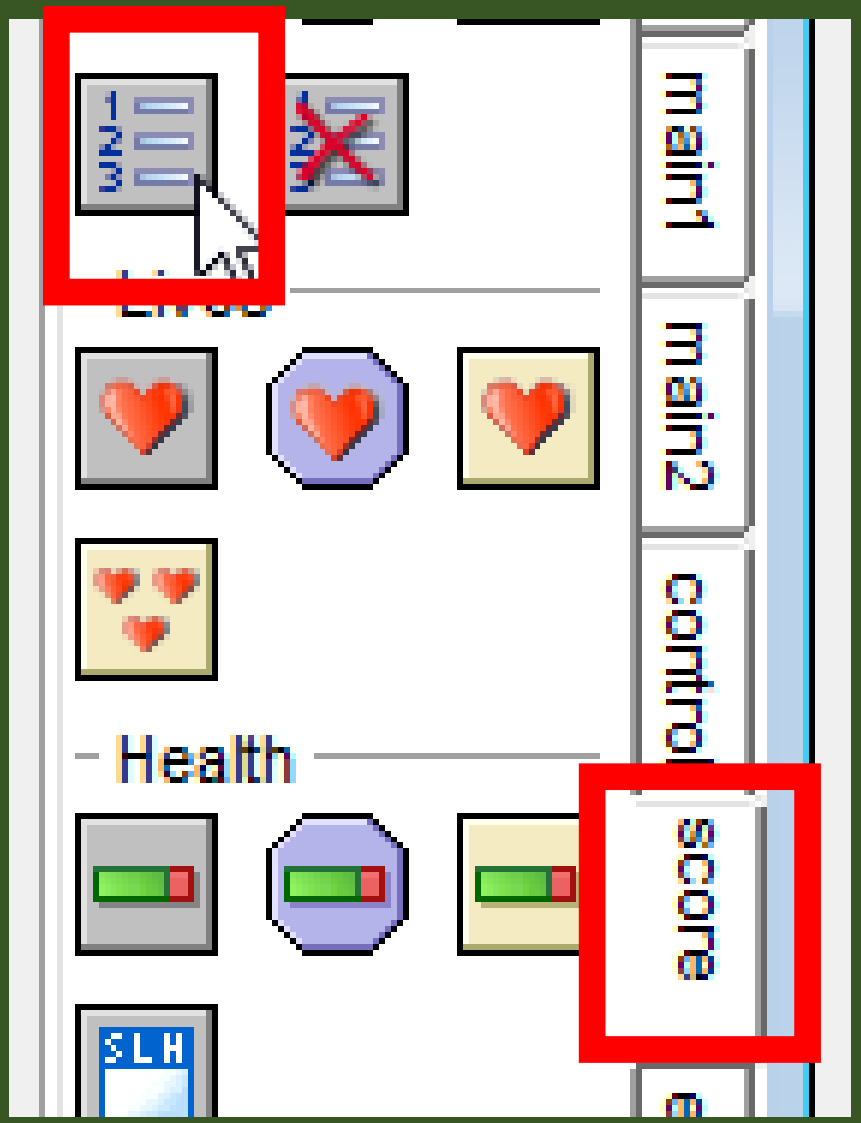
Under main2 tab, there is an option to display a message.

Drag this onto your actions area.

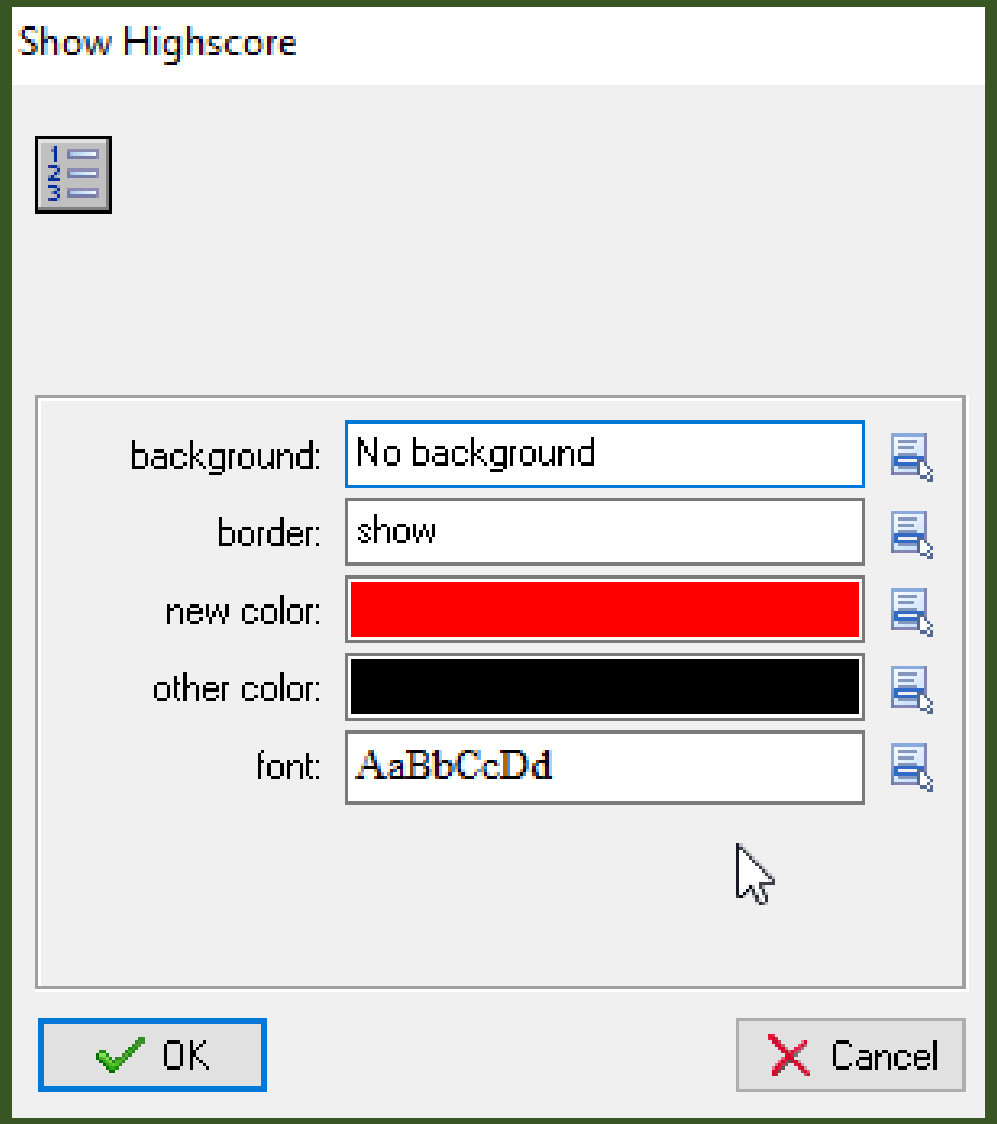
Set up the message as You Win!

Then press OK





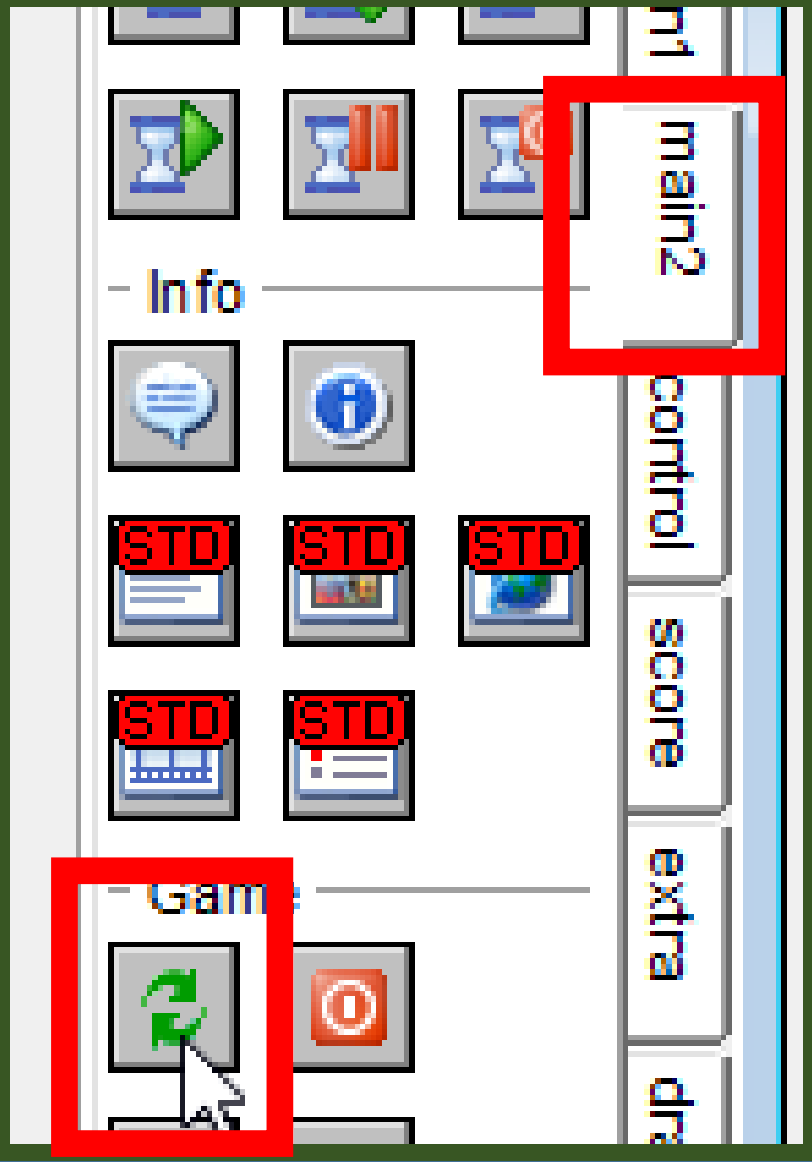
Under the score tab, drag the ‘show highscore’ option into your actions.



This will allow you to customise what the highscore table will look like.

We can come back to this but let's skip it for now.

Hit OK.



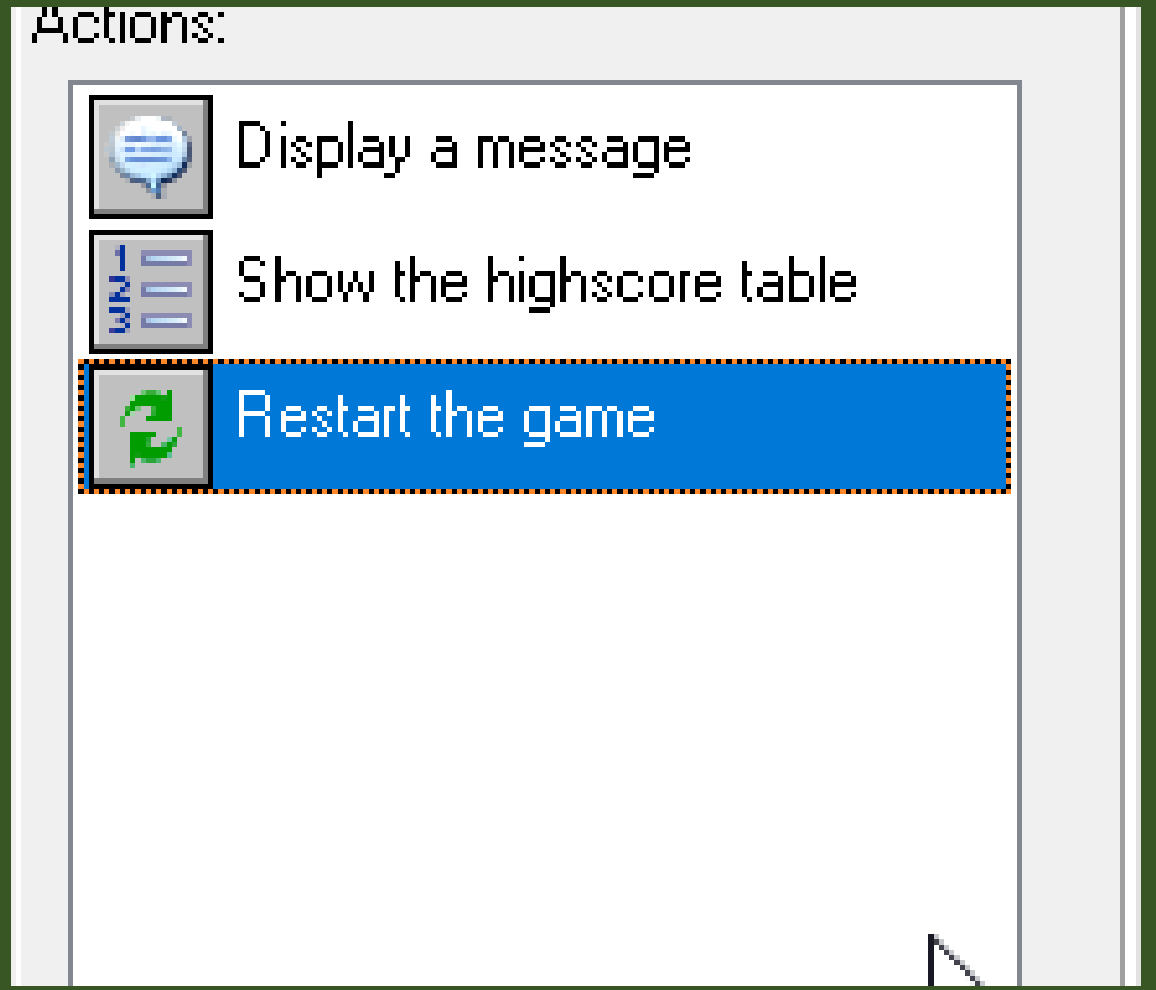
Click on the main2 tab.

Then use the ‘Restart Game’ action.

Is this what your code now looks like?

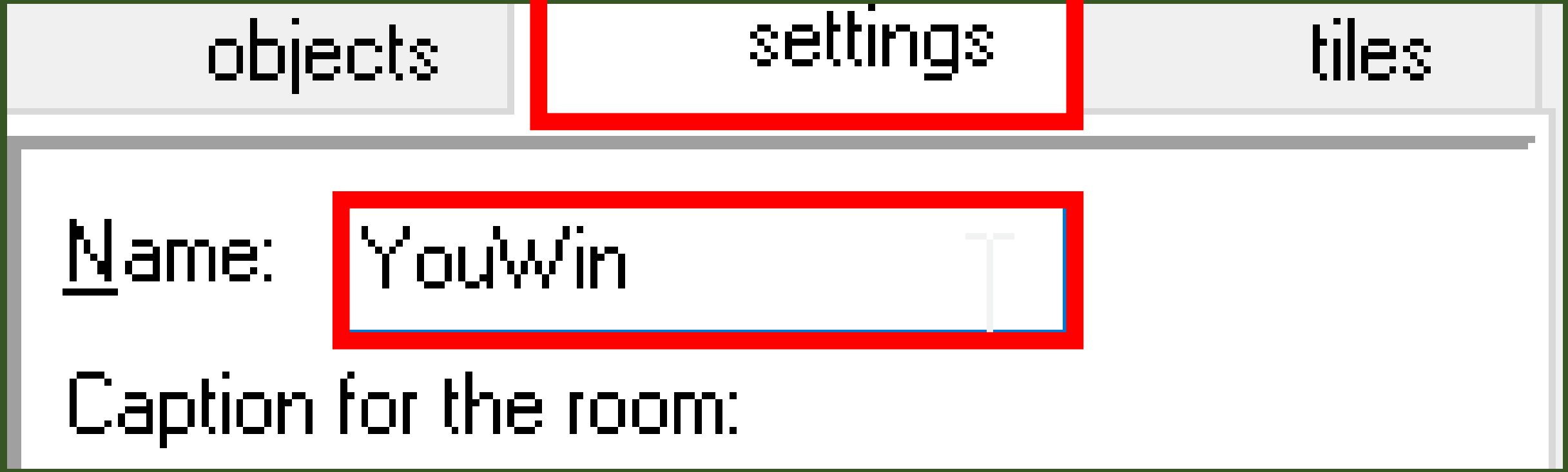
Perfect!

Click the Green Tick to save



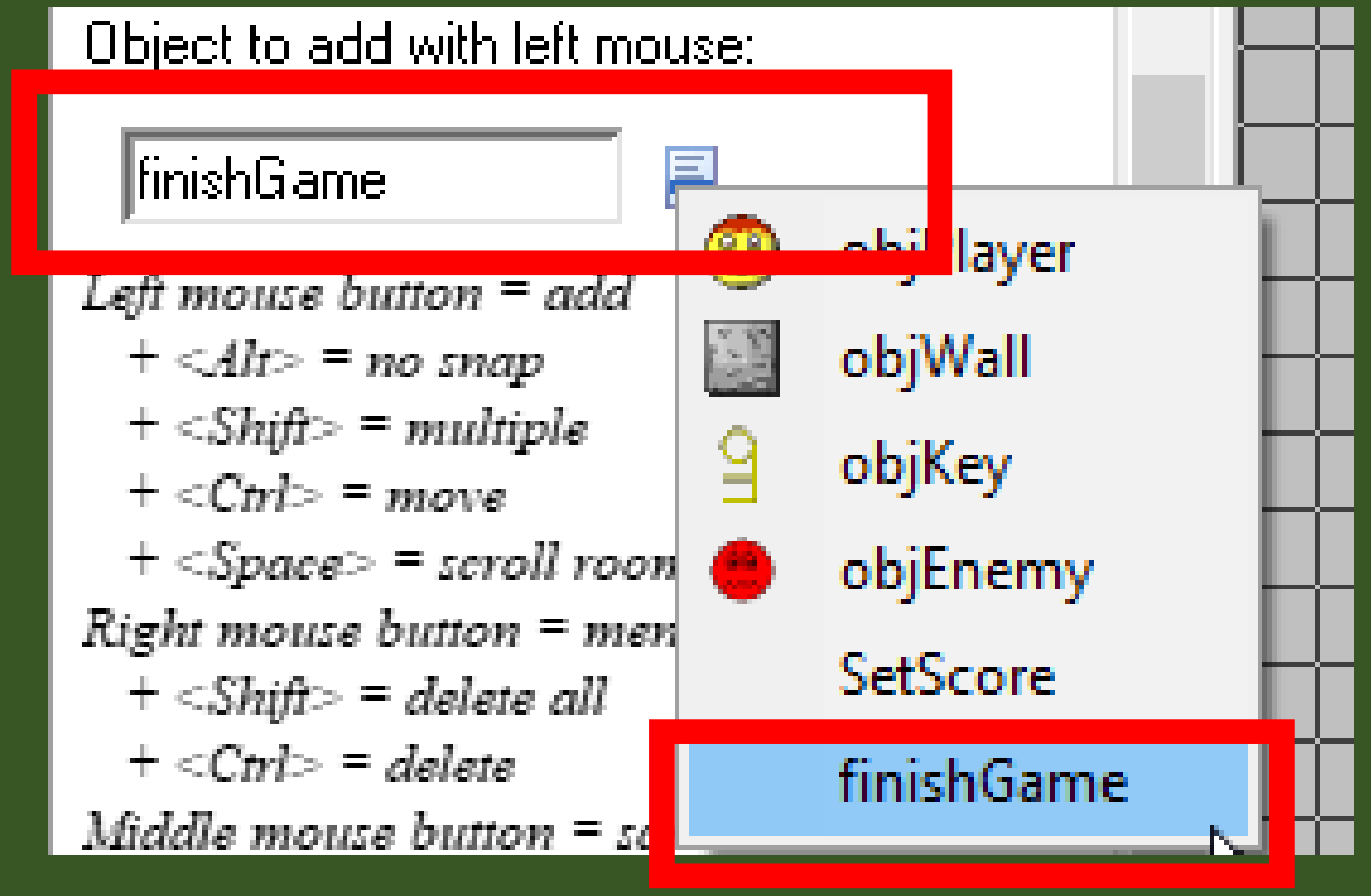


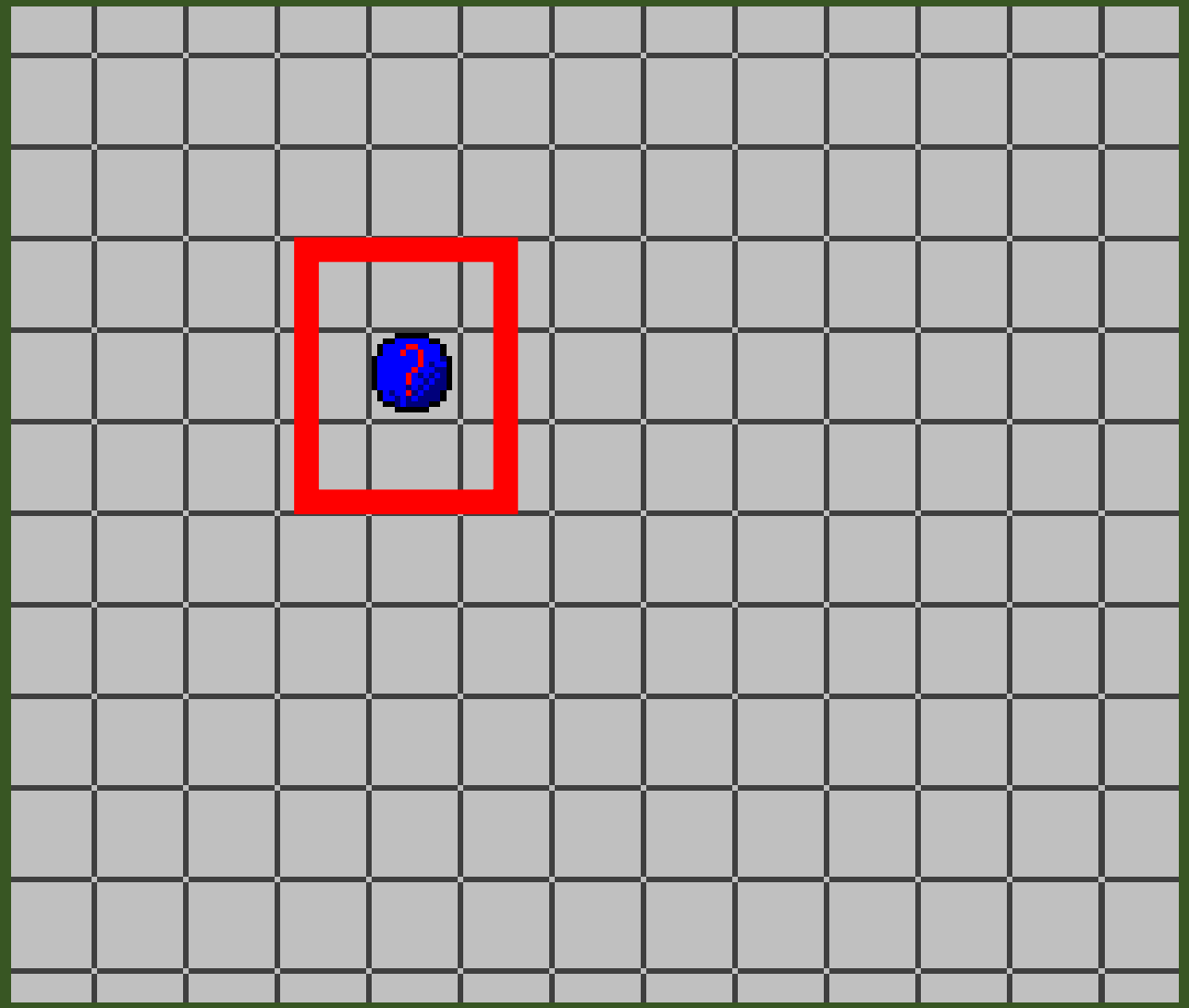
Right Click on Rooms > Create Room



Under the Settings tab, set the name as YouWin

Go now to the objects tab. Select the object finishGame we have just created.





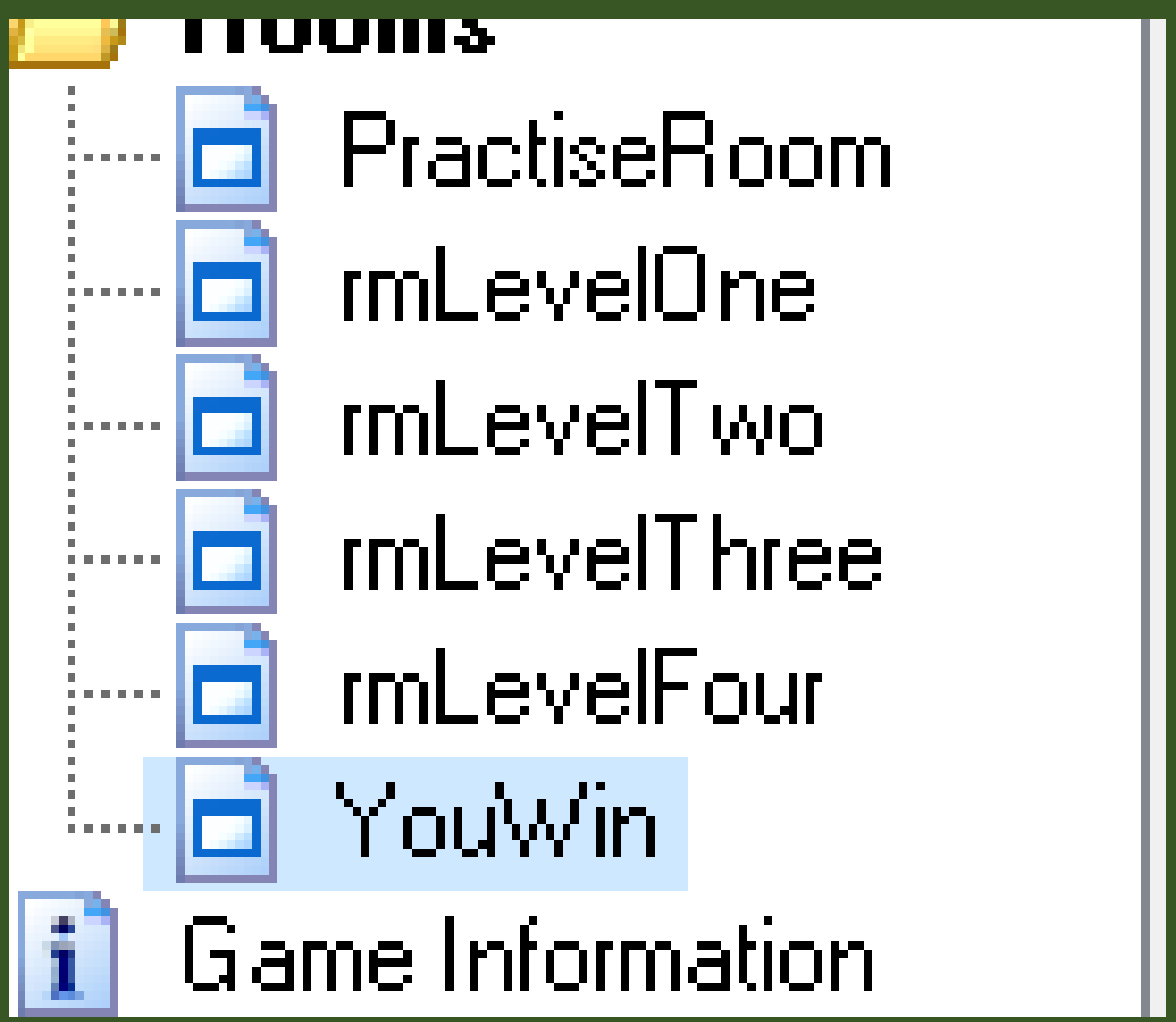
Click anywhere in the room.

Q. Why do we get the ? Again?

Save your work by clicking on the green tick.

Now run your game. Does everything work?





You can add new rooms/levels.

Make sure they appear between the PractiseRoom and YouWin!

Extensions

Can you add a front page to your game (Press space to start?)

Add in some collectibles (coins) into your game.

On collision, you need to:

- Destroy Instance
- Increase Score By 10 (relative)



Scoring and its Importance in Gaming

- Scoring refers to **an abstract quantity associated with a player or team**
- Most games with score use it as a quantitative indicator of success in the game, and in competitive games, a goal is often made of attaining a better score than one's opponents in order to win.
- Scoring is very different from game to game
  - Seen as a ‘way to win’ a game
  - Fortnite for example uses a ‘last man standing’ approach instead of a high score table.
  - Some games include no score at all, yet you can still ‘win’



Target Audience

- The ‘target audience’ can be defined as **a particular group at which a product such as a film or advertisement is aimed.**
- When developing a game, it is very important to keep the target audience in mind as it will ultimately be those players who will be playing your game.
- A game such as Fortnite will be aimed a majority young teens, yet a game like Chess will be aimed at logical thinkers and a more mature generation of gamer.

TASK:
Target Audience

- In your booklets there are two photos of games. Decide what would the target audience.
- Explain **why** you think this is the case.
  - Hint: Think about the colour schemes, icons, gameplay you think there is.





Practical Assessment

Magic Amusements

Magic Amusements

- Magic Amusements are an up and coming Game Development company, situated in Belfast. They specialise in 3D games, however are wishing to expand their company by developing a 2D game aimed at teenagers aged 10-14.

- The Game should be developed using the Game Maker software, and should have a continuous theme throughout. The title of the Game has been left to you to decide, but should be suitable to the target audience.

Magic Amusements

- The aim is to create an engaging first product which will allow the user to:
- View an introductory Title Screen, which includes a company logo and information of how to access the main menu
- A Main menu, which allows the user to:
  - Start the Game
  - Access A help feature
  - View Highscore
  - Load the previously saved game
  - View Credits of the Game
- At least three levels, each of which get slightly harder. All levels should be possible to complete
- A Scoring system
- Collectables

Magic Amusements:
Planning

Storyboarding

Within your game, you will be required to storyboard your:

- Front/Opening Page | Your loading Screen | The Different Levels | The final/End Credits Screen | Help Page

Use the space below to design these Screens. You will need to label the different colours, fonts and layouts for each screen.

Opening Screen:	**Loading Screen:**	**Level 1**
Level 2	**End Credits Screen**	**Help Page**

Magic Amusements: Creation

Let’s Get Creating!

I will of course help you as you go along.

Just remember to use your storyboarding sheet to help!